Exploiting Structural Synergy in post-SIRT imaging: Anatomically Guided Joint Variational Reconstruction of Yttrium-90 PET/SPECT/CT

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I, Sam Porter, confirm that the work presented in this thesis is my own. Where information has been derived from other sources, I confirm that this has been indicated in the work.

Abstract

Selective Internal Radiotherapy (SIRT) uses Yttrium-90 (^{90}Y) microspheres to treat unresectable liver tumours. Accurate post-treatment imaging is required to verify microsphere distribution and perform dosimetric evaluations. However, imaging with ^{90}Y presents unique challenges. Bremsstrahlung SPECT offers high count statistics but suffers from poor spatial resolution and difficulties in correcting for scatter and other additive terms. ^{90}Y PET offers superior resolution, but suffers from high noise due to the extremely low internal pair branching ratio ($\sim 0.003\%$). Traditional reconstruction fails to exploit complementary information in these paired datasets.

This thesis proposes a synergistic framework that uses anatomically guided joint variational regularisation to exploit shared underlying structure. We introduce the Directional Total Nuclear Variation Prior, which promotes edge alignment between PET and SPECT without enforcing intensity correlation, while incorporating CT edge-guidance and suppressing unwanted noise.

Further to this, we introduce a simple method of bremsstrahlung range modelling approach and residual correction routine to improve quantification.

To ensure computational efficiency, we propose a synergy-aware preconditioner which, in conjunction with a variance reduced stochastic gradient algorithms, results in fast convergence.

We validate our method against the state-of-the-art sequential hybrid kernel expectation maximisation scheme (HKEM). We demonstrate that dTNV shows improved quantitative accuracy and increased reproducibility over noise realisations compared to existing methods.

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Acknowledgements

So, it turns out that a PhD is quite an undertaking. It's been a long old journey, but the end is in sight!

I've had an extraordinary four years and would like to thank some extraordinary people. Firstly, I'd like to thank my supervisors, Kris, Daniel, and Simon, who have been a near-bottomless source of knowledge, inspiration and ideas. Secondly, my colleagues in the INM, especially my fellow students. This would have been a very difficult (and much lonelier) few years without you all. I'm so grateful for our conversations in and out of the workplace. I'd like to also thank all of my colleagues in the scientific and medical spheres, whom I've found to be endlessly generous with their time and patience.

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***w*-dTNV** weighted directional total nuclear variation.

^{90}Y Yttrium-90.

^{90}Zr Zirconium-90.

BH Benjamini–Hochberg.

BSREM block-sequential regularised expectation maximisation.

CDR collimator–detector response.

CoV coefficient of variation.

CT computed tomography.

CTAC computed tomography attenuation correction.

DTV directional total variation.

EM expectation–maximisation.

ET emission tomography.

FDG ^{18}F-fluoro-2-deoxyglucose.

FoV field of view.

FWHM full width at half maximum.

GATE Geant4 Application for Tomographic Emission.

HCC hepatocellular carcinoma.

HEGP high-energy general-purpose.

HKEM hybrid kernelised expectation maximisation.

HKRL Hybrid Kernelised Richardson–Lucy.

HU Hounsfield units.

IEC International Electrotechnical Commission.

IPP internal pair production.

IRLS iteratively reweighted least squares.

JTV joint total variation.

KEM kernelised expectation maximisation.

KRL Kernelised Richardson–Lucy.

LoR line of response.

LUT look-up table.

MAA macroaggregated albumin.

MAP maximum a posteriori.

MC Monte Carlo.

MEGP medium-energy general-purpose.

ML maximum-likelihood.

MLEGP medium–low-energy general-purpose.

MLEM maximum-likelihood expectation–maximisation.

MM majorisation–minimisation.

MR magnetic resonance.

MRI magnetic resonance imaging.

NaI(Tl) thallium-doped sodium iodide.

NEMA National Electrical Manufacturers Association.

NRMSE normalised root-mean-square error.

OSEM ordered subsets expectation–maximisation.

PDF probability density function.

PET positron emission tomography.

PLS parallel level sets.

PSF point spread function.

PVC partial volume correction.

RC recovery coefficient.

RDP relative difference prior.

RL Richardson–Lucy.

RL–DTV Richardson–Lucy with Directional Total Variation.

SAGA stochastic average gradient amélioré.

SHKEM sequential hybrid kernelised expectation maximisation.

SIMIND Simulation of Imaging Nuclear Detectors.

SIRF Synergistic Image Reconstruction Framework.

SIRT selective internal radiotherapy.

SPD symmetric positive-definite.

SPECT single photon emission computed tomography.

STIR Software for Tomographic Image Reconstruction.

SVD singular value decomposition.

SVRG stochastic variance–reduced gradient.

TARE transarterial radioembolisation.

TBR tumour-to-background ratio.

TNV total nuclear variation.

ToF time of flight.

TV total variation.

VOI volume of interest.

Notation

x Emission activity image (voxel intensities), $x \in \mathbb{R}_+^J$.

y Measured projection data (counts), $y \in \mathbb{N}^I$.

$\bar{y}(x)$ Expected measured data under the forward model, $\bar{y}(x) = Ax + b$.

A System (forward-projection) matrix, $A = [a_{ij}] \in \mathbb{R}_+^{I \times J}$.

b Additive term (scatter, randoms, and other background contributions).

C Image-domain point-spread-function convolution (blurring) operator used in deconvolution.

μ Linear attenuation coefficient / CTAC-derived attenuation (μ-)map.

h Source-to-collimator distance (SPECT depth-dependent resolution).

I Number of measurement bins (lines of response / projection bins).

J (J_m) Number of voxels in the native / modality-m image space.

N Number of voxels on the shared reconstruction grid.

M Number of coupled modalities.

d Number of directional finite-difference channels (gradient dimension).

i Measurement-bin index.

j Voxel index (native grid).

k Iteration / epoch index (written as a superscript, $x^{(k)}$).

m Modality index.

n Voxel index (shared reconstruction grid).

s Data-subset index.

v Sphere / volume-of-interest index.

c Reconstruction-method (comparison) index; c_0 denotes the reference method.

bed PET bed-position index, $\text{bed} = 1, \ldots, N_{\text{bed}}$.

bs Bootstrap-realisation index, $\text{bs} = 1, \ldots, N_{\text{bs}}$.

D Discrete (directional) finite-difference operator, $(Dx)_n \in \mathbb{R}^d$.

div Discrete divergence, the negative adjoint of D.

∇ Gradient of a functional (e.g. $\nabla_x \mathcal{L}$, $\nabla_{\mathcal{G}} \phi$).

$\Pi_{m,n}$ Anisotropic directional projection operator built from the guidance field.

W_m Registration / resampling operator from native grid to shared grid; W_m^* its adjoint.

K Kernel matrix (kernelised image model $x = K\alpha$).

α Kernel coefficient image.

$\mathcal{P}$ Preconditioner (diagonal or voxel-wise block).

$\mathbb{I}$ Identity matrix ($\mathbb{I}_d$ when the dimension is made explicit).

$\mathbf{1}$ All-ones vector.

R_{bed} Restriction from the combined PET image to a bed position's axial support.

$\mathcal{D}$ Data-fidelity term (Poisson negative log-likelihood).

$\mathcal{L}$ Negative Poisson log-likelihood (up to an additive constant).

$\mathcal{R}$ Regularisation functional (prior).

$\mathcal{O}$ Full optimisation objective, $\mathcal{O} = \mathcal{D} + \mathcal{R} + \iota_{\mathbb{R}_+}$.

$\mathcal{C}$ Feasible set (typically the non-negative orthant).

$\iota_{\mathbb{R}_+}$ Indicator function of the non-negative orthant.

β Regularisation strength (prior weight).

$\rho_{m,n}$ Modality weight; collected in the diagonal matrix $B_n = \mathrm{diag}(\rho_{1,n}, \rho_{2,n})$.

γ, γ_m RDP edge-preservation parameter γ; modality-specific directional-guidance strength $\gamma_m \in [0,1]$.

η Direction-field stabilisation parameter (distinct from the relaxation parameter η_{relax}).

ε Numerical stabiliser in the RDP and Charbonnier / smoothed-TV parameter (subscripted variants denote other local stabilisers).

ξ_n Normalised guidance direction field.

u Anatomical guidance image (for the ^{90}Y study, $u = \mu$).

z_m, $\boldsymbol{z}$ Modality-m image on the shared grid; $\boldsymbol{z} = (z_1, \ldots, z_M)$ the stacked images.

$\mathcal{G}_n$ Voxel-wise multi-modality Jacobian (columns are modality gradients).

$\tilde{\mathcal{G}}_n$ Weighted directional Jacobian, $\tilde{\mathcal{G}}_n = \mathcal{G}_n^{\mathrm{dir}} B_n$.

ϕ Smoothed spectral potential applied to the Jacobian, $\phi(Y) = \sum_\ell g(\sigma_\ell(Y))$.

g Charbonnier smoothing function of a singular value.

σ_ℓ ℓ-th singular value.

$\|\cdot\|_*$ Nuclear norm (sum of singular values).

V_{vox} Voxel volume.

$\varsigma^{(k)}$ (ς_k) Step size at iteration k (relaxed / diminishing over iterations).

$\tilde{x}$, $\tilde{z}$ Stored snapshot iterate used by the variance-reduced estimators (SVRG/SAGA).

$g^{(k)}$ (Variance-reduced) stochastic gradient estimator at iteration k.

$H_{\mathcal{R}}$ Hessian of the prior (used for prior-aware preconditioning).

p Matrix-Lehmer combination order ($p = 0$ gives the parallel sum).

S Number of data subsets.

N_{d} Number of stochastic data functions in a reconstruction.

s_m Modality-m sensitivity image on the shared reconstruction grid.

η_{relax} Step-size relaxation / decay parameter.

ε_x Small positive image-domain stabiliser used in data preconditioning.

$\tau_{m,n}$ Finite-difference participation factor in the diagonal prior-curvature approximation.

$\hat{x}$ Reconstruction estimate.

$x^{\star}$ Digital ground truth or long-run reference image.

$\mathcal{S}_v$ Set of voxel indices belonging to sphere v.

$\mathcal{B}$ Background region of interest.

$\mathcal{V}$ Volume of interest (VOI).

$\bar{x}$ Regional / inter-bootstrap sample mean.

SD Sample standard deviation.

$\Delta_x, \Delta_y, \Delta_z$ Voxel dimensions; Δ_ℓ is the length of finite-difference offset ℓ.

Chapter 1

Introduction

Yttrium-90 (^{90}Y) selective internal radiotherapy (SIRT), also referred to as transarterial radioembolisation (TARE), is an established treatment for primary and secondary liver malignancies [**sundramSelectiveInternalRadiation2017**]. In this procedure, Yttrium-90 (^{90}Y)-labelled microspheres are administered into the hepatic artery to preferentially irradiate tumour tissue while sparing healthy liver parenchyma. The administered activities are high (typically of the order 2 GBq), and clinically relevant absorbed-dose heterogeneity can arise at multiple spatial scales due to vascular delivery, microsphere clustering, and patient-specific hepatic anatomy [**strigariEvidenceBaseUse2014**]. Tumour doses of clinical interest often exceed 200 Gy [**salemClinicalDosimetricConsiderations2019**], and recent evidence has shown potential benefit from additional subsequent treatment. Additional treatments could lead to more dose to healthy liver tissue without proper treatment planning and verification [**reinckeHepaticDecompensationTransarterial2022**]. As a consequence, post-therapy quantitative imaging and dosimetry are central to treatment verification, outcome modelling, and the optimisation of any subsequent therapy.

Despite its importance, routine post-selective internal radiotherapy (SIRT) dosimetry is not universally performed in clinical practice, primarily due to practical workflow constraints and limitations in quantitative accuracy [**kappadathQuantitativeEvaluation90YPET2024**, **gearEANMEnablingGuide2023**]. Post-therapy imaging of ^{90}Y is inherently challenging: image quality is often degraded by low effective count statistics and by physics that is less favourable than standard diagnostic radionuclide imaging, leading to increased noise and modelling error [**kappadathQuantitativeEvaluation90YPET2024**]. More broadly, emission tomography is vulnerable to substantial degradation from Poisson counting statistics as well as inaccuracies in system modelling and hardware limitations [**baiTumorQuantificationClinical2013**]. Improving reconstruction quality in this low-count, model-mismatch regime is therefore a key enabling step for accurate absorbed-dose estimation and for robust patient-specific outcome prediction [**davisPersonalisationMolecularRadiotherapy2020**].

In post-SIRT imaging, the two clinically relevant emission modalities are complementary. ^{90}Y positron emission tomography (PET) offers comparatively high spatial resolution, but is severely count-limited because of the extremely low internal pair-production branch. Bremsstrahlung single photon emission computed tomography (SPECT), by contrast, provides higher count statistics but suffers from poor spatial resolution and more severe modelling challenges, particularly for scatter, septal penetration, and additive corrections [**kappadathQuantitativeEvaluation90YPET2024**]. In contemporary workflows, these emission images are routinely paired with computed tomography (CT), which provides attenuation correction and anatomical context. This naturally motivates reconstruction approaches in which information is shared across modalities, with CT contributing structural guidance, and PET and SPECT contributing complementary functional information. Synergistic methods, where such information is incorporated during reconstruction rather than only after image formation, have gained prominence in related hybrid imaging contexts [**arridgeOverviewSynergisticReconstruction2021**]. Within post-^{90}Y SIRT imaging, these approaches have the potential to improve lesion-level quantification and absorbed-dose estimation without increasing acquisition time or administered activity.

The central aim of this thesis is therefore to develop and evaluate a synergistic reconstruction framework tailored to post-^{90}Y SIRT, with a particular focus on improving the reliability of quantitative emission imaging and therefore downstream dosimetry under clinically realistic count levels and modelling limitations. To this end, the thesis develops practical improvements to the bremsstrahlung single photon emission computed tomography (SPECT) system model, introduces a weighted directional total nuclear variation prior for joint PET/SPECT reconstruction with CT-derived directional guidance, and investigates optimisation and preconditioning strategies for this coupled penalised-likelihood problem.

The thesis is organised as follows:

- **Chapter 1** motivates synergistic post-SIRT imaging and outlines the thesis contributions.
- **Chapter 2** introduces the physics and measurement models of PET, SPECT, and CT, with emphasis on post-^{90}Y imaging.
- **Chapter 3** formulates penalised-likelihood emission reconstruction and reviews reconstruction priors and optimisation algorithms.
- **Chapter 4** develops practical improvements to the ^{90}Y bremsstrahlung SPECT system model.
- **Chapter 5** studies anatomy-informed regularisation in a controlled post-reconstruction deconvolution setting.
- **Chapter 6** introduces the weighted directional total nuclear variation prior and its differentiable formulation.
- **Chapter 7** investigates the optimisation of w-dTNV, including multi-bed PET reconstruction, subset organisation, and prior-aware preconditioning.
- **Chapter 8** evaluates w-dTNV for joint post-^{90}Y PET/SPECT reconstruction with CTAC-derived guidance.
- **Chapter 9** discusses the findings, limitations, and directions for future work, and concludes the thesis.
- **The Appendix** documents additional software and reproducibility material that supports the thesis workflow.

Chapter 2

Physics and Measurement Models in Emission Tomography

This chapter introduces the physical principles and measurement models underpinning emission tomography, with emphasis on the effects that dominate reconstruction quality and quantitative reliability. In both positron emission tomography (PET) and single photon emission computed tomography (SPECT), the reconstruction task is to infer a non-negative radiotracer activity distribution from projection data generated by stochastic radioactive decay and corrupted by attenuation, scatter, and blurring effects. These are naturally captured by a forward model that maps the image to expected measurement counts with additive components representing background contributions.

We first summarise the detection physics of positron emission tomography (PET) and SPECT and present convenient factorisations of the system matrix that separate geometry, attenuation, and detector effects. We then focus on the particularly challenging case of post-SIRT ^{90}Y imaging, where quantification is limited by extreme count starvation in internal pair production (IPP) PET and by spectral and resolution degradations in bremsstrahlung SPECT. Finally, we describe the role of computed tomography (CT) in this thesis as the source of attenuation information required for physically accurate emission modelling and a high-resolution anatomical reference for reconstruction regularisation in low-count regimes.

2.1 Principles of Emission Tomography

PET fundamentally differs from anatomical modalities such as computed tomography (CT). Rather than mapping the structural properties, emission tomography (ET) measures the spatio-temporal distribution of a radiolabelled substance introduced into the physiological system. A representative example is ^{18}F-fluoro-2-deoxyglucose (FDG)[**boellaardFDGPETCT2015**]. FDG is a glucose-analogue, meaning energy-hungry cells take up an increased amount of the molecules, and therefore radionuclide. Cells are unable to metabolise FDG, meaning it accumulates. The distribution of FDG can be imaged using a PET scanner, giving a functional view of glucose metabolism and highlighting any unusual uptake caused by overactive cells such as cancer.

While the instrumentation differs between SPECT and PET, both involve a similar inverse problem: estimating the activity distribution from projection measurements governed by Poisson counting statistics, attenuation, scatter, and detector resolution. The continuous activity distribution is represented on a finite voxel grid as a non-negative image vector $x \in \mathbb{R}_+^J$. The primary contribution to the expected measurements is modelled using the non-negative system matrix $A \in \mathbb{R}_+^{I \times J}$. Including additive background contributions gives the affine forward model

$$\bar{y} = Ax + b, \tag{2.1}$$

where $\bar{y} = \mathbb{E}[y] \in \mathbb{R}_+^I$ denotes the total expected measurement counts and $b \in \mathbb{R}_+^I$ contains contributions not described by the primary emission model. The precise discretisation and statistical measurement model are introduced later.

2.2 Positron Emission Tomography (PET)

PET primarily uses neutron-deficient isotopes (see Section **??** for a counter-example) that stabilise via β^+ decay. Within the nucleus, a proton converts into a neutron, emitting a positron (e^+) and an electron neutrino (ν_e) to conserve lepton number:

$$p \to n + e^+ + \nu_e. \tag{2.2}$$

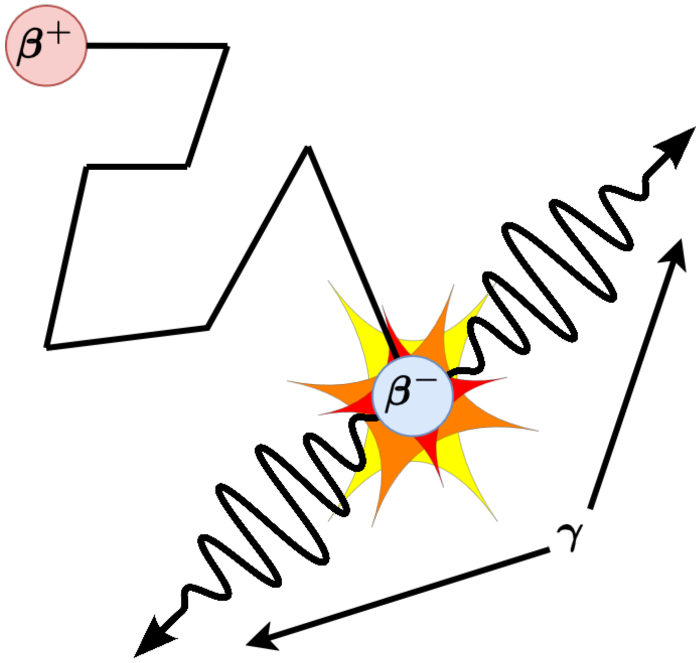


Figure 2.1: A positron traverses tissue, losing kinetic energy, before annihilating with an electron, resulting in two (nearly) anti-parallel photons.

The decay energy is shared between the positron and the neutrino, resulting in a continuous positron energy spectrum. This positron traverses the tissue losing kinetic energy via Coulomb interactions (a distance known as the positron range) before thermalising. This range imposes an effective blurring limit on spatial resolution, with an effective positron range of approximately 0.5–0.7 mm for ^{18}F and several millimetres for high-energy emitters such as ^{82}Rb [**cherryPhysicsNuclearMedicine2003**].

Upon thermalisation, the positron interacts with an electron to form positronium, which rapidly decays via annihilation:

$$e^+ + e^- \rightarrow \gamma + \gamma \tag{2.3}$$

The rest mass energy of the pair (2×511 keV) is converted into two[1] (near) anti-parallel[2] photons (Figure **??**).

[1] Positron/electron pairs can also annihilate to produce more than two photons but this is much less likely ($p(3\gamma)/p(2\gamma) \approx$1/378)[**bassColloquiumPositroniumPhysics2023**]

[2] In order to conserve momentum, a certain a-collinearity will exist between the photons, introducing additional blurring [**shibuyaAnnihilationPhotonAcollinearity2007**].

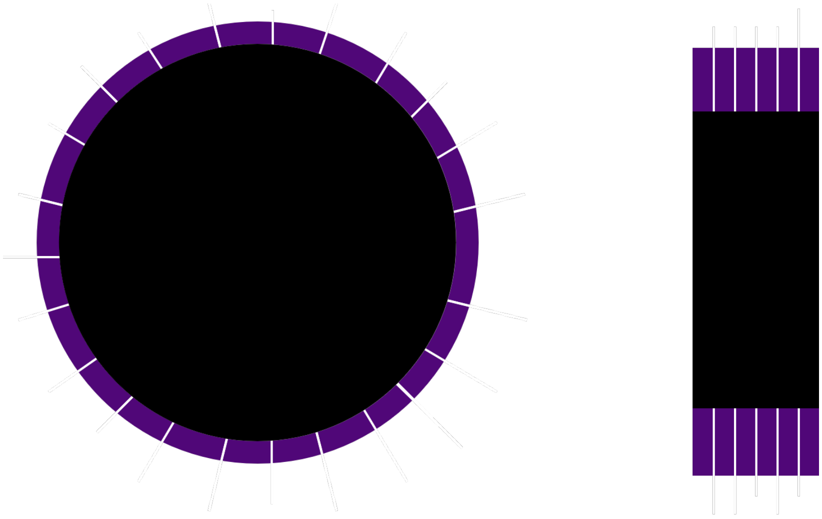


Figure 2.2: Simple diagram of a PET scanner. Left, axial view showing detectors arranged in a ring. Right, sagittal/coronal view showing multiple rings forming the axial FoV.

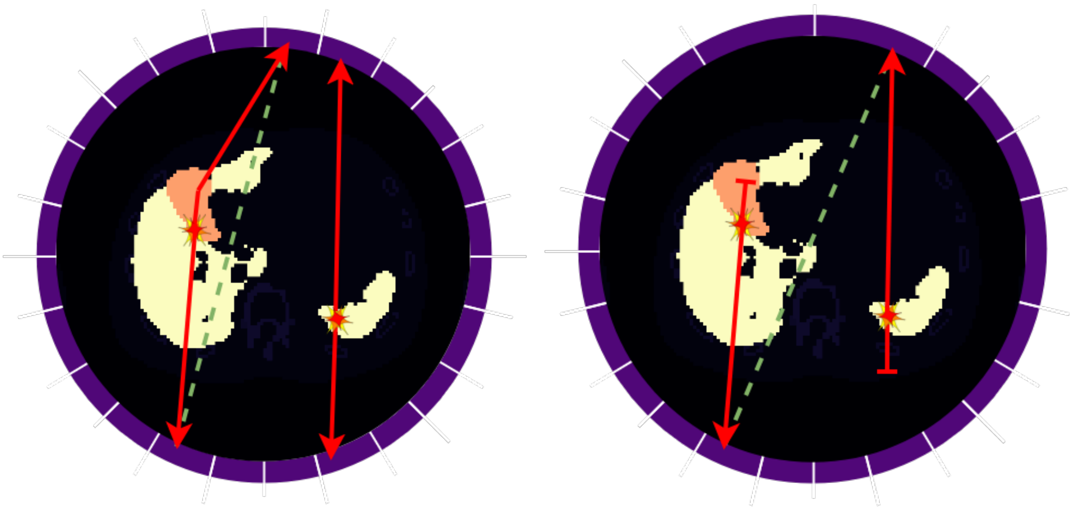
2.2.1 Coincidence Detection and Data Acquisition

PET detectors are typically assembled into modular detector blocks, which are then arranged circumferentially to form a ring around the patient. Modern clinical systems comprise multiple such rings stacked along the axial direction, increasing axial coverage and sensitivity by enlarging the axial field of view (FoV). Figure **??** illustrates this geometry showing a single ring in the axial view, and multiple rings in the sagittal view.

PET imaging relies on a technique known as electronic collimation. A valid event is recorded only if two detectors register photons within a coincidence timing window 2τ (typically 6–12 ns) [**cherryPhysicsNuclearMedicine2003**]. The measured coincidences (or counts) along a line of response (LoR), connecting detectors 1 and 2, denoted as $y_{i_{1,2}}$, is a superposition of three components:

$$y_{i_{1,2}} = t_{i_{1,2}} + s_{i_{1,2}} + r_{i_{1,2}} \tag{2.4}$$

where t_{ij} represents true coincidences (Figure **??**, right), $s_{i_{1,2}}$ represents scatter (where one or both photons have undergone at least one Compton interaction; Figure **??**, left)), and $r_{i_{1,2}}$ represents random coincidences (Figure **??**).



(a) (b)

Figure 2.3: Possible coincidences under PET electronic collimation. Panel (a) shows a scattered coincidence (the green dashed line is the effective LoR) and a true coincidence; panel (b) shows a random coincidence (green dashed lines are the effective LoRs).

Random coincidences occur when two unrelated annihilation events are detected within the timing window. The rate of randoms on a specific LoR is governed by the singles rates of the individual detectors (R_1, R_2), often approximated as:

$$r_{i_{1,2}} = 2\tau R_1 R_2. \tag{2.5}$$

2.2.2 The PET model

In PET, the system matrix may be factorised as

$$A = A_{\text{det,sens}} A_{\text{det,blur}} A_{\text{att}} A_{\text{geom}} A_{\text{ToF}} A_{\text{positron}}. \tag{2.6}$$

Here, A_{positron} models positron-range blurring, A_{ToF} models the timing localisation accuracy for time of flight (ToF) PET (and is omitted for non-ToF PET), A_{geom} is the geometric projection operator mapping voxels to lines of response in the absence of attenuation, A_{att} is a diagonal operator containing attenuation factors along each line of response, $A_{\text{det,blur}}$ models detector-related resolution effects in reporting the true line-of-response positions, and $A_{\text{det,sens}}$ is a diagonal sensitivity factor

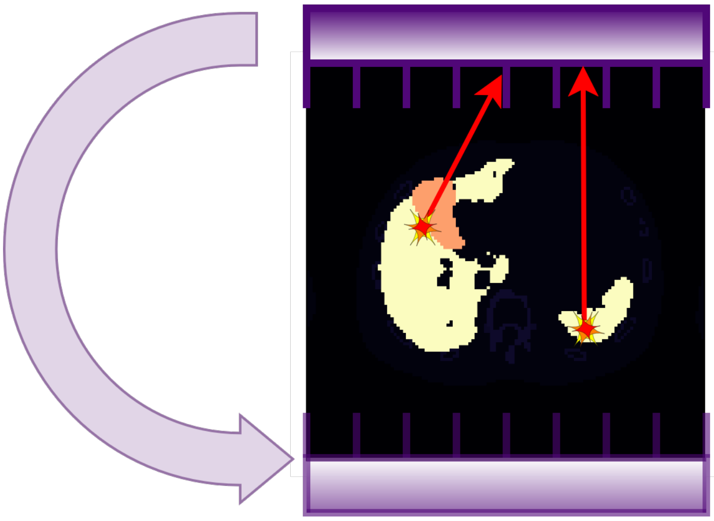


Figure 2.4: Left, non-parallel photon rejected by physical collimation. Right, photon parallel to the collimator can be detected, giving a line of response containing the emission source.

incorporating multiplicative effects such as detector efficiencies and normalisation [**baileyNuclearMedicinePhysics2014**]. The additive term is commonly written as $b = s + r$, corresponding to scatter and randoms, respectively.

2.3 Single Photon Emission Computed Tomography (SPECT)

SPECT imaging relies on single gamma emissions. As there is no coincident pair to define the LoR, physical collimation is required. A lead collimator, typically a parallel-hole geometry, allows only photons travelling through the collimator holes. A single (or multiple, most often a dual-head system) collimated gamma camera is rotated around the patient (Figure **??**).

The sensitivity (g) of a parallel-hole collimator depends on the hole diameter (d), septal thickness (t), and the effective hole length (L_e). A commonly used geometric-efficiency scaling is

$$g \propto \frac{d^4}{L_e^2\,(d+t)^2}, \tag{2.7}$$

which makes explicit the trade-off between resolution and sensitivity: reducing d improves geometric resolution but decreases sensitivity [**cherryPhysicsNuclearMedicine2003**]. In practice, the geometric efficiency of a (high-resolution) parallel-hole collimator can be very low (often $\lesssim 0.05\%$) [**angerScintillationCamera1958**]. In the context of bremsstrahlung imaging (Section **??**), this low efficiency is compounded by the continuous energy spectrum of the emissions, which limits the effectiveness of standard photopeak-based energy windowing for scatter rejection.

2.3.1 The SPECT model

For SPECT, a convenient factorisation is

$$A = A_{\text{det,sens}} A_{\text{det,blur}} A_{\text{geom,att}} A_{\text{bremss}}, \tag{2.8}$$

where $A_{\text{det,sens}}$ collects multiplicative sensitivity factors (e.g. detector efficiency and calibration/normalisation terms), $A_{\text{det,blur}}$ represents intrinsic resolution effects within the gamma camera, and $A_{\text{geom,att}}$ is the geometric projection operator that encapsulates collimator effects (including depth-dependent resolution) together with depth- and view-dependent attenuation. The operator A_{bremss} is specific to bremsstrahlung SPECT and models the additional blurring associated with bremsstrahlung range (discussed in Section **??**).

For a parallel-hole collimator, septal penetration can be incorporated via an energy-dependent effective hole length,

$$L_e(E) = l - \frac{2}{\mu(E)}, \tag{2.9}$$

where l is the physical hole length and $\mu(E)$ is the linear attenuation coefficient of the collimator material at photon energy E [**cherryPhysicsNuclearMedicine2003**]. The combined intrinsic detector blur and collimator blur is commonly approximated by a depth-dependent Gaussian point-spread function embedded within $A_{\text{geom,att}}$, with standard deviation $\sigma(h,E)$ increasing with source-to-collimator distance h (and depending on E through penetration).

2.4 Yttrium-90 Radioembolisation (SIRT)

SIRT uses ^{90}Y microspheres for the locoregional treatment of hepatic malignancies. It is an established treatment for unresectable hepatocellular carcinoma (HCC) and liver metastases [**sundramSelectiveInternalRadiation2017**], recommended by NICE for unresectable HCC and used for neuroendocrine and colorectal liver metastases in the UK [**nationalinstituteforhealthandcareexcellenceniceSelectiveInternalRadiation2021**, **nationalinstituteforhealthandcareexcellenceniceSelectiveInternalRadiation2024**]. The reconstruction of post-SIRT microsphere distribution presents a severe ill-posed inverse problem, as ^{90}Y is designed for therapy, not imaging.

2.4.1 Decay Characteristics

^{90}Y is an almost pure β^- emitter decaying to the stable ground state of Zirconium-90 (^{90}Zr) with a half-life of 64.1 hours:

$$^{90}\mathrm{Y} \rightarrow ^{90}\mathrm{Zr} + e^- + \bar{\nu}_e \tag{2.10}$$

The electrons have a high endpoint energy of 2.28 MeV. The primary challenge for quantitative imaging is the absence of primary gamma photons. Imaging must therefore exploit two secondary, low-probability physical interactions.

2.4.2 Bremsstrahlung SPECT

As high-energy β^- particles decelerate in tissue through Coulomb interactions with charged particles, they emit Bremsstrahlung photons. These photons can be imaged with SPECT systems. However, the continuous spectrum of bremsstrahlung photons (Figure **??**) hugely reduces the suitability of traditional energy-window scatter-correction methods, while bremsstrahlung range blurring (Figure **??**) reduces resolution. Consequently, bremsstrahlung SPECT is often characterised by poor spatial resolution and limited quantitative accuracy [**simpkinSpatialEnergyDependence1992**].

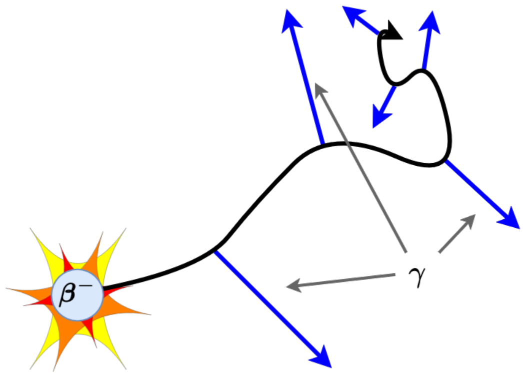


Figure 2.5: Representative diagram of bremsstrahlung production in tissue for an emitted β^- particle.

2.4.3 Internal Pair Production PET

A minor decay branch (31.86×10^{-6}) allows the nucleus to transition to an excited 0^+ state of ^{90}Zr, which de-excites via internal pair production (IPP):

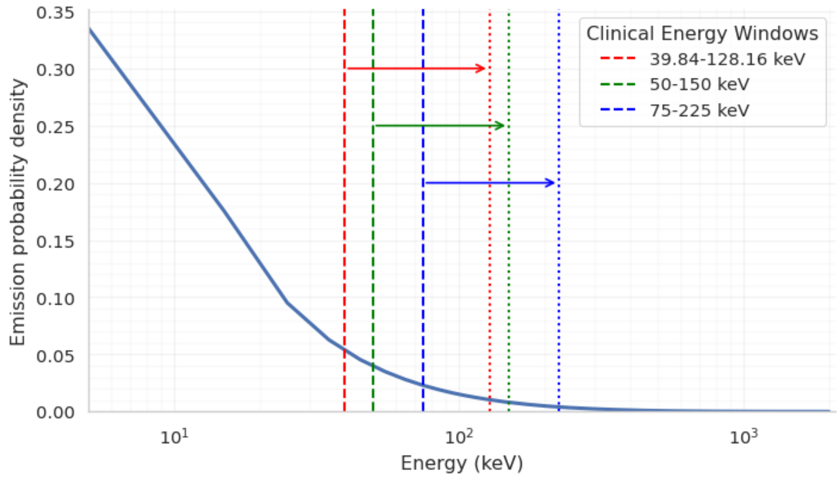
$$^{90}\mathrm{Y} \rightarrow ^{90}\mathrm{Zr}^* + \cdots \rightarrow ^{90}\mathrm{Zr} + e^-_{IPP} + e^+_{IPP} \tag{2.11}$$

The resulting positron annihilates to produce standard 511 keV photon pairs [**selwynNewInternalPair2007**]. While this enables quantitative PET imaging, the acquisition is "count-starved" due to the extremely low branching ratio and noise-dominated due to the Bremsstrahlung background (Figure **??**).

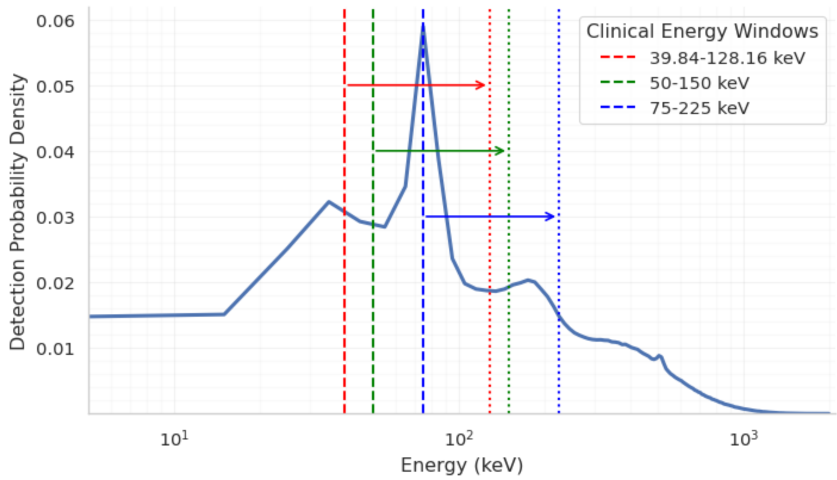
2.5 X-ray CT

In this work, CT is used primarily to provide anatomical context and attenuation information for emission tomography reconstruction. X-ray CT measures the transmitted intensity of a photon beam after passage through an object or patient. In an ideal monoenergetic setting, transmission follows the Beer–Lambert law, so that the log-normalised measurements are proportional to line integrals of the linear attenuation coefficient. In practice, polychromaticity introduces effects such as beam hardening, which are mitigated by scanner calibration and correction steps to yield images that approximate an attenuation map.

Reconstructed CT images are typically reported in Hounsfield units (HU),



(a) Bremsstrahlung emission spectrum for ^{90}Y.



(b) Simulated detected spectrum (10 cm, AnyScan).

Figure 2.6: Bremsstrahlung spectra: (a) emitted spectrum from Geant4 Application for Tomographic Emission (GATE) FastY90 model [**raultFastSimulationYttrium902010**, **janGATESimulationToolkit2004**] and (b) simulated detected spectra using AnyScan Trio GATE scanner model [**pellsQuantitativeValidationMonte2023**] with a visible k-edge at $\sim 90\,\text{keV}$ resulting from photoelectric absorption in the collimator

calibrated such that water is 0 HU and air is -1000 HU. The high spatial resolution of CT provides structurally reliable edge information (e.g. liver boundaries and lesion margins) that can be used to guide regularisation in the ill-posed reconstruction of post-SIRT ^{90}Y PET and SPECT.

In addition, CT is routinely used for computed tomography attenuation correction (CTAC) in both PET and SPECT. The reconstructed HU image is first converted to an estimate of the linear attenuation coefficient map $\mu(E) \in \mathbb{R}_+^J$ at the relevant

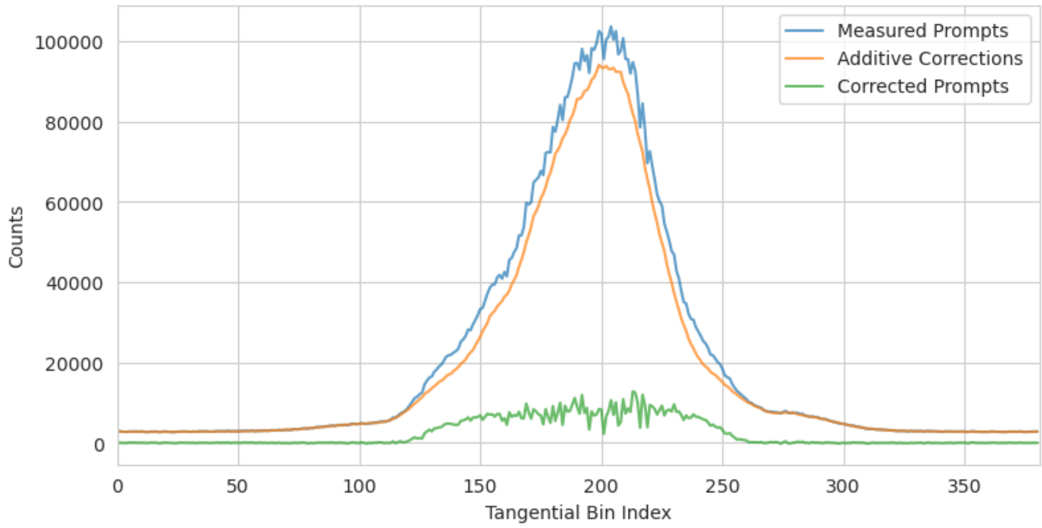


Figure 2.7: Tangential profile of measured prompt counts and estimated additive terms (randoms+scatter), with the corresponding corrected prompts for an ^{90}Y PET scan. The data are summed over all views and segments, leaving counts per tangential bin.

photon energy E (e.g. $E = 511$ keV for PET, or the photopeak energy for SPECT) [**schneider˙calibration˙1996**, **carney˙method˙2006**].

Misregistration (e.g. respiratory motion), metal artefacts, and beam-hardening residuals can all propagate into CTAC and therefore bias quantitative PET/SPECT activity estimates [**singhAttenuationArtifactAttenuation2007**, **pattonSPECTCTPhysical2008**].

Chapter 3

Penalised-Likelihood Reconstruction: Priors and Optimisation Algorithms

Emission tomography reconstruction is an ill-posed inverse problem: the unknown activity must be inferred from noisy Poisson count data through a highly smoothing and poorly conditioned forward operator. While maximum-likelihood (ML) approaches remain the clinical workhorse [**dempsterMaximumLikelihoodIncomplete1977**, **sheppMaximumLikelihoodReconstruction1982**], their practical success relies heavily on early stopping, which acts as an implicit regulariser but can introduce iteration-dependent bias and compromise quantification [**veklerovStoppingRuleMLE1987**, **wuConvergencePropertiesEM1983**]. For quantitative, low-count, and multi-modality settings, precisely those that arise in modern PET/SPECT practice, explicit regularisation and fast, robust optimisation become essential.

This chapter reviews the modelling and algorithmic foundations that underpin the remainder of the thesis. We first formalise the Poisson statistical model and the penalised-likelihood reconstruction objective (often termed maximum a posteriori (MAP)), highlighting where ill-conditioning and noise amplification enter the problem. We then present some regularisation strategies used in emission tomography: from classical quadratic smoothness priors and total variation (TV) [**rudinNonlinearTotalVariation1992**], through clinically deployed edge-preserving models such as the relative difference prior (RDP) [**nuytsConcavePriorPenalizing2002**], to anatomically guided methods that use

aligned structural images (e.g. Bowsher [**bowsherUtilizingMRIInformation2004**], parallel level sets (PLS) [**ehrhardtVectorValuedImageProcessing2014**], and kernelised representations [**wangPETImageReconstruction2015**, **hutchcroftAnatomicallyaidedPETR**
We then focus on synergistic regularisation, where multiple co-registered modalities are coupled during reconstruction, and motivate Jacobian-based vectorial priors such as joint total variation (JTV) and total nuclear variation (TNV) [**holtTotalNuclearVariation2014**, **rigieJointReconstructionMultichannel2015**, **knollJointMRPETReconstruction2017**], which will form the basis of the synergistic models developed later in the thesis.

Finally, we review optimisation algorithms suitable for large-scale maximum a posteriori (MAP) reconstruction. After discussing EM-type updates and their subset-accelerated variants, we highlight why subset methods can exhibit limit-cycle behaviour and why convergent penalised algorithms typically require relaxation (e.g. block-sequential regularised expectation maximisation (BSREM) [**ahnGloballyConvergentOrdered2002**]). We then introduce variance-reduced subset methods (e.g. stochastic average gradient amélioré (SAGA) and stochastic variance–reduced gradient (SVRG)) as a mechanism to retain low per-iteration cost while suppressing subset-induced stochasticity, and discuss the role of preconditioning, including "prior-aware" diagonal curvature approximations, in accelerating and stabilising convergence [**ehrhardtFastPETReconstruction2025**, **twymanInvestigationStochasticVariance2022**].

3.1 The Inverse Problem in Emission Tomography

To discretely represent the spatial distribution of tracer concentration within a patient, the continuous activity distribution $f(r)$ is approximated using a basis expansion $f(r) \approx \sum_{j=1}^{J} x_j h_j(r)$. Typically, $\{h_j\}$ are chosen to be indicator functions on uniform cuboid voxels. In that case, it is customary to use a non-negative activity image vector $x = (x_1, \dots, x_J)^\top \in \mathbb{R}_+^J$.

The measured data consist of coincidence counts along lines of response (PET) or projection bins (SPECT). We denote the observed count in bin i as y_i, and model it as a realisation of a Poisson counting process,

$$y_i \sim \mathrm{Poisson}(Y_i), \qquad i = 1, \ldots, I, \tag{3.1}$$

where $Y_i = \mathbb{E}[y_i]$ is the (unknown) mean count, which we model as $Y_i \approx \bar{y}_i(x)$ with

$$\bar{y}_i(x) = \sum_{j=1}^{J} a_{ij} x_j + b_i, \qquad i = 1, \ldots, I. \tag{3.2}$$

For notational simplicity, we will use the vector form

$$\bar{y}(x) = Ax + b, \qquad \bar{y}_i(x) = (Ax + b)_i, \tag{3.3}$$

where $A = [a_{ij}] \in \mathbb{R}_+^{I \times J}$ and $b \in \mathbb{R}_+^I$. Assuming independence of measurement bins, we obtain

$$p(y \mid x) = \prod_{i=1}^{I} \frac{\bar{y}_i(x)^{y_i} \exp(-\bar{y}_i(x))}{y_i!}. \tag{3.4}$$

Taking logarithms yields the Poisson log-likelihood

$$\log p(y \mid x) = \sum_{i=1}^{I} \left(y_i \log \bar{y}_i(x) - \bar{y}_i(x) - \log(y_i!) \right). \tag{3.5}$$

Since the term $\sum_{i=1}^{I} \log(y_i!)$ does not depend on x, it can be dropped for optimisation purposes. Thus, maximising the likelihood w.r.t. x is equivalent to minimising

$$\mathcal{L}(x) := -\log p(y \mid x) + \text{const} = \sum_{i=1}^{I} \left(\bar{y}_i(x) - y_i \log \bar{y}_i(x) \right). \tag{3.6}$$

Unconstrained maximum-likelihood (ML) solutions are dominated by noise due to the poor conditioning of A and the Poisson noise inherent in y. While iterative expectation–maximisation (EM) methods such as maximum-likelihood expectation–maximisation (MLEM) and ordered subsets expectation–maximisation (OSEM) [**dempsterMaximumLikelihoodIncomplete1977**, **sheppMaximumLikelihoodReconstruction1982** are the clinical standard, they rely on early stopping to act as an implicit regulariser.

This yields clinically acceptable images but introduces solution-dependent bias that may not give reliable quantitative metrics [**veklerovStoppingRuleMLE1987**]. In addition, finding the ML solution for quantification can be prohibitively slow [**wuConvergencePropertiesEM1983**].

A robust alternative to early stopping is the Bayesian formulation. By Bayes' rule, we have

$$p(x \mid y) = \frac{p(y \mid x)\, p(x)}{p(y)}, \tag{3.7}$$

where $p(y \mid x)$ is the Poisson likelihood induced by the forward model, $p(x)$ encodes prior information about the unknown image, and the evidence

$$p(y) = \int p(y \mid x)\, p(x)\, \mathrm{d}x \tag{3.8}$$

acts as a normalising constant independent of x.

A maximum a posteriori (MAP) estimate is then defined as

$$\hat{x}_{\mathrm{MAP}} \in \arg\max_{x \in \mathcal{C}} p(x \mid y) = \arg\max_{x \in \mathcal{C}} p(y \mid x)\, p(x), \tag{3.9}$$

where $\mathcal{C}$ denotes a feasible set (typically enforcing non-negativity). Taking negative logarithms, as above, gives an equivalent minimisation problem:

$$\hat{x}_{\mathrm{MAP}} \in \arg\min_{x \in \mathcal{C}} \left\{ -\log p(y \mid x) - \log p(x) \right\}, \tag{3.10}$$

since $-\log p(y)$ is constant with respect to x.

It is common to express the prior in Gibbs form,

$$p(x) \propto \exp(-\beta\, \mathcal{R}(x)), \tag{3.11}$$

where $\mathcal{R}(x)$ is a regularisation functional and $\beta > 0$ controls the strength of the prior. Substituting this into the MAP objective yields the penalised-likelihood form

$$\hat{x}_{\mathrm{MAP}} \in \arg\min_{x \in \mathcal{C}} \left\{ \mathcal{L}(x) + \beta\, \mathcal{R}(x) \right\}. \tag{3.12}$$

3.2 Regularisation Strategies

The choice of $\mathcal{R}(x)$ determines the texture and quantitative characteristics of the reconstructed image. Historically, this has evolved from simple smoothing constraints to sophisticated, anatomically guided models.

3.2.1 Image-based Regularisation

One of the simplest forms of regularisation is Tikhonov regularisation, which imposes an ℓ_2-norm penalty on local voxel-wise differences. This yields the quadratic prior

$$\mathcal{R}_{\text{Quad}}(x) = \sum_{j=1}^{J} \sum_{k \in \mathcal{N}_j} w_{jk} \, (x_j - x_k)^2, \tag{3.13}$$

where $\mathcal{N}_j$ denotes a local neighbourhood of voxel j (e.g. 6, 18, or 26-connected in 3D) and $w_{jk} \geq 0$ are weights (often distance-based and symmetric). While strongly convex and computationally efficient, the quadratic penalty strongly discourages large local differences, which tends to oversmooth edges and suppress high-contrast structures [**gemanStochasticRelaxationGibbs1984**].

To better preserve edges, total variation (TV) [**rudinNonlinearTotalVariation1992**] replaces the squared penalty with an ℓ_1-type penalty on local differences. A neighbourhood-difference form that makes the connection to $\mathcal{R}_{\text{Quad}}$ explicit is

$$\mathcal{R}_{\text{TV}}(x) = \sum_{j=1}^{J} \sum_{k \in \mathcal{N}_j} w_{jk} \, |x_j - x_k|. \tag{3.14}$$

Relative to $\mathcal{R}_{\text{Quad}}$, the linear growth of $|x_j - x_k|$ reduces the marginal cost of large voxel-wise jumps in intensity, thereby promoting piecewise-constant reconstructions with sharper edges.

In much of the imaging literature, TV is instead written in terms of a discrete gradient operator. Defining $(Dx)_j \in \mathbb{R}^d$ to collect finite differences at voxel j along the d coordinate directions, a common choice on a regular grid is

$$\big((Dx)_j\big)^{(\ell)} = \frac{x_{j+\delta_\ell} - x_j}{\Delta_\ell}, \qquad \ell = 1, \ldots, d, \tag{3.15}$$

where Δ_ℓ denotes the Euclidean distance between voxels $j+\delta_\ell$ and j. The discrete TV functional is then

$$\mathcal{R}_{\mathrm{TV}}(x) = \sum_{j=1}^{J} |(Dx)_j|, \tag{3.16}$$

with an isotropic alternative, more common in emission tomography [**jonssonTotalVariationRegulariz**; **paninTotalVariationRegulated1999**],

$$\mathcal{R}_{\mathrm{TV}}(x) = \sum_{j=1}^{J} \|(Dx)_j\|_2. \tag{3.17}$$

For optimisation, it is common to use a smooth approximation to avoid nondifferentiability at zero, for example

$$\tilde{\mathcal{R}}_{\mathrm{TV}}(x) = \sum_{j=1}^{J} \sqrt{\|(Dx)_j\|_2^2 + \varepsilon^2}, \qquad \varepsilon > 0. \tag{3.18}$$

In clinical practice, adaptable edge-preserving priors are often preferred. The relative difference prior (RDP) [**nuytsConcavePriorPenalizing2002**], the core component of the GE “Q.Clear” algorithm, modifies the penalty to depend on the absolute intensity sums, penalising relative differences, as opposed to absolute ones.

$$\mathcal{R}_{\mathrm{RDP}}(x) = \frac{1}{2} \sum_{j=1}^{J} \sum_{k \in \mathcal{N}_j} w_{jk} \frac{(x_j - x_k)^2}{x_j + x_k + \gamma |x_j - x_k| + \varepsilon}, \tag{3.19}$$

with γ controlling the amount of edge preservation and $\varepsilon \geq 0$ the STIR RDP stabiliser. Positive ε regularises zero-valued voxel pairs; when $\varepsilon = 0$, STIR assigns the zero–zero contribution its limiting value of zero. Note that if $(x_j + x_k + \varepsilon) \gg \gamma |x_j - x_k|$ then $\mathcal{R}_{\mathrm{RDP}}$ behaves like a relative quadratic prior, whereas if $\gamma |x_j - x_k| \gg (x_j + x_k + \varepsilon)$ it behaves like an anisotropic TV penalty (up to a constant factor).

3.2.2 Anatomically Guided Regularisation

In low-count regimes, single-modality priors struggle to distinguish between noise and legitimate small structures. Anatomically guided reconstruction uses a high-resolution structural image u (e.g., CT) to modulate the regularisation of the emission image x.

Geometric Alignment methods assume that if an edge exists in the emission image, it should be supported by corresponding boundaries in a registered anatomical guidance image. The seminal Bowsher prior [**bowsherUtilizingMRIInformation2004**] implements this idea by defining, for each voxel j, a data-adaptive and dynamic neighbourhood $\mathcal{B}_j(u)$ using the guidance image u. Concretely, from a fixed candidate neighbourhood $\mathcal{N}_j$, the Bowsher construction selects the B voxels whose intensities in u are most similar to u_j, i.e. those minimising a similarity metric such as

$$d_{jk}(u) := |u_j - u_k| \quad (k \in \mathcal{N}_j), \tag{3.20}$$

and sets $\mathcal{B}_j(u)$ to be the indices of the B smallest values of $d_{jk}(u)$. Regularisation is then applied only over this anatomically consistent subset, for example via a quadratic penalty

$$\mathcal{R}_{\text{Bow}}(x;u) = \sum_{j=1}^{J} \sum_{k \in \mathcal{B}_j(u)} w_{jk} \, (x_j - x_k)^2, \tag{3.21}$$

(or an alternative edge-preserving potential). By restricting the penalty to neighbours that are locally homogeneous in u, the prior encourages smoothing within anatomical regions while reducing smoothing across anatomical edges, thereby allowing sharp boundaries to form in x when they are supported by the guidance image.

An even more geometric framework is parallel level sets (PLS) [**ehrhardtVectorValuedImagePro**
These methods encourage structural consistency by encouraging the gradient vectors of the emission image and the guidance image to be aligned, i.e. $(Dx)_j \parallel (Du)_j$. A common formulation (often termed PLS1) penalises the magnitude of the component of $(Dx)_j$ orthogonal to $(Du)_j$, which is proportional to the sine of the enclosed angle θ_j:

$$R_{\text{PLS1}}(x,u) = \sum_{j=1}^{J} \|(Dx)_j\|_2 \, \|(Du)_j\|_2 \, |\sin\theta_j|. \tag{3.22}$$

This term vanishes when $(Dx)_j$ and $(Du)_j$ are parallel (or when either gradient is zero), and increases with the component of $(Dx)_j$ orthogonal to $(Du)_j$. While numerically stable, PLS1 scales with $\|(Du)_j\|_2$ and therefore inherits the spatially varying gradient magnitude of the anatomical image, which can lead to over-weighting at strong anatomical edges and under-weighting in low-contrast regions.

To decouple the directional information from the anatomical gradient magnitude, directional total variation (DTV) [**ehrhardtPETReconstructionAnatomical2016**] introduces a normalised direction field

$$\xi_j = \frac{(Du)_j}{\sqrt{\|(Du)_j\|_2^2 + \eta^2}}, \tag{3.23}$$

where $\eta > 0$ is a small stabilisation parameter that suppresses unreliable directions induced by noise or very fine-scale anatomical texture. Using ξ_j, one defines a local anisotropic projection operator

$$\Pi_j := \mathbb{I} - \gamma_j \, \xi_j \xi_j^\top, \tag{3.24}$$

where $\mathbb{I}$ is the $d \times d$ (number of dimensions, d, squared) identity matrix and $\gamma_j \in [0,1]$ controls the strength of directional guidance. The resulting DTV prior is

$$R_{\mathrm{dTV}}(x;u) = \sum_{j=1}^{J} \|\Pi_j (Dx)_j\|_2, \tag{3.25}$$

which penalises predominantly the gradient component orthogonal to the anatomical level-set direction, thereby encouraging edges in x to align with edges in u. In regions where u is locally flat (so $\xi_j \approx 0$), $\Pi_j \approx \mathbb{I}$ and DTV reduces to standard isotropic TV.

Regularisation strategies also exist outside the standard Bayesian (penalised-likelihood) framework. A prominent example is the kernelised expectation maximisation (KEM), in which prior information is encoded directly in the image representation rather than through an explicit penalty term. Following Wang and Qi [**wangPETImageReconstruction2015**] and its anatomically-aided extension [**hutchcroftAnatomicallyaidedPETReconstruction2016**], the emission image is modelled as a linear combination of kernel-derived basis functions:

$$x = K\alpha, \tag{3.26}$$

where $\alpha \in \mathbb{R}^J$ is a coefficient image and $K \in \mathbb{R}^{J\times J}$ is a (typically sparse) kernel matrix whose entries encode similarity between voxels based on features extracted from a guidance image [**hutchcroftAnatomicallyaidedPETReconstruction2016**]. Each voxel j is assigned a feature vector f_j (commonly a patch of intensities in the high-resolution guidance image), and a radial Gaussian kernel is often used,

$$\kappa(f_j, f_k) = \exp\left(-\frac{\|f_j - f_k\|_2^2}{2\sigma^2}\right), \tag{3.27}$$

with σ controlling edge sensitivity. The kernel matrix is then defined entrywise by $K_{jk} = \kappa(f_j, f_k)$, with additional sparsification for tractability: for each voxel j, only a small set of the most similar neighbours (e.g. via a local k-nearest-neighbour search within a window) are allowed to contribute, yielding a sparse K. To preserve overall intensity scaling (counts), K is typically row-normalised.

Substituting $x = K\alpha$ into the Poisson forward model gives a kernelised expectation for the data,

$$\bar{y}(\alpha) = A(K\alpha) + b, \tag{3.28}$$

so that reconstruction reduces to maximum-likelihood estimation of α under the same Poisson model, followed by $\hat{x} = K\hat{\alpha}$. Importantly, because the model remains linear in the unknown coefficients, an EM-type update can be derived with essentially the same structure as conventional MLEM/OSEM, but with K embedded within the projector/backprojector operations. This yields a procedure that is straightforward to implement in existing EM pipelines, while often delivering improved bias–variance trade-offs at low counts by enforcing anatomically-informed, spatially adaptive smoothing in the image representation [**wangPETImageReconstruction2015**, **hutchcroftAnatomicallyaidedPETReconstruction2016**].

A known limitation of purely anatomy-derived kernels is PET-unique feature suppression: structures present in the emission image but absent (or weak) in the guidance image can be over-smoothed or biased towards anatomically plausible but functionally incorrect solutions [**deiddaHybridPETMR2019**]. To mitigate this, hybrid kernelised expectation maximisation (HKEM) augments the kernel with synergistic information from the evolving PET estimate itself, yielding an iteration-dependent kernel that combines anatomical and emission-driven similarity . In the HKEM formulation of Deidda *et al.*, the hybrid kernel entry factorises as

$$k_{fj}^{(n)} = k_m(v_f, v_j)\, k_p(z_f^{(n)}, z_j^{(n)}), \tag{3.29}$$

where v_j denotes the (fixed) anatomical feature vector at voxel j and $z_j^{(n)}$ is a feature derived from the current PET update (updated each iteration) [**deiddaHybridPETMR2019**]. Both factors are commonly chosen as Gaussian similarities modulated by spatial proximity,

$$k_m(v_j, v_k) = \exp\left(-\frac{\|v_j - v_k\|_2^2}{2\sigma_a^2}\right) \exp\left(-\frac{\|x_j - x_k\|_2^2}{2\sigma_{da}^2}\right), \tag{3.30}$$

$$k_p(z_j^{(n)}, z_k^{(n)}) = \exp\left(-\frac{\|z_j^{(n)} - z_k^{(n)}\|_2^2}{2\sigma_p^2}\right) \exp\left(-\frac{\|x_j - x_k\|_2^2}{2\sigma_{df}^2}\right), \tag{3.31}$$

where x_j denotes the voxel coordinates and the parameters $\sigma_a, \sigma_f, \sigma_{da}, \sigma_{df}$ control the relative influence of anatomical similarity, functional-driven (emission) similarity, and spatial locality. The functional term helps address emission–guidance mismatch and, crucially, reduces the tendency of anatomy-only kernels to suppress PET-unique lesions by allowing the neighbourhood coupling to adapt to features emerging in the emission estimate. Empirically, Deidda *et al.* report that this hybridisation can reduce negative bias under low-count conditions and improve lesion contrast while retaining the denoising benefits of kernelised reconstruction [**deiddaHybridPETMR2019**].

Kernel approaches extend naturally to multiple guidance sources by enlarging the feature vectors (e.g. concatenating features from CT and/or additional functional channels [**wangPETImageReconstruction2015**]) and defining the kernel on the resulting joint feature space; this provides a flexible mechanism for sequential or multi-modality guidance without introducing an explicit Bayesian penalty term.

3.2.3 Synergistic Regularisation

If multiple modalities are available that depict the same underlying anatomy or functional distribution, it can be advantageous to reconstruct them synergistically, allowing information to be shared across modalities throughout the reconstruction process [**arridgeOverviewSynergisticReconstruction2021**]. A prerequisite for such coupling is that the images are defined in a common spatial coordinate system. In practice, this requires geometric registration between modalities and resampling onto a common reconstruction grid, so that voxel index n refers to the same physical location across all channels.

When the native modalities, indexed using m, are acquired on different grids or coordinate frames, we introduce spatial transforms W_m (registration + resampling operators) and define the co-registered images on a shared grid, with $W_m : \mathbb{R}_+^{J_m} \to \mathbb{R}_+^N$:

$$z_m := W_m x_m \in \mathbb{R}_+^N, \qquad m = 1, \dots, M, \tag{3.32}$$

and collect them as the modality tuple

$$\boldsymbol{z} := (z_1, \dots, z_M). \tag{3.33}$$

Coupling priors are then naturally expressed in terms of $\boldsymbol{z}$, since neighbourhoods, gradients, and voxel-wise comparisons are only meaningful when images are spatially aligned.

In the Bayesian setting, a generic multi-modality reconstruction is given by

$$\hat{\boldsymbol{x}} \in \arg\min_{\boldsymbol{x} \in \mathcal{C}} \left\{ \sum_{m=1}^{M} \mathcal{L}_m(x_m) + \beta\, \mathcal{R}(\boldsymbol{W}\boldsymbol{x}) \right\}, \tag{3.34}$$

where $\boldsymbol{x} := (x_1, \dots, x_M)$ and $\boldsymbol{W} := \mathrm{diag}(W_1, \dots, W_M)$, and $\mathcal{R}(\boldsymbol{W}\boldsymbol{x})$ promotes cross-modality consistency (e.g. co-located edges or shared support) in the co-registered domain.

A practical way to exploit multiple modalities without solving a fully joint inverse problem is sequential reconstruction, in which one modality is first reconstructed, and the resulting image is then used as side-information to guide the reconstruction of another modality. In the Bayesian regime, this is

$$\hat{x}_m \in \arg\min_{x_m \in \mathcal{C}} \left\{ \mathcal{L}_m(x_m) + \beta\, \mathcal{R}(x_m; W_{m\leftarrow(m-1)} x_{m-1}) \right\}, \tag{3.35}$$

where $W_{m\leftarrow(m-1)}$ maps the image reconstructed in the previous stage, $\hat{x}_{m-1}$, into the spatial grid of modality m.

Deidda *et al.* have previously investigated a similar method for ^{90}Y bremsstrahlung SPECT–informed PET reconstructions [**deiddaTripleModalityImage2023**]. Bremsstrahlung SPECT images are first reconstructed using HKEM with CT and SPECT image-update information to obtain an enhanced SPECT estimate. This SPECT image is then resampled to the PET grid and used as functional guidance for PET reconstruction.

$$\hat{\alpha}_m \in \arg\min_{\alpha_m \in \mathcal{C}} \left\{ \mathcal{L}_m\big(W_{m\leftarrow(m-1)}\, K_m \alpha_m\big) \right\}. \tag{3.36}$$

The paper reports that this sequential design is particularly beneficial for low-count ^{90}Y PET: SPECT provides complementary denoising/uptake support within PET–SPECT-consistent regions, while CT contributes primarily through resolution/edge information injected at the SPECT stage, and the PET image-update features reduce suppression of PET-unique structures and mitigate inter-modality mismatch.

Sequential guidance is limited by one-way dependency: errors in u may propagate directly to x and the shared information may not be fully utilised. Truly synergistic reconstruction estimates both images simultaneously, allowing information to flow bi-directionally. Perhaps the most obvious way to extend single modality regularisation to achieve a synergistic regulariser is joint total variation (JTV) [**blomgrenColorTVTotal1998**], a simple extension of TV which sums gradient magnitudes:

$$\mathcal{R}_{\mathrm{JTV}}(\boldsymbol{z}) = \sum_{n=1}^{N} \sqrt{\|Dz_{1,n}\|^2 + \|Dz_{2,n}\|^2 + \cdots + \|Dz_{M,n}\|^2}. \tag{3.37}$$

This encourages joint sparsity, encouraging co-located edges. A convenient way to formalise multi-modality edge co-localisation is to view the set of images as a vector-valued field. For each voxel n, the voxel-wise Jacobian is defined by stacking image gradients (Equation **??**) in columns:

$$\mathcal{G}_n(\boldsymbol{z}) \coloneqq \begin{bmatrix} Dz_{1,n} & Dz_{2,n} & \cdots & Dz_{M,n} \end{bmatrix} \in \mathbb{R}^{d \times M}. \tag{3.38}$$

When viewed in this way, JTV can be redefined as a Schatten 2-norm (also known as a Frobenius norm) applied to this jacobian [**holtTotalNuclearVariation2014**],

$$\mathcal{R}_{\text{JTV}}(\boldsymbol{z}) = \sum_{n=1}^{N} \|\mathcal{G}_n(\boldsymbol{z})\|_F \tag{3.39}$$

where the Schatten p-norm is defined as

$$\|Y\|_{S_p} \coloneqq \left(\sum_{r=1}^{R} \sigma_r(Y)^p \right)^{1/p} \tag{3.40}$$

for singular values $\sigma_\ell(Y)$ of a rank-R matrix Y. Note that $R = \min\{d, M\}$. The Schatten 1-norm, also known as the nuclear norm is the sum of singular values

$$\|Y\|_* = \sum_{r=1}^{R} \sigma_r(Y). \tag{3.41}$$

The nuclear norm is the convex envelope of the Rank function. Applied to the voxel-wise jacobians, it promotes singular value sparsity and has the effect of either driving functionally misaligned gradients to zero or parallel/anti-parallel, promoting edge-alignment. Applying this nuclear norm to the voxel-wise jacobians is known as total nuclear variation (TNV) [**holtTotalNuclearVariation2014, rigieJointReconstructionMultichannel2015**].

$$\mathcal{R}_{\text{TNV}}(\boldsymbol{z}) = \sum_{n=1}^{N} \|\mathcal{G}_n(\boldsymbol{z})\|_* \tag{3.42}$$

It has been applied to synergistic PET/MR reconstruction through its extended variant, total generalised nuclear variation [**knollJointMRPETReconstruction2017**].

The benefits of using the nuclear norm over the Frobenius norm can be visualised in Figure **??**, where the nuclear norm exhibits sharp grooves along the coordinate axes that encourage one singular value to vanish (promoting low-rank structure), whereas the Frobenius norm is smooth and tends to shrink singular values jointly without explicitly favouring rank reduction.

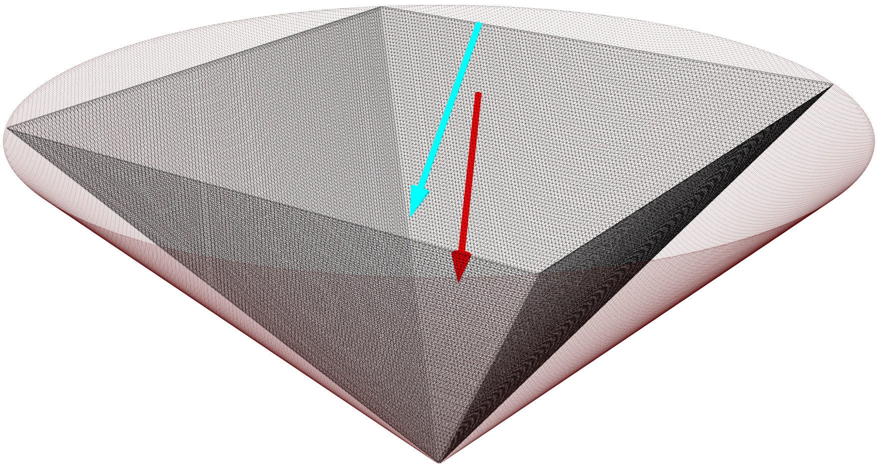


Figure 3.1: Visualisation of the Frobenius and nuclear norms (vertical axis) of a matrix X as functions of its (two) singular values (horizontal axis). The Frobenius norm is $\|X\|_F = \sqrt{\sigma_1(X)^2 + \sigma_2(X)^2}$ (red mesh), whereas the nuclear norm is $\|X\|_* = |\sigma_1(X)| + |\sigma_2(X)|$ (black mesh). For visualisation purposes we extend the singular-value axes to negative values. The nuclear norm exhibits sharp grooves along the coordinate axes, encouraging one singular value to vanish (i.e. promoting low-rank structure), while the Frobenius norm is smooth and does not preferentially drive a singular value exactly to zero. This behaviour can also be understood via subgradients: the nuclear-norm subgradient (cyan) points towards rank-1 grooves, preferentially shrinking the smaller singular value, whereas the Frobenius-norm subgradient (red) points towards the origin, shrinking both singular values equally.

Total Nuclear variation will form the basis for the synergistic prior demonstrated later on in this thesis.

3.3 Optimisation Algorithms

The reconstruction of MAP (and, even more so, synergistic images) requires more nuanced algorithms. MLEM can be relatively simply extended for penalised reconstruction via Green's one-step-late-method [**greenBayesianReconstructionsEmission1990**]. The one-step-late method constructs an EM-type multiplicative update by evaluating the penalty derivative at the current iterate $x^{(n)}$ rather than at $x^{(n+1)}$. The resulting OSL-MAP update is

$$x_j^{(n+1)} = \frac{x_j^{(n)}}{\displaystyle\sum_{i=1}^{I} a_{ij} + \beta \left.\frac{\partial R(x)}{\partial x_j}\right|_{x=x^{(n)}}} \sum_{i=1}^{I} a_{ij} \frac{y_i}{\bar{y}_i\big(x^{(n)}\big)}, \qquad j = 1, \dots, J, \tag{3.43}$$

The term $\frac{\partial R}{\partial x_j}\Big|_{x^{(n)}}$ acts as an iteration-dependent modification of the MLEM denominator, yielding an algorithm that is typically simple to implement for a broad class of differentiable priors. Unfortunately, the one-step-late method is not convergent and special attention must be paid to initialisation conditions for a reasonable reconstruction [**zengGreensOnesteplateAlgorithm2019**].

Although too fragile for low count / synergistic reconstruction, the one-step-late algorithm has some useful takeaways. Primarily the joint update of data and prior functions with some kind of voxel-wise scaling. This is otherwise known as a preconditioner. MLEM has the multiplicative update equation for iteration k:

$$x^{(k+1)} = \left(x^{(k)} \oslash A^\top \mathbf{1}_I\right) \odot A^\top (y \oslash \bar{y}), \tag{3.44}$$

where $\odot$ and $\oslash$ denote elementwise multiplication and division, respectively, $\mathbf{1}_I \in \mathbb{R}^I$ is an all-ones vector and k denotes the iteration number. For notational simplicity, we drop the explicit elementwise operators; all divisions are understood componentwise. Then

$$x^{(k+1)} = \operatorname{diag}\left(\frac{x^{(k)}}{A^\top \mathbf{1}_I}\right) A^\top \left(\frac{y}{\bar{y}}\right). \tag{3.45}$$

This can be rearranged to form the additive update equation

$$\begin{aligned} x^{(k+1)} &= x^{(k)} + \operatorname{diag}\left(\frac{x^{(k)}}{A^\top \mathbf{1}_I}\right)\left(A^\top\left(\frac{y}{\bar{y}(x^{(k)})}\right) - \mathbf{1}_J\right) \\ &= x^{(k)} + \operatorname{diag}\left(\frac{x^{(k)}}{A^\top \mathbf{1}_I}\right)\left(A^\top\left(\frac{y}{\bar{y}(x^{(k)})}\right) - A^\top \mathbf{1}_I\right) \\ &= x^{(k)} + \operatorname{diag}\left(\frac{x^{(k)}}{A^\top \mathbf{1}_I}\right)A^\top\left(\frac{y}{\bar{y}(x^{(k)})} - \mathbf{1}_I\right) \\ &= x^{(k)} - \operatorname{diag}\left(\frac{x^{(k)}}{A^\top \mathbf{1}_I}\right)\nabla_x \mathcal{L}(x^{(k)}). \end{aligned} \tag{3.46}$$

where $\mathbf{1}_J \in \mathbb{R}^J$ is an all-ones vector. This form highlights that the MLEM update can be interpreted as a preconditioned gradient descent algorithm with preconditioner $\mathcal{P}_{\text{EM}}^{(k)} = \operatorname{diag}\left(\frac{x^{(k)}}{A^\top \mathbf{1}_I}\right)$, i.e.

$$x^{(k+1)} = x^{(k)} - \varsigma \mathcal{P}_{\text{EM}}^{(k)} \nabla_x \mathcal{L}(x^{(k)}), \tag{3.47}$$

with a step size $\varsigma = 1$.

A preconditioner aims to recondition the reconstruction space so that gradient steps are appropriately scaled across voxels, improving numerical conditioning and convergence. Intuitively, we seek larger steps in poorly sensitive directions and smaller steps in highly sensitive directions, thereby approximating curvature information. Since curvature is governed by the inverse Hessian, it is often approximated in a diagonal form for computational efficiency.

For the Poisson negative log-likelihood, the expected Hessian (Fisher information) is

$$\mathcal{I}(x) \;:=\; \mathbb{E}\left[\nabla_x^2 \mathcal{D}(y \mid x)\right] \;=\; A^\top \operatorname{diag}\left(\frac{1}{\bar{y}(x)}\right) A. \tag{3.48}$$

Its diagonal entries satisfy, for each voxel index j,

$$\mathcal{I}_{jj}(x) = \sum_i \frac{a_{ij}^2}{\bar{y}_i(x)} \;\leq\; \sum_i \frac{a_{ij}^2}{a_{ij} x_j} \;=\; \frac{(A^\top \mathbf{1})_j}{x_j}, \tag{3.49}$$

where we used $\bar{y}_i(x) = \sum_\ell a_{i\ell}x_\ell + r_i \geq a_{ij}x_j$ and $r_i \geq 0$. Consequently,

$$\mathcal{I}_{jj}(x) \;\leq\; \frac{(A^\top \mathbf{1})_j}{x_j}. \tag{3.50}$$

This motivates the diagonal scaling

$$\mathcal{P}_{\mathrm{EM}}^{(k)} \;:=\; \mathrm{diag}\left(\frac{x^{(k)}}{A^\top \mathbf{1}}\right), \tag{3.51}$$

which can be interpreted as the inverse of a conservative diagonal upper bound on the per-voxel curvature of the Poisson negative log-likelihood (in expectation).

Subsets are commonly used to accelerate early convergence by partitioning the data into S subsets and updating the image using only the contribution from a single subset (or a small batch of subsets) at each iteration. Writing the negative log-likelihood as a sum of subset terms,

$$\mathcal{L}(x) = \sum_{s=1}^{S} \mathcal{L}_s(x), \tag{3.52}$$

a generic subset (stochastic / incremental) scaled-gradient update takes the form

$$x^{(k+1)} = x^{(k)} - \varsigma\, \mathcal{P}^{(k)}\, \nabla_x \mathcal{L}_s\left(x^{(k)}\right). \tag{3.53}$$

If multiple subsets are used per iteration, $\nabla_x \mathcal{L}_s$ is replaced by an average (or sum) over the chosen subset indices. The most common algorithm found in clinical systems is the OSEM algorithm. For subset s at iteration k, a standard OSEM update can be written as

$$x^{(k+1)} = \mathrm{diag}\left(\frac{x^{(k)}}{A_s^\top \mathbf{1}_I}\right) A_s^\top \left(\frac{y_s}{\bar{y}_s\left(x^{(k)}\right)}\right), \tag{3.54}$$

where A_s denotes the system matrix and data restricted to subset s.

OSEM typically shows a marked improvement in early-stage convergence by providing a cheaper, higher-variance approximation of the full-data EM update.

However, because the subset gradient is only an approximation to the full gradient, OSEM is not, in general, a convergent algorithm for the maximum-likelihood problem: as the iterates approach a solution, the stochasticity introduced by cycling through subsets can dominate, leading to limit-cycle behaviour in which successive subset updates fail to settle to a fixed point (Figure **??**).

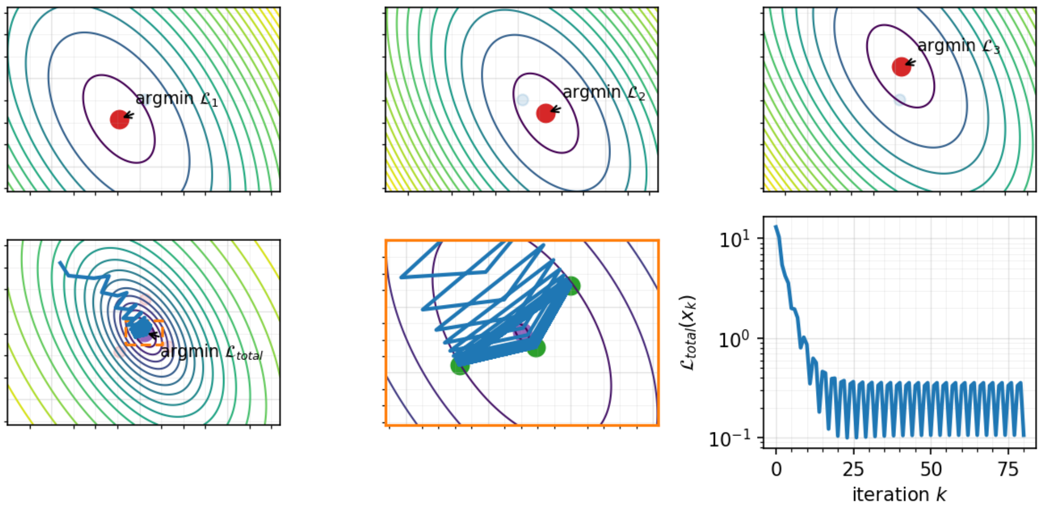


Figure 3.2: Demonstration of limit cycle behaviour with an ordered subset gradient descent algorithm. Top row: 3 different simple sub-functions representing subset likelihoods. Bottom row: left, full function with sub-gradient steps; centre, zoomed version of the full function demonstrating limit cycle behaviour (green dots show approximate points of limit cycle); right, likelihood value per iteration, demonstrating limit cycle precession.

We seek a fast and convergent algorithm for MAP reconstruction. A widely used approach is the (modified) block-sequential regularised expectation maximisation (BSREM) framework [**ahnGloballyConvergentOrdered2002**], which combines EM-type preconditioning and subset acceleration with a relaxation (diminishing step-size) to recover convergence while incorporating an explicit penalty. In a scaled-gradient form, a generic BSREM-type update can be written as

$$x^{(k+1)} = x^{(k)} - \varsigma_k \mathcal{P}_{\mathrm{EM}}^{(k)} \left(\nabla_x \mathcal{L}_s(x^{(k)}) + \frac{\beta}{S} \nabla_x \mathcal{R}(x^{(k)}) \right), \tag{3.55}$$

where $\varsigma_k > 0$ is an iteration dependent step size. Convergence is promoted by choosing $\{\varsigma_k\}$ to diminish appropriately (e.g. $\varsigma_k \to 0$ with $\sum_k \varsigma_k = \infty$), which suppresses the subset-induced limit-cycle behaviour while retaining rapid early progress (see Figure **??**).

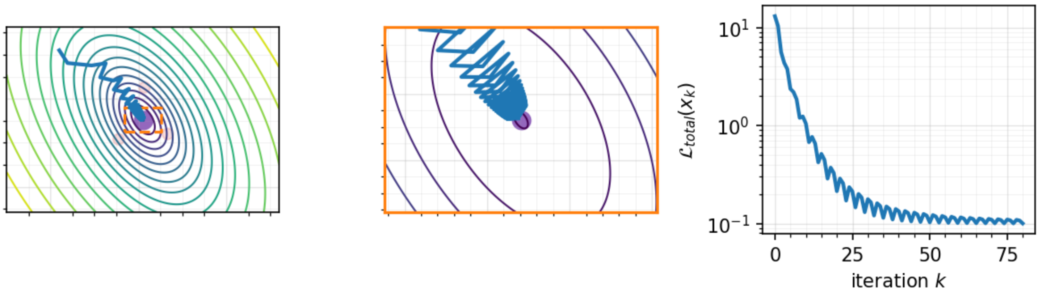


Figure 3.3: Demonstration of reduction in limit cycle behaviour with step size relaxation $\varsigma_k = \varsigma_0/(1+0.05k)$. Left, full function with sub-gradient steps; centre, zoomed version of the full function demonstrating relaxed limit cycle behaviour; right, likelihood value per iteration, demonstrating relaxed limit cycle precession.

However, we can do better. Variance-reduced subset methods have become increasingly popular in the PET reconstruction literature because they retain the low per-iteration cost of subset updates while substantially reducing the subset-induced stochasticity that drives OSEM limit cycles. The key idea is to replace the raw subset gradient, which is a high-variance estimator of the full gradient, with a variance-reduced estimator that is (approximately) unbiased and whose variance decreases as the iterates approach a solution. This allows the use of larger (or non-vanishing) relaxation parameters than classical diminishing-step BSREM-type schemes, while maintaining stable convergence behaviour in practice [**ehrhardtFastPETReconstruction2025**, **twymanInvestigationStochasticVariance2022**].

A generic subset (stochastic) scaled-gradient step uses $g_{s_k}(x^{(k)})$ in place of $\nabla\mathcal{L}(x^{(k)}) = \sum_s g_s(x^{(k)})$, i.e.

$$x^{(k+1)} = x^{(k)} - \varsigma_k \mathcal{P}^{(k)}\left(g_{s_k}(x^{(k)}) + \beta\nabla_x\mathcal{R}(x^{(k)})\right), \tag{3.56}$$

which is cheap but noisy. Variance reduction augments this step by incorporating additional information (stored gradients or periodic full-gradient computations) to form a corrected estimator $\tilde{g}_k$ with reduced variance. Two widely used examples are stochastic average gradient amélioré (SAGA) and stochastic variance–reduced gradient (SVRG):

SAGA maintains a table of the most recently computed subset gradients $\{g_s(\cdot)\}_{s=1}^S$ and updates using

$$\tilde{g}_k = g_{s_k}(x^{(k)}) - g_{s_k}(\tilde{x}_{s_k}) + \frac{1}{S}\sum_{s=1}^{S} g_s(\tilde{x}_s), \tag{3.57}$$

where $\tilde{x}_s$ denotes the iterate at which the stored gradient for subset s was last evaluated.

SVRG periodically computes a reference full gradient at a snapshot $\tilde{x}$ and then performs inner iterations using

$$\tilde{g}_k = g_{s_k}(x^{(k)}) - g_{s_k}(\tilde{x}) + \nabla_x \mathcal{L}(\tilde{x}). \tag{3.58}$$

In both cases, $\tilde{g}_k$ is designed so that $\mathbb{E}[\tilde{g}_k] \approx \nabla_x \mathcal{L}(x^{(k)})$ while its variance decreases as $x^{(k)}$ approaches a stationary point, thereby mitigating the oscillatory behaviour seen in OSEM and improving stability at high iteration counts (see Figure **??**).

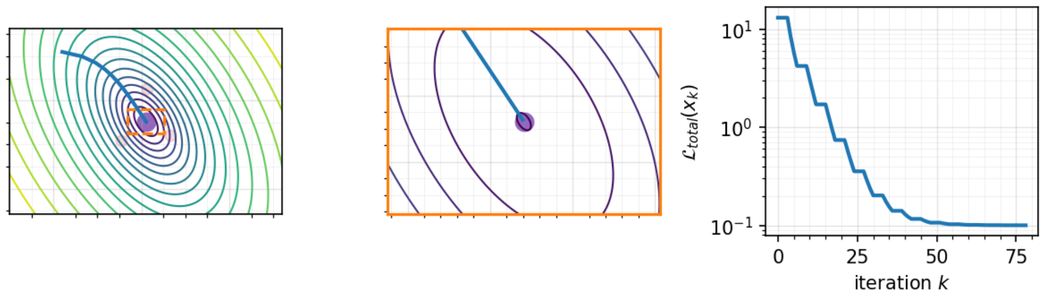


Figure 3.4: Demonstration of removal of limit cycle behaviour with an SVRG-type (Ordered VRG) algorithm. Left, full function with sub-gradient steps; centre, zoomed version of the full function demonstrating removal of limit cycle behaviour; right, likelihood value per iteration, demonstrating removal of limit cycle precession

A variance-reduced, prior-regularised update can then be written compactly as

$$x^{(k+1)} = x^{(k)} - \varsigma \mathcal{P}^{(k)} \left(\tilde{g}_k + \beta \nabla_x \mathcal{R}(x^{(k)}) \right). \tag{3.59}$$

Recent work indicates that incorporating the effect of the prior into $\mathcal{P}^{(k)}$ ("prior-aware preconditioning") can be crucial for fast and robust convergence. Ehrhardt *et al.* [**ehrhardtFastPETReconstruction2025**] showed substantially improved convergence rates with RDP-regularised PET reconstruction when using harmonic averaging of the EM-type preconditioner and a scaled diagonal estimate of the inverse Hessian of the prior. Using only the diagonal elements of the RDP Hessian

$$[\mathcal{H}_{\mathrm{RDP}}(x)]_{jj} = \sum_{k \in \mathcal{N}_j} \frac{2w_{jk}(2x_k + \varepsilon)^2}{(x_j + x_k + \gamma|x_j - x_k| + \varepsilon)^3}, \tag{3.60}$$

they constructed a preconditioner as

$$\mathcal{P}^{(k)} = \mathrm{diag}\left(\frac{x^{(k)} + \delta}{A^\top \mathbf{1} + \alpha \ \mathrm{diag}\big(\beta \mathcal{H}_{\mathrm{RDP}}\left(x^{(k)}\right)\big)\left(x^{(k)} + \delta\right)} \right) \tag{3.61}$$

with scalar α tuned to correct for the loss of off-diagonal information. Although this is not emphasised explicitly in [**ehrhardtFastPETReconstruction2025**], the harmonic averaging viewpoint can also be interpreted as a majorisation-preserving way to combine curvature information from the likelihood and the prior. In particular, the voxel-wise harmonic blend can be rearranged as

$$\left(\frac{1}{a} + \frac{1}{b}\right)^{-1}, \tag{3.62}$$

so that combining two diagonal scalings a and b harmonically is equivalent to inverting the sum of the corresponding inverse-scalings (i.e. two diagonal Hessian surrogates). Consequently, when these inverse-scalings are chosen as upper-bounding approximations of the respective Hessians, their sum remains a valid upper bound, and the resulting harmonic combination inherits this majorising property while still incorporating both data-fidelity and prior-induced curvature.

Preconditioning and stochastic variance reduction for synergistic MAP reconstruction has not been well investigated in the literature so far and will form a part of this thesis.

A widely used alternative to EM-type and subset-gradient schemes is the limited-memory BFGS algorithm with bound constraints (L-BFGS-B) [**broydenConvergenceClassDoublerank1970**, **fletcherNewApproachVariable1970**, **goldfarbFamilyVariablemetricMethods1970**, **shannoConditioningQuasiNewtonMethods1970**, **liuLimitedMemoryBFGS1989**, **boydDistributedOptimizationStatistical2010**], a quasi-Newton method that targets $\min_{x\in\mathcal{C}}\{\mathcal{L}(x)+\beta\mathcal{R}(x)\}$ by iteratively building a low-rank approximation to the inverse Hessian from successive iterate and gradient differences. The "limited-memory" construction stores only a small number of correction pairs, making it viable for high-dimensional images, while the "B" variant enforces simple bounds (e.g. non-negativity) via an active-set strategy and projected search directions. In practice, L-BFGS-B can exhibit rapid local convergence on smooth penalised objectives, but each iteration requires one or more full objective and gradient evaluations (coupled with a line search), which is computationally burdensome in large-scale emission tomography when the likelihood is evaluated over all subsets. For this reason it is not used in the main reconstruction experiments in this thesis; nevertheless, it provides a useful reference optimiser and is employed later for a small demonstrative deconvolution problem in Chapter **??**.

A practical limitation of vanilla L-BFGS-B in emission tomography is its sensitivity to poor variable scaling: the line search and the quasi-Newton curvature updates can be dominated by a small subset of high-sensitivity voxels, leading to slow progress. Tsai *et al.* propose a diagonally preconditioned variant (L-BFGS-B-PC) that embeds analytic curvature information into the optimisation via a change of variables [**tsaiFastQuasiNewtonAlgorithms2018**]. We can choose a diagonal preconditioner $\mathcal{P}$ from a cheap curvature surrogate (e.g. an EM-type diagonal approximation of the Hessian), and run L-BFGS-B on the transformed problem $\mathcal{P}x$. This rescales the objective so that gradients are more isotropic in the transformed coordinates, while bound constraints are preserved by transforming the box constraints accordingly. The resulting method combines low-cost, problem-specific diagonal curvature (from $\mathcal{P}$) with the low-rank curvature learned by L-BFGS-B, typically yielding substantially faster and more robust convergence than unpreconditioned L-BFGS-B in penalised reconstructions.

3.4 Summary and Research Gap

This chapter reviewed the statistical and algorithmic foundations of emission tomography reconstruction under a Poisson measurement model. Starting from maximum-likelihood estimation, we motivated penalised-likelihood (MAP) formulations as a principled alternative to early stopping, particularly in low-count and quantitatively demanding settings. We then surveyed regularisation strategies ranging from quadratic smoothness and total variation to clinically deployed edge-preserving models such as the relative difference prior, before focusing on anatomically guided approaches (e.g. Bowsher, PLS/DTV) and kernelised representations that incorporate structural side-information through the image model.

A central theme in modern PET/SPECT practice is the availability of multiple co-registered modalities. While sequential (one-way) guidance can be effective, it is inherently limited by asymmetric information flow and by the risk of propagating errors or mismatches from the guidance image into the emission reconstruction. This motivates synergistic reconstruction, in which modalities are coupled through joint priors. In this setting, Jacobian-based penalties such as JTV and TNV provide a natural mechanism to promote co-located, directionally consistent edges across modalities while allowing modality-specific contrast.

From an optimisation perspective, clinical subset-accelerated EM methods achieve fast early progress but can exhibit limit-cycle behaviour and lack convergence guarantees in penalised problems without relaxation. Convergent frameworks such as BSREM recover stability via diminishing step sizes, but may converge slowly when the curvature induced by strong, spatially varying priors dominates the data term. Variance-reduced subset methods (e.g. SAGA/SVRG) offer a promising compromise, retaining low per-iteration cost while suppressing subset-induced stochasticity. However, their practical performance depends critically on effective preconditioning.

Research gap. Despite substantial progress in both synergistic regularisation and variance-reduced optimisation, two gaps remain directly relevant to quantitative ^{90}Y imaging:

1. **Synergistic, anatomy-informed coupling:** existing Jacobian-based priors (e.g. TNV) are typically formulated to promote cross-modality edge consistency, but are not routinely combined with explicit directional guidance derived from high-resolution CT edges in a way that remains robust in low-count regimes.

2. **Synergy-aware optimisation:** variance-reduced subset algorithms and prior-aware diagonal preconditioners have mainly been studied in single-modality PET with clinically popular priors (e.g. RDP). Their extension to synergistic MAP reconstruction, where cross-modality coupling induces strong and spatially heterogeneous curvature, is comparatively underexplored.

Thesis direction. Motivated by these gaps, the remainder of this thesis develops a directionally guided, differentiable total nuclear variation prior that uses CT-derived edge information while coupling functional modalities through voxel-wise Jacobians. We then design and evaluate a variance-reduced subset optimisation scheme (SVRG-type) equipped with a synergy-aware preconditioner that accounts for both data fidelity and coupled prior curvature, targeting fast, stable convergence in low-count ^{90}Y reconstruction.

Chapter 4

Improvements to the ^{90}Y Bremsstrahlung System Model

Before tackling the problem of synergistic reconstruction with ^{90}Y, we need to properly model the physics of out

4.1 Challenges in Bremsstrahlung Imaging

As introduced in Section **??**, unlike diagnostic radionuclides such as ^{99m}Tc that emit gamma photons at discrete photopeaks, ^{90}Y has no practically useful prompt gamma emission for SPECT. Imaging therefore relies on detecting the continuous spectrum of Bremsstrahlung photons generated as emitted β-particles decelerate in tissue [**woernerAdvancesY90Radioembolization2022**]. This emission physics gives rise to several fundamental challenges for quantitative reconstruction:

- **Continuous energy spectrum:** The emitted photon energies span a continuum from (near) zero up to the maximum β-energy of 2.28 MeV [**beTableRadionuclides2004**] (see Figure **??**). In conventional SPECT, scatter may be estimated using energy windows adjacent to a photopeak; for ^{90}Y, the absence of a photopeak makes it difficult to separate primary photons from those that have undergone Compton scatter within the patient. As a result, more sophisticated scatter-modelling and correction strategies are typically required [**rongDevelopmentEvaluationImproved2012**].
- **Septal penetration and collimator design:** The high-energy tail of the

Bremsstrahlung spectrum increases the probability of septal penetration, which degrades spatial resolution and contrast. Collimator choice and optimisation are therefore critical to limit penetration while maintaining adequate sensitivity [**rongCollimatorOptimizationMethod2013**].

- **Bremsstrahlung-induced blur:** Bremsstrahlung photons are produced along the electron track rather than at a single point, so the effective photon emission location is distributed relative to the ^{90}Y decay site. This introduces an additional range-dependent blurring component that is not present in photopeak imaging and can further degrade reconstructed resolution [**rongDevelopmentEvaluationImproved2012**] (see Figure.

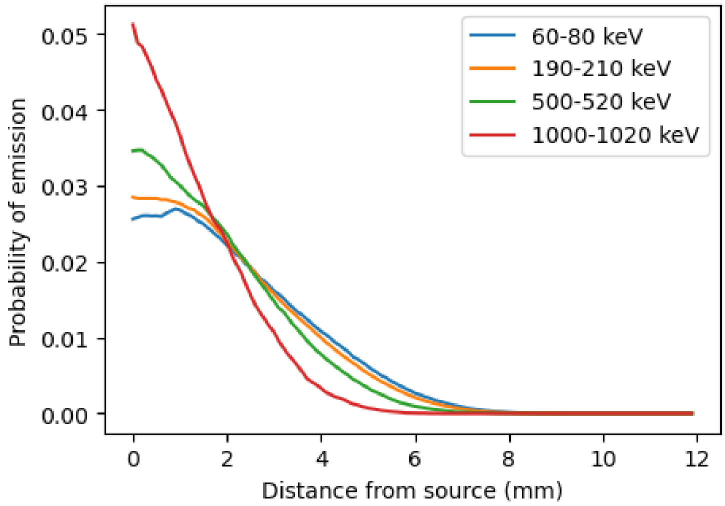


Figure 4.1: Plots showing probability of bremsstrahlung gamma emission versus the distance from ^{90}Y decay site. Data from Rault *et al.*[**raultFastSimulationYttrium902010**]

4.2 Optimisation Strategies and Monte Carlo Modeling

To overcome these limitations, optimisation must occur at both the acquisition and reconstruction stages. While hardware choices, such as the preference for medium-energy general-purpose (MEGP) or high-energy general-purpose (HEGP) collimators, can mitigate septal penetration [**taherparvarEvaluationEffectDifferent2019**,

kimEffectsCollimatorImaging2019, **inceCollimatorEnergyWindow2021**], substantial gains in quantitative accuracy are achieved through advanced computational modeling [**dewarajaImprovedQuantitative90Y2017**].

Monte Carlo (MC) simulation codes, such as Simulation of Imaging Nuclear Detectors (SIMIND) [**ljungbergSIMINDMonteCarlo2012**] and GATE [**janGATESimulationToolkit2004**, **raultFastSimulationYttrium902010**], play a pivotal role in this optimisation. These tools allow for the generation of accurate scatter kernels, as well as other effects that are specific to the continuous ^{90}Y spectrum. Some methods have integrated modelling of the spatial offset between the β-decay location and the point of Bremsstrahlung emission into MC forward projectors to refine the system matrix [**rongDevelopmentEvaluationImproved2012**].

Furthermore, novel reconstruction techniques such as joint spectral reconstruction have been proposed to exploit the information contained across multiple energy windows, rather than relying on a single wide integration window [**chunAlgorithmsAnalysesJoint2020**]. By combining multi-band acquisition with accurate Monte Carlo-based forward modeling, it is possible to improve the precision of activity estimation and contrast recovery in post-SIRT dosimetry Can I fit this in somehow? Probably not... No time.

This chapter explores these advanced reconstruction methodologies. We also introduce a simple Gaussian image-based blurring operation, in order to account for bremsstrahlung range blur. In addition, we investigate the suitability of Fu & Qi's residual correction method [**fuResidualCorrectionMethod2010**], originally designed for correcting for positron range blur, on bremsstrahlung reconstruction.

4.3 Additive Corrections

A key difficulty in quantitative ^{90}Y Bremsstrahlung SPECT is that a substantial fraction of the measured counts arise from photons that are not well modelled by a purely "primary-only" forward projector. In this work we treat these contributions as *additive terms* in the Poisson measurement model, with a particular focus on (patient) scatter.

4.3.1 Statistical model with an additive background term

Let $y \in \mathbb{N}^I$ denote the measured projection data (all bins across all views) and let $x \in \mathbb{R}_+^J$ denote the unknown activity image. We use the standard Poisson model

$$y_i \sim \text{Poisson}((Ax)_i + b_i), \qquad i = 1, \ldots, I, \tag{4.1}$$

where A is the system matrix modelling the (primary) detection process (including attenuation and detector response as implemented in our reconstruction pipeline), and $b \in \mathbb{R}_+^I$ is an *additive* component. In Bremsstrahlung SPECT, b is typically dominated by Compton-scattered photons, with additional contributions possible from other backgrounds (e.g. stray radiation or modelling mismatch). The role of additive correction is therefore to provide an estimate of b that is consistent with the acquisition protocol and reconstruction model.

Given a *fixed* b, OSEM yields an approximate maximum-likelihood estimate of x under this Poisson model. However, in ^{90}Y Bremsstrahlung imaging, b is large and impossible to estimate from neighbouring energy-window , motivating model-based strategies.

4.3.2 Initial scatter estimate via a pre-trained deep network

To initialise b we use a deep-learning scatter estimator that predicts scatter projections directly from the measured emission projections (and CT-derived attenuation information), following the approach by Xiang *et al.* who trained a convolutional neural network to regress MC-derived “true” scatter [**xiangDeepNeuralNetwork2020**].

In our case this initialisation is explicitly heuristic: the available pre-trained network was developed and trained under a different hardware/protocol configuration (scanner, collimation, energy windowing, and number of views) than the Siemens and Mediso datasets used in this thesis [**xiangDeepNeuralNetwork2020**]. In particular, the network was trained on data acquired on a Symbia Intevo SPECT/CT system with high-energy collimation and a 105–195 keV window with 80 projections, whereas our reconstructions were performed on Siemens or Mediso systems with different energy windows and either MEGP (Siemens) or MLEGP (Mediso) collimation.

Despite this domain shift, we found that the predicted scatter term fits the measured data sufficiently well to provide a sensible starting point for subsequent model-based refinement (Figure **??**).

4.3.3 Iterative OSEM–SIMIND refinement loop

Starting from the deep-learning initialisation $b^{(0)}$, we then refine the additive term using an alternating procedure:

1. **Reconstruction step:** update x using OSEM and the current additive estimate $b^{(k)}$. We used 12 subsets and 10 epochs (total 120 subiterations), warm-started from the previous outer iterate.

2. **Regularisation of the intermediate image:** apply an isotropic Gaussian post-smoothing of 5 mm full width at half maximum (FWHM) to the intermediate reconstruction (used only for the subsequent scatter simulation step).

3. **Scatter simulation step:** run a SIMIND-based Monte Carlo scatter simulation using the (smoothed) current activity estimate to produce a new scatter estimate $\widehat{b}^{(k+1)}$.

We repeat this OSEM $\rightarrow$ SIMIND $\rightarrow$ OSEM+5 mm smoothing $\rightarrow \ldots$ loop for 10 outer iterations. The 5 mm smoothing is used to suppress high-frequency noise that would otherwise be amplified by repeatedly re-simulating scatter from noisy intermediate reconstructions.

4.3.4 Offline versus online additive correction strategies

We consider two practical strategies for how the scatter term is used throughout the full reconstruction:

Offline (fixed) additive correction. In the *offline* strategy, we compute a scatter estimate once (after the initialisation/refinement loop) and then keep it fixed for the remainder of the reconstruction. Concretely, we set $b \leftarrow b^{(\star)}$ and run OSEM to convergence (or for a fixed number of iterations) with a constant additive term. This approach is typically stable, but may be biased if the fixed scatter estimate is inconsistent with the final activity estimate.

Online (intermittently updated) additive correction. In the *online* strategy, we update the scatter estimate during reconstruction. That is, after some number of OSEM iterations, we re-run the SIMIND scatter simulation using the current image estimate and update b. This is conceptually aligned with hybrid Monte-Carlo/analytical reconstruction approaches in which the scatter estimate is updated a small number of times until it stabilises [**dewarajaImprovedQuantitative90Y2017**, **elschotQuantitativeComparisonPET2013**]. In practice, the update cadence is a tunable choice: too frequent updates increase computational cost and can introduce instability; too infrequent updates reduce the benefit relative to the offline approach.

4.3.5 Full replacement versus damped scatter updates

For the online strategy, we investigate two update rules.

(1) Full replacement. We set

$$b^{(k+1)} \leftarrow \widehat{b}^{(k+1)}. \tag{4.2}$$

Although appealingly direct, we observed that full replacement can be unstable in some cases, with the behaviour varying across phantoms. A practical interpretation is that once b is changed, the likelihood model being optimised by OSEM is also changed; therefore the overall alternating procedure is not guaranteed to behave monotonically. An

(2) Damped (relaxed) updates. To improve stability, we use a convex combination of the old and new scatter estimates:

$$b^{(k+1)} \leftarrow (1-\alpha)\, b^{(k)} + \alpha\, \widehat{b}^{(k+1)}, \qquad \alpha \in (0,1]. \tag{4.3}$$

This can be viewed as a relaxation of the fixed-point iteration $b = \widehat{b}(x)$, reducing sensitivity to noise and modelling mismatch at the cost of slower adaptation. In all experiments here we set $\alpha = 0.2$. This value was chosen heuristically; we did not perform a dedicated study to optimise α, but empirically it provided a good stability–convergence trade-off across datasets.

Overall, these offline/online and replacement/damping variants provide a practical framework for controlling the bias–variance and stability properties of Bremsstrahlung additive correction, while keeping the reconstruction pipeline compatible with Monte Carlo scatter modelling.

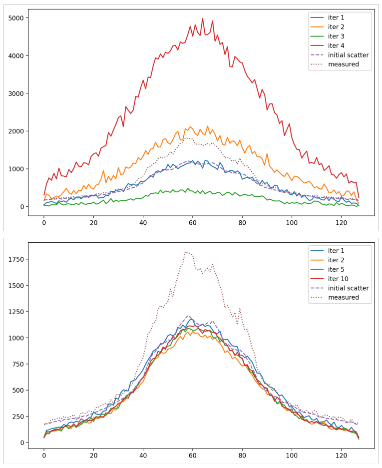


Figure 4.2: Figures showing profile through projections for NEMA phantom. Summed over projections, axial slice is through NEMA spheres. Horizontal axis show tangential coordinate on gamma camera, Vertical axis shows counts. Top, replacing additive data; bottom, damped addition of additive data ($\alpha = 0.2$).

4.4 Bremsstrahlung Range Blur

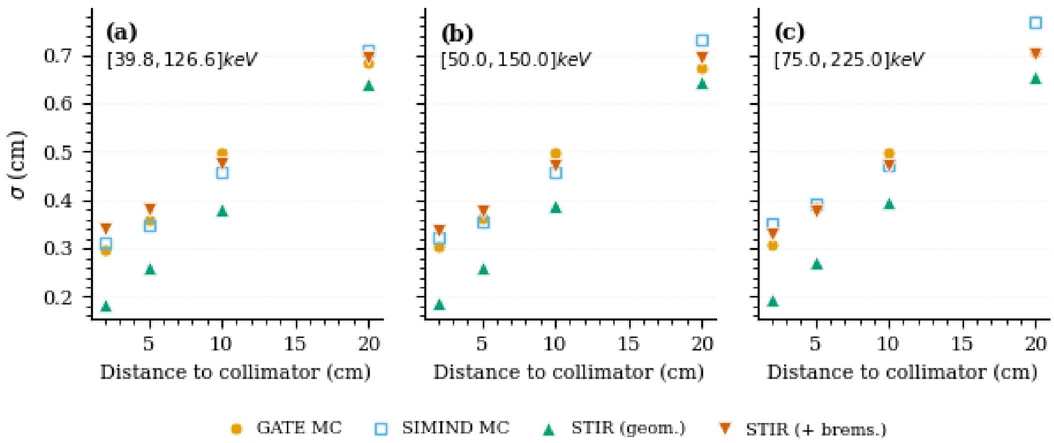


Figure 4.3: Example fits of an isotropic Gaussian proxy to the Monte Carlo source-to-emission displacement distribution for ^{90}Y Bremsstrahlung in water (clinical energy windows).

As mentioned previously, bremsstrahlung photons are generated along the

β-particle track, inducing an additional *range blur* that is absent in photopeak imaging. Several MC frameworks can model this effect using detailed, energy-dependent probability density functions (PDFs) for the displacement between the decay location and the photon emission location. However, in practice, most general-purpose reconstruction toolkits are not designed to ingest arbitrary, precomputed displacement PDFs within the forward projector and its adjoint.

Here we adopt a pragmatic approximation: for clinically relevant energy windows and in water-equivalent tissue, the marginal displacement distribution is observed to be close to Gaussian. We therefore model the Bremsstrahlung range blur as an *isotropic image-space Gaussian* whose variance is selected to match MC-derived displacement statistics (Figure **??**). This is simple to implement (a single 3D Gaussian smoothing) and is compatible with standard projector architectures.

4.4.1 Projector point-spread function

We model the collimator–detector response (CDR) by a Gaussian point spread function (PSF) whose width increases with the source–collimator distance h and depends on energy E. In the projector we represent the depth-dependent blur by a spatially variant isotropic Gaussian with standard deviation

$$\sigma_{\text{lin}}(h;E) = c_0(E) + c_1(E)\,h. \tag{4.4}$$

This linear model approximates the standard quadrature combination of intrinsic detector resolution and geometric collimator resolution,

$$\sigma_{\text{true}}(h;E) = \left[\ \sigma_{\text{intr}}(E)^2 + \sigma_{\text{geom}}(h;E)^2\ \right]^{1/2}, \tag{4.5}$$

while avoiding repeated evaluation of a square root inside the projector. The coefficients $c_0(E)$ and $c_1(E)$ are obtained by a least-squares fit of $\sigma_{\text{lin}}(h_i;E)$ to $\sigma_{\text{true}}(h_i;E)$ over a clinically relevant distance range ($h \in [2,20]\,\text{cm}$). Writing

$$\bar{h} = \frac{1}{N}\sum_{i=1}^{N} h_i, \qquad \bar{\sigma} = \frac{1}{N}\sum_{i=1}^{N} \sigma_{\text{true}}(h_i;E), \tag{4.6}$$

the closed-form solution (assuming $\sum_i (h_i - \bar{h})^2 > 0$) is

$$c_1(E) = \frac{\sum_{i=1}^{N} (h_i - \bar{h}) \left(\sigma_{\text{true}}(h_i; E) - \bar{\sigma}\right)}{\sum_{i=1}^{N} (h_i - \bar{h})^2}, \qquad c_0(E) = \bar{\sigma} - c_1(E)\,\bar{h}. \tag{4.7}$$

Intrinsic detector spatial response. Intrinsic resolution is measured using a standard source [**pellsQuantitativeValidationMonte2023**] and modelled as an isotropic Gaussian,

$$\mathrm{PSF}_{\text{intr}}(\mathbf{r}; E) = \mathcal{N}\big(\mathbf{r};\ \mathbf{0},\ \sigma_{\text{intr}}^2(E)\,\mathbb{I}\big)\,. \tag{4.8}$$

When intrinsic resolution is reported as $\mathrm{FWHM}_{\text{intr}}(E)$, we convert using

$$\sigma_{\text{intr}}(E) = \frac{\mathrm{FWHM}_{\text{intr}}(E)}{2\sqrt{2\ln 2}}. \tag{4.9}$$

Geometric collimator response with penetration via effective length. For a parallel-hole collimator, the (FWHM) geometric resolution at distance h is commonly approximated by

$$\mathrm{FWHM}_{\text{geom}}(h; E) \approx d_{\text{eff}} \left(1 + \frac{h}{l_{\text{eff}}(E)}\right), \tag{4.10}$$

where d_{eff} is the effective hole diameter and $l_{\text{eff}}(E)$ is an *effective* hole length used to account for septal penetration. We use

$$l_{\text{eff}}(E) = l - \frac{2}{\mu_{\text{Pb}}(E)}, \qquad \mu_{\text{Pb}}(E) \text{ interpolated from NIST XCOM tables}, \tag{4.11}$$

where l is the physical hole length and $\mu_{\text{Pb}}(E)$ is the linear attenuation coefficient of lead at energy E [**XCOMPhotonCross2009**]. Converting from FWHM to standard deviation gives

$$\sigma_{\text{geom}}(h; E) = \frac{\mathrm{FWHM}_{\text{geom}}(h; E)}{2\sqrt{2\ln 2}}. \tag{4.12}$$

4.4.2 Bremsstrahlung-specific blur

Gaussian proxy for range blur. Let Δ denote the displacement between the ^{90}Y decay location and the Bremsstrahlung photon emission location. Rather than using a detailed displacement PDF inside the projector, we approximate the induced range blur by an isotropic Gaussian kernel

$$\mathrm{PSF}_{\mathrm{brem}}(\mathbf{r};E) = \mathcal{N}\big(\mathbf{r};\ \mathbf{0},\ \sigma^2_{\mathrm{brem}}(E)\,\mathbb{I}\big)\,. \tag{4.13}$$

For each clinical energy window, $\sigma_{\mathrm{brem}}(E)$ is obtained from a coarse look-up table (LUT) by fitting a zero-mean Gaussian proxy to MC displacement curves for ^{90}Y Bremsstrahlung in water from GATE fast-simulation studies [**raultFastSimulationYttrium902010**] (Figure **??**).

Composite energy windows. For a set of energy windows $\mathcal{W}$ combined into a single projection dataset, we collapse the corresponding Gaussian mixture by second-moment matching:

$$\sigma^2_{\mathrm{clin}} = \frac{\sum_{i\in\mathcal{W}} w_i\,\sigma_i^2}{\sum_{i\in\mathcal{W}} w_i}, \qquad w_i \propto \text{per-window counts.} \tag{4.14}$$

Implementation in image space. Because the convolution of isotropic Gaussians is Gaussian, incorporating Bremsstrahlung range blur is equivalent to adding variances. In practice we implement this as an image-space convolution by $\mathrm{PSF}_{\mathrm{brem}}$ (i.e. a 3D Gaussian smoothing) applied to the current activity estimate before forward projection. This keeps the forward model compatible with standard projector implementations while capturing the dominant range-broadening effect in water-equivalent tissue.

4.4.3 Validation against SIMIND and GATE forward projections

We assessed how well the isotropic Gaussian proxy reproduces the broader displacement PDFs (including angular effects in GATE) by comparing projected point-source responses produced by:

1. Software for Tomographic Image Reconstruction (STIR) forward projections with geometric CDR modelling only;
2. STIR forward projections with geometric CDR plus the proposed isotropic

Bremsstrahlung range blur;

3. SIMIND simulations (which use an approximate isotropic Bremsstrahlung emission model); and

4. GATE simulations (used here as a higher-fidelity reference that includes angular dependence of Bremsstrahlung emission).

Point-source-in-sphere configuration. A true ^{90}Y point source is not physically realisable at clinical activity levels because the β-particles have a finite range. To approximate a point source for spatial response measurements, we embedded the activity within a small water-equivalent sphere of radius 1.2 cm, consistent with the maximum β-particle range in soft tissue/water (approximately 11 mm) [**pasciakRadioembolizationDynamicRole2014**]. For each configuration we measured the detector-plane FWHM at source–collimator distances of 2, 5, 10, 15, and 20 cm.

Clinical energy windows. Comparisons were performed for three representative clinical windows:

$$[38.24, 126.16]\,\mathrm{keV}, \qquad [50.0, 150.0]\,\mathrm{keV}, \qquad [75.0, 225.0]\,\mathrm{keV}. \tag{4.15}$$

Scope and limitations. This validation focuses on water-equivalent media, which is appropriate for the liver (the primary application in post-radioembolization imaging). Differences in range and scatter properties in lung or cortical bone are not explicitly modelled here and are not expected to dominate performance for liver-focused dosimetry.

4.5 Residual Correction

Fu and Qi proposed a residual-correction (RC) strategy to accelerate reconstructions when an *accurate* forward model is too expensive to apply at every iteration [**fuResidualCorrectionMethod2010**]. In their positron-range setting, the accurate system model can be expressed as a fast projector composed with an additional blurring operator. RC then intermittently estimates the modelling discrepancy and folds it into an additive term, allowing subsequent iterations to proceed using the computationally cheaper projector.

In this work we investigate whether RC can be applied analogously to ^{90}Y Bremsstrahlung SPECT reconstruction. Empirically, we find that RC behaves well for high-count, low-background phantom data, but can become unreliable in low-count clinical-like regimes where regularisation is necessarily strong. We attribute this behaviour to the fact that RC corrects the *forward prediction* but does not, in general, restore a *matched* forward/adjoint pair inside a regularised reconstruction.

4.5.1 Residual correction update

Let $A_\star$ denote the fast (computationally cheap) system matrix used in the iterative updates and let A_{MC} denote a more accurate (expensive) system model (here, Monte Carlo based). We consider the Poisson model

$$y_i \sim \mathrm{Poisson}\big((A_\star x)_i + b_{E,i}\big)\,, \tag{4.16}$$

where b_E is an additive correction term (e.g. scatter). At spaced iterations, we estimate the discrepancy between the accurate and fast forward predictions at the current iterate by approximately solving

$$\hat{x} \approx \operatorname*{arg\,min}_{x\geq 0} \mathcal{O}(x \mid y\,; b_E, A_\star)\,, \tag{4.17}$$

and then forming the residual

$$(\Delta A)\hat{x} \;=\; A_{\mathrm{MC}}\hat{x} - A_\star\hat{x}. \tag{4.18}$$

This residual is incorporated into the additive term,

$$b_E^{(\mathrm{new})} \;=\; b_E^{(\mathrm{old})} + (\Delta A)\hat{x}, \tag{4.19}$$

after which reconstruction proceeds again under the updated additive correction. In favourable regimes, this produces forward predictions closer to those of A_{MC} while retaining the per-iteration cost of $A_\star$.

4.5.2 Regularised objective with a Relative Difference prior

In Bremsstrahlung SPECT, low-count data frequently necessitate comparatively strong regularisation. In all RC experiments reported here, we reconstruct using a Relative Difference prior (RDP), i.e. we solve a MAP problem of the form

$$\hat{x} = \arg\min_{x \geq 0} \{\Phi_{A_\star}(x) + \beta R_{\mathrm{RDP}}(x)\}, \tag{4.20}$$

where $\Phi_{A_\star}(x)$ is the negative Poisson log-likelihood (up to constants),

$$\Phi_{A_\star}(x) = \sum_{i=1}^{I} \left[(A_\star x)_i + b_{E,i} - y_i \log\big((A_\star x)_i + b_{E,i}\big)\right], \tag{4.21}$$

and the Relative Difference prior is defined over a neighbourhood system $\mathcal{N}$ by

$$R_{\mathrm{RDP}}(x) = \frac{1}{2} \sum_{(j,k)\in\mathcal{N}} w_{jk} \frac{(x_j - x_k)^2}{x_j + x_k + \gamma|x_j - x_k| + \varepsilon}. \tag{4.22}$$

Here, x_j and x_k denote neighbouring voxels, $w_{jk} \geq 0$ are neighbourhood weights, γ controls edge preservation, and $\varepsilon \geq 0$ is the STIR RDP stabiliser; STIR assigns a zero contribution to a zero–zero voxel pair when $\varepsilon = 0$. The hyperparameter $\beta > 0$ controls the overall strength of regularisation.

4.5.3 Interpretation via matched versus unmatched projectors

To understand why RC can be fragile in low-count, strongly regularised settings, it is helpful to view the method through the lens of *matched* versus *unmatched* projector pairs.

Set-up. Suppose the accurate forward model can be written as a fast projector composed with an additional blur (or resolution) operator K,

$$P := A_{\mathrm{fast}}K, \qquad \bar{y}(x) = Px + r = A_{\mathrm{fast}}Kx + r, \tag{4.23}$$

with r an additive background term and Poisson data $y \sim \mathrm{Poisson}(\bar{y}(x))$.

Maximum likelihood (no regularisation). The negative log-likelihood for forward model P has gradient

$$\nabla \Phi_P(x) = P^\top \left(1 - \frac{y}{Px + r}\right). \tag{4.24}$$

For $P = A_{\text{fast}} K$, the *matched* stationarity condition is

$$K^\top A_{\text{fast}}^\top \left(1 - \frac{y}{A_{\text{fast}} K x + r}\right) = 0. \tag{4.25}$$

If instead one uses the fast backprojector $A_{\text{fast}}^\top$ with the same forward model (an *unmatched* pair), the stationarity condition would be

$$A_{\text{fast}}^\top \left(1 - \frac{y}{A_{\text{fast}} K x + r}\right) = 0. \tag{4.26}$$

In the special case where $K^\top$ is injective (and numerically well-conditioned on the relevant subspace), (**??**) and (**??**) share the same stationary points, since $K^\top z = 0 \Rightarrow z = 0$. This explains why unmatched backprojection can appear benign in some unregularised, high-count regimes.

How RC fits. Residual correction aims to make the *forward prediction* behave like the accurate model without applying K at every iteration, i.e.

$$A_{\text{fast}} K x \;\approx\; A_{\text{fast}} x + \mathrm{RC}(x), \tag{4.27}$$

where $\mathrm{RC}(x)$ approximates $(A_{\text{fast}} K - A_{\text{fast}})x$ (often using lagged/intermittent updates). If the correction is consistent at a limit point $x^\star$, then the ratio term $y/(A_{\text{fast}} K x + r)$ is effectively "accurate-forward". However, the backprojection used by the iterative update remains $A_{\text{fast}}^\top$, not $(A_{\text{fast}} K)^\top = K^\top A_{\text{fast}}^\top$. In this sense, RC can be interpreted as correcting the forward model while still behaving like an *unmatched* method on the adjoint side.

Why low-count + RDP regularisation is high-risk. With an RDP prior, the MAP stationarity condition (ignoring the non-negativity constraint) is

$$P^\top \left(1 - \frac{y}{Px + r}\right) + \beta \nabla R_{\text{RDP}}(x) = 0. \tag{4.28}$$

For $P = A_{\text{fast}}K$, the matched condition becomes

$$K^\top A_{\text{fast}}^\top \left(1 - \frac{y}{A_{\text{fast}}Kx + r}\right) + \beta \nabla R_{\text{RDP}}(x) = 0, \tag{4.29}$$

whereas an unmatched analogue would be

$$A_{\text{fast}}^\top \left(1 - \frac{y}{A_{\text{fast}}Kx + r}\right) + \beta \nabla R_{\text{RDP}}(x) = 0. \tag{4.30}$$

As total counts decrease, the likelihood term becomes flatter and noisier, so the relative influence of $\beta \nabla R_{\text{RDP}}(x)$ increases even when β is held fixed. Unlike the ML case, the factor $K^\top$ multiplies only the data-term gradient in (**??**) and does not act on $\nabla R_{\text{RDP}}(x)$. Therefore, for $\beta > 0$, matched and unmatched fixed points generally differ. In this prior-dominated regime, using an unmatched backprojector while correcting only the forward prediction can substantially alter the balance between data fidelity and regularisation, leading to marked divergence from the solution associated with the fully matched accurate model.

4.5.4 Empirical observations

For high-count, low-background phantom acquisitions (e.g. NEMA-style datasets), RC substantially improves agreement with accurate forward projections and remains stable. For low-count Bremsstrahlung acquisitions, however, we observe that RC can become unreliable when combined with strong RDP regularisation. These observations are consistent with the interpretation above: in prior-dominated regimes, mismatched adjoints change the effective weighting of the data term relative to the RDP, and RC does not in general restore a matched forward/adjoint pair.

4.6 Experimental Evaluation

The preceding sections introduced four modelling choices that can be layered onto the ^{90}Y Bremsstrahlung forward model: an image-space Gaussian approximation to the range blur (Section **??**); an additive scatter term that is either fixed (*offline*) or updated on-the-fly (*online*) during reconstruction (Section **??**); a damped rather than full update of that additive term (Section **??**); and a Fu & Qi-style residual correction of the resolution model (Section ??). This section evaluates each of these on measured data, and asks not whether any single option is universally "best", but how each behaves as the regularisation strength is varied and how they interact.

4.6.1 Datasets and reconstruction protocol

We evaluated the corrections on three measured phantom acquisitions and one patient study, chosen to span the range of background activity encountered in post-radioembolization imaging:

- **NEMA (cold background):** a NEMA IEC body phantom with six equal-concentration hot spheres (10–37 mm) in a (nominally) cold background.
- **NEMA (warm background):** the same phantom geometry with an 8:1 sphere-to-background activity ratio. The 17 mm sphere was unfilled and is excluded; the two smallest spheres (≤ 13 mm) are noise-limited at these count levels and are excluded from the quantitative analysis.
- **Anthropomorphic liver:** an anthropomorphic phantom with a single hot lesion insert in a warm liver compartment ($\approx 4{:}1$).
- **Patient (SIRT-3):** a clinical post-SIRT ^{90}Y acquisition with five liver lesions delineated in the co-registered ^{90}Y PET space.

Each reconstruction was performed with OSEM using 12 staggered subsets and 10 epochs (120 subset updates), warm-started from the previous outer iterate. The prior was included through a one-step-late gradient step applied after each subset update, using the Relative Difference prior (RDP, Equation ??) with weight $\beta \in \{0, 10^{-3}, 10^{-2}, 10^{-1}\}$; unless a β-sweep is shown, all other results use $\beta = 10^{-2}$. The SPECTUB system matrix modelled attenuation (from the co-registered CT) and a depth-dependent Gaussian collimator–detector response. The Bremsstrahlung range blur, when enabled, was the isotropic 6.9 mm-FWHM image-space Gaussian of Section ??, applied to the activity estimate inside the acquisition model before forward projection. This reconstruction was embedded in the outer OSEM → SIMIND additive-update loop of Section ??, run for 10 outer iterations;

between outer iterations the reconstruction was smoothed with a 5 mm-FWHM isotropic Gaussian before being used as the source for the next Monte Carlo simulation.

Additive-term construction (reproducibility). At each outer iteration k, a single SIMIND PENETRATE simulation (Section **??**) was run on the current, body-masked reconstruction, yielding two projection components: the total detected fluence $p_{\text{tot}}^{(k)}$ (all interactions) and the geometrically collimated, attenuated primary $p_{\text{geo}}^{(k)}$. The Monte Carlo output was calibrated to the fast SPECT forward projector $A_\star$ by a scalar $s^{(k)}$, computed as a ratio-of-sums of $A_\star x^{(k)}$ to $p_{\text{geo}}^{(k)}$ after trimming the 10% most extreme per-bin ratios for robustness. The two corrections were then formed as

$$\widehat{b}_{\text{sc}}^{(k)} = s^{(k)}\big(p_{\text{tot}}^{(k)} - p_{\text{geo}}^{(k)}\big) \quad \text{(scatter: all non-geometric-primary counts)}, \tag{4.31}$$

$$(\Delta A)x^{(k)} = s^{(k)}\, p_{\text{geo}}^{(k)} - A_\star x^{(k)} \quad \text{(residual, cf. Equation ??)}. \tag{4.32}$$

Each term was folded into the running additive estimate by the damped update of Section **??** — the scatter term with $\alpha = 0.2$, the residual term with $\alpha \in \{0.2, 1\}$ — and the effective additive term applied to the OSEM step, $b^{(k)} = b_{\text{sc}}^{(k)} + (\Delta A)x^{(k)}$ (with only the component(s) active for a given strategy), was projected to non-negativity before each forward projection. The initial additive term $b^{(0)}$ was the pre-trained deep-learning scatter estimate of Section **??**; for the fixed-scatter strategies it was held constant. The body mask used to suppress out-of-patient activity before each simulation was obtained by thresholding the attenuation map (at $0.05\,\text{cm}^{-1}$), with a morphological opening (radius 2) and retention of the largest connected component. Crossing the two scatter options (fixed/updated), the residual option (off/on), the range blur (off/on) and the four values of β yields the four correction *strategies* — fixed scatter, fixed scatter + residual, updated scatter, and updated scatter + residual — each evaluated across the blur and β axes.

Monte Carlo (SIMIND) configuration. All simulations used MC code SIMIND [**ljungbergSIMINDMonteCarlo2012**] with the PENETRATE scoring routine (which resolves geometric collimation, septal penetration and collimator scatter as separate components) and the Y-90 Frey Bremsstrahlung source model. Common settings were a 9.5 mm thallium-doped sodium iodide (NaI(Tl)) crystal, a 10 keV energy-bin width up to 2.08 MeV, a photon-history multiplier of 1, and a low-energy photon-history cutoff at 75% of the lower window threshold. The scanner-, collimator- and window-specific settings, which differ between the Mediso and Siemens acquisitions, are listed in Table **??**.

Table 4.1: Dataset-specific SIMIND acquisition settings. The collimator routine selects the penetration/collimator-scatter model (1) or a simpler geometric model (0).

Dataset	Nominal E (keV)	Energy windov
National Electrical Manufacturers Association (NEMA) (cold)	150	75–225
NEMA (warm)	83	39.8–126
Anthropomorphic	150	75–225
Patient (SIRT-3)	100	50–150

The regularisation strength β is not itself an object of study here: it is well established that increasing the RDP weight suppresses noise and, beyond a point, also suppresses contrast recovery. We instead sweep β so that each correction strategy traces out a recovery–noise *operating curve*, allowing the strategies to be compared fairly at matched noise rather than at a single arbitrary β.

4.6.2 Figures of merit

No spatially co-registered ground-truth image exists for any of these acquisitions (the only available ^{90}Y PET volumes are not registered to the SPECT grid). We therefore avoid any image-to-reference comparison and report only scale-invariant quantities:

- **Recovery coefficient (RC):** the measured activity ratio between a hot volume of interest (VOI) and a reference region, divided by the *known* true ratio, so that perfect recovery gives RC $= 1$. For the *cold-background* NEMA phantom the spheres are all filled to the *same* concentration, so the reference is taken to be the eroded *centre of the largest (*37 mm*) sphere* and each sphere's recovery is RC $= \bar{x}_{\text{sphere}}/\bar{x}_{37\,\text{mm}\,\text{centre}}$, with an ideal value of 1 for every sphere (the 37 mm sphere thus measures its own whole-VOI mean against its centre).

Recovery is reported for the six spheres ($10, 13, 17, 22, 28, 37\,\text{mm}$). For the *warm-background* NEMA phantom the sphere-to-background ratio is normalised by the known 8:1 (the unfilled 17 mm sphere is excluded, and the two smallest spheres, $\leq 13\,\text{mm}$, are noise-limited and excluded from the quantitative summaries); for the anthropomorphic phantom the lesion-to-liver ratio is normalised by the known $\approx 4{:}1$ concentration.

- **Noise:** the voxel-wise coefficient of variation (CoV, standard deviation over mean) in a uniform region (the 37 mm-sphere centre for cold NEMA; the background compartment for warm NEMA; the liver for the anthropomorphic phantom; a background region for the patient).

- **Patient lesion contrast:** as no known activity ratio is available *in vivo*, we use the lesion-to-background uptake ratio as a contrast proxy.

All quantities are read at the converged tenth outer iteration unless a convergence trajectory is shown explicitly. To keep comparisons fair, the reported paired effects hold every other factor fixed and change only the correction under test. Except where recovery is resolved by sphere size (Figure **??**), a single reported recovery coefficient is the mean over the evaluated spheres of the phantom.

4.6.3 On-the-fly scatter update: gains, cost and stability

Updating the additive scatter term during reconstruction, rather than fixing it, is the most aggressive of the four modelling choices, and its behaviour is strongly dataset-dependent (Figure **??**). On the warm-background phantom the online update raises recovery substantially (mean $\Delta\text{RC} \approx +0.10$ relative to a fixed scatter term) but nearly *doubles* the noise (mean $\Delta\text{CoV} \approx +0.9$, an increase in every matched run). On the cold-background phantom, where there is little patient scatter to re-estimate, the update is essentially inert in both recovery and noise. Pooled across all datasets the recovery benefit is not statistically reliable (recovery increased in only $\sim 53\%$ of matched runs); the update helps precisely where there is substantial scatter to model, and destabilises the reconstruction as it does so. The convergence trajectories in Figure **??** make the mechanism explicit: on the warm phantom the updated-scatter recovery oscillates violently in the early outer iterations while the noise climbs monotonically, whereas the fixed-scatter reconstruction is smooth and stable.

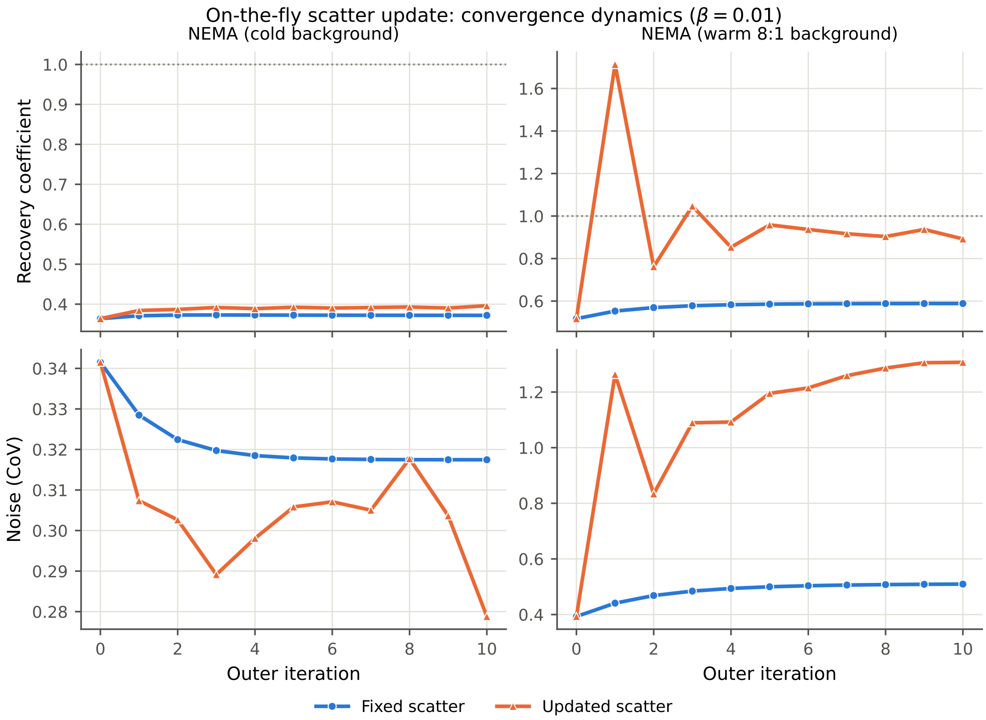


Figure 4.4: Convergence dynamics of the on-the-fly scatter update ($\beta = 10^{-2}$, range blur on). Recovery coefficient (top) and noise (bottom) versus outer iteration for a fixed scatter term (blue) and an on-the-fly updated term (orange), on the cold-background (left) and warm-background (right) NEMA phantoms. On the warm phantom the updated term drives an early recovery overshoot and a monotonic increase in noise; on the cold phantom, where there is little scatter to re-estimate, the two strategies are almost indistinguishable.

4.6.4 Damping the online update

The instability introduced by the online update motivates the damped update rule of Section **??**. Replacing the full update ($\alpha = 1$) with a damped one ($\alpha = 0.2$) reduces noise consistently — in every matched run on the warm-background and anthropomorphic phantoms (mean ΔCoV ≈ -0.09 overall). Crucially, in the most unstable configuration — the combined updated-scatter + residual strategy on the warm phantom — damping improves *both* recovery and noise simultaneously (mean ΔRC $\approx +0.15$, ΔCoV ≈ -0.28), because it suppresses the iteration-to-iteration oscillation of the undamped update rather than merely trading recovery for smoothness

(Figure **??**). In the already-stable configurations damping behaves as expected, exchanging a small amount of recovery for reduced noise. The role of damping is therefore best understood not as a generic smoother but as the stabiliser that renders the aggressive online update usable.

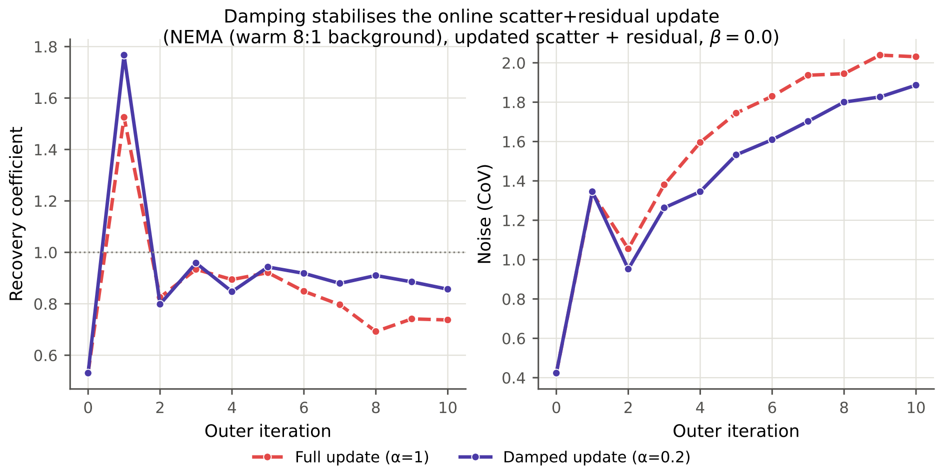


Figure 4.5: Effect of damping the online update on the warm-background NEMA phantom, for the (most unstable) updated-scatter + residual strategy at $\beta = 0$. The full update ($\alpha = 1$, red dashed) settles to lower recovery and higher noise than the damped update ($\alpha = 0.2$, violet solid); damping improves both figures of merit by suppressing the inter-iteration oscillation.

4.6.5 Bremsstrahlung range blur

Including the 6.9 mm Gaussian range-blur term in the forward model behaves as a near-“free” regulariser rather than a resolution-recovery tool. Across all datasets its dominant, reliable effect is a reduction in noise (noise fell in every matched run on the warm-background and anthropomorphic phantoms, mean $\Delta\text{CoV} \approx -0.07$ to -0.15), while the accompanying change in recovery is small and only weakly consistent (mean $\Delta\text{RC} \approx +0.01$, positive in $\sim 61\%$ of runs). Figure **??** shows this as off $\to$ on displacement arrows in the recovery–noise plane: on the warm phantoms the arrows point predominantly leftward (lower noise) at little cost in recovery, whereas on the cold phantom — where counts are low and scatter is minimal — the blur has almost no effect. Modelling the range blur is thus worthwhile because it reduces noise essentially for free, not because it restores contrast.

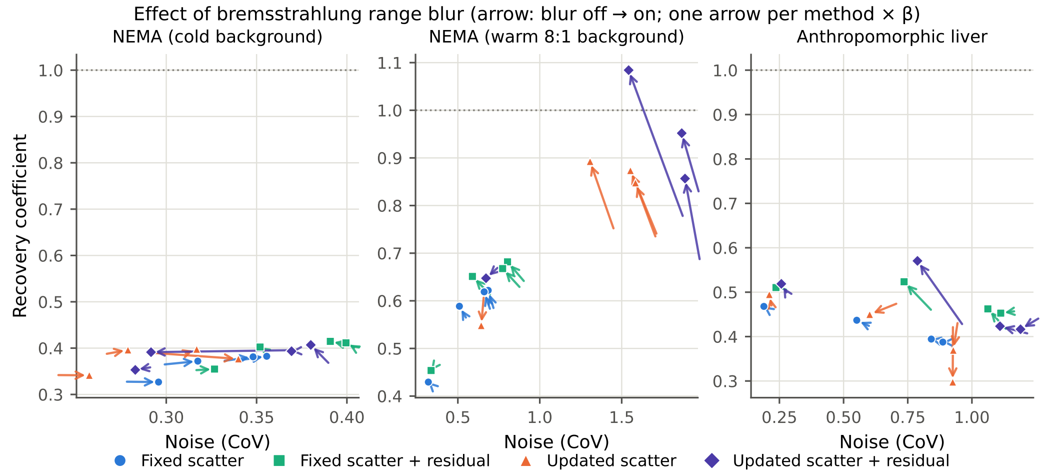


Figure 4.6: Effect of the Bremsstrahlung range-blur model in the recovery–noise plane. Each arrow runs from "blur off" to "blur on" for one correction strategy at one value of β; colour denotes strategy. Leftward arrows indicate a noise reduction. The effect is consistent and noise-reducing on the warm-background and anthropomorphic phantoms and negligible on the cold-background phantom.

4.6.6 Residual correction and resolution recovery

Of the four choices, the Fu & Qi-style residual correction is the most dependable route to improved recovery. Added to a stable (fixed-scatter) base it raises recovery in 86% of matched runs (mean ΔRC $\approx +0.035$, with a small spread), at a modest and universal noise cost (mean ΔCoV $\approx +0.09$) — a textbook resolution-recovery / noise trade-off. The recovery gain is largest on the warm-background phantom ($\approx +0.10$) and smallest on the cold-background phantom ($\approx +0.02$), mirroring the scatter-update behaviour. Figure **??** resolves the effect by lesion size: the residual correction and the online update both lift the recovery-versus-diameter curve above the fixed-scatter baseline, with the largest gains on the mid-to-large spheres. When the residual correction is instead stacked on the *unstable* online update its average benefit persists but becomes erratic (its run-to-run spread grows roughly eight-fold), consistent with the interpretation in Section **??** that the correction is well behaved only when the underlying reconstruction is stable.

4.6.7 Overall trade-off and method comparison

Sweeping β turns each correction strategy into an operating curve in the recovery–noise plane (Figure **??**). Read this way, the strategies separate cleanly on the

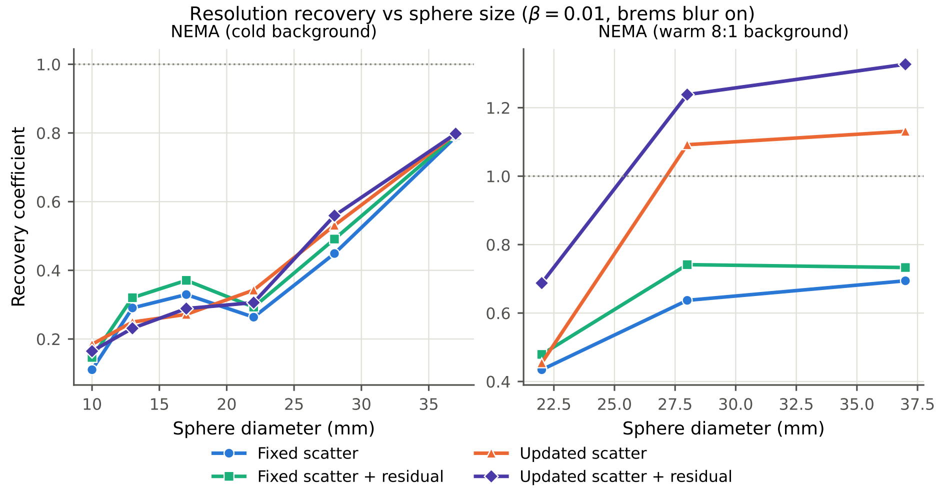


Figure 4.7: Recovery coefficient versus sphere diameter at $\beta = 10^{-2}$ (range blur on) for the cold-background (left) and warm-background (right) NEMA phantoms. Recovery increases monotonically with size; adding residual correction and/or an online scatter update raises the curve, most clearly on the warm phantom. Values above unity on the large warm-phantom spheres reflect edge enhancement. The two smallest warm-phantom spheres are noise-limited and omitted.

warm-background and anthropomorphic phantoms but collapse onto one another on the cold-background phantom, where recovery saturates near 0.4 irrespective of the correction applied. On the warm phantoms the ordering is consistent: the updated-scatter + residual strategy reaches the highest recovery, followed by the residual-only and updated-only strategies, with fixed scatter lowest — but the higher-recovery strategies also sit at higher noise, so no strategy dominates across the whole curve. The paired-effect summary of Figure **??** condenses the four mechanisms into a single view: residual correction gives a reliable recovery gain at a fixed noise cost; the online scatter update offers a large but dataset-dependent recovery gain at a large noise cost; damping and the range blur are noise-reducers; and every effect is modulated by whether the background is cold or warm.

The same behaviour is visible directly in the reconstructions (Figure **??**): on the cold phantom the four strategies are almost indistinguishable by eye, whereas on the two warm phantoms the updated-scatter strategies visibly sharpen the hot inserts against a noticeably noisier background.

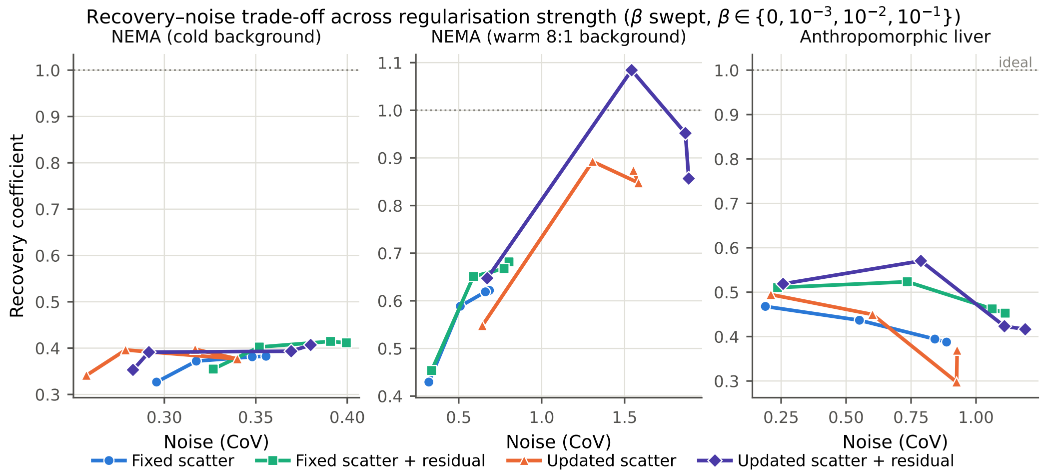


Figure 4.8: Recovery–noise operating curves for the four correction strategies, obtained by sweeping the RDP weight β (range blur on, converged iteration). Each marker is one value of β; the dotted line marks ideal recovery. The strategies separate on the warm-background (centre) and anthropomorphic (right) phantoms and are indistinguishable on the cold-background phantom (left).

4.6.8 Clinical example

The single patient study is consistent with the phantom findings and, in particular, behaves like the *warm*-background phantom, as expected for a liver with diffuse uptake surrounding the lesions (Figure **??**). The online scatter update markedly increases lesion-to-background contrast relative to a fixed scatter term, again at the price of higher background noise, reproducing on measured clinical data the trade-off seen in the warm-background phantom. This should be read qualitatively — it is a single subject with no ground-truth activity ratio — but it demonstrates that the phantom-derived behaviour transfers to the clinical setting. The effect is apparent in the reconstructions themselves (Figure **??**): the online update suppresses the diffuse liver background and lifts the delineated lesion out of it, at the price of a grainier background, exactly as in the warm-background phantom.

4.6.9 Summary

Taken together, the four modelling choices form a hierarchy of aggressiveness and risk. The range blur is a low-risk, noise-reducing addition that is worth including by default. Residual correction is the dependable way to improve contrast recovery, provided the underlying reconstruction is stable. The on-the-fly scatter update

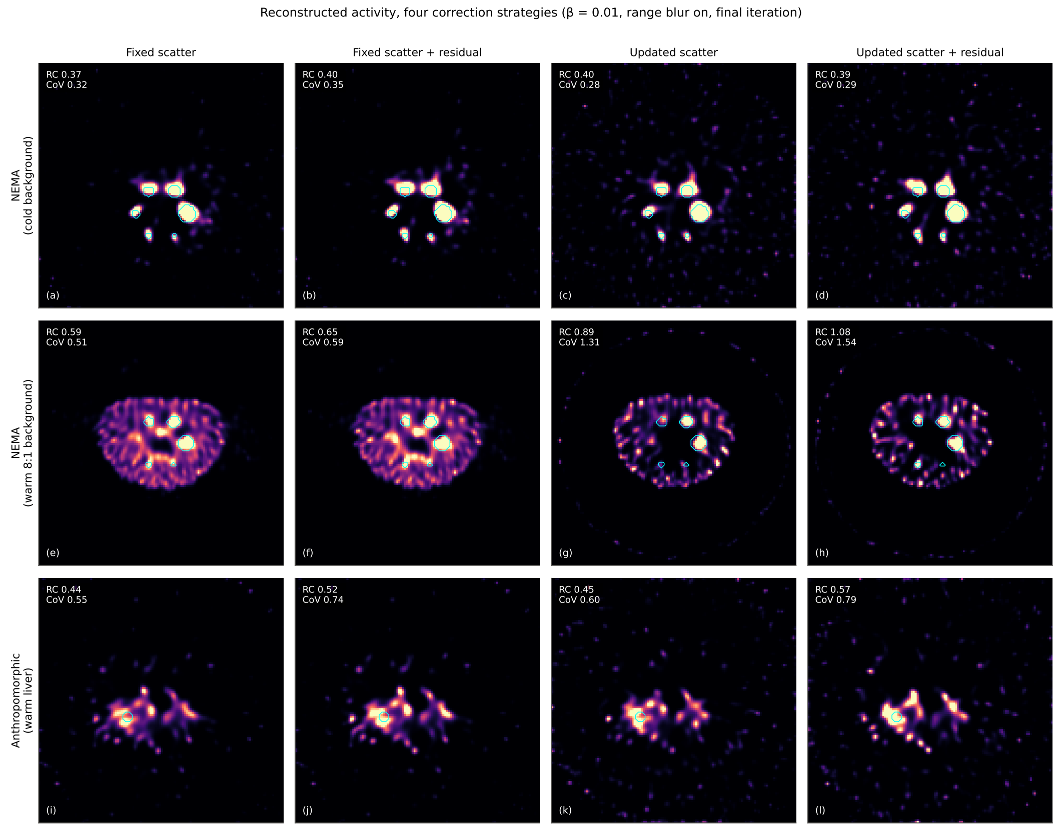


Figure 4.9: Reconstructed activity for the four correction strategies on the three phantoms: cold-background NEMA (top), warm-background NEMA (middle) and the anthropomorphic warm-liver phantom (bottom), at $\beta = 10^{-2}$ with the range blur enabled and at the converged tenth outer iteration (single axial slice through the hot inserts). All panels within a row share an intensity window; cyan outlines mark the hot VOIs and the inset reports the mean recovery coefficient (RC) and background noise (CoV). On the cold phantom the strategies are visually near-indistinguishable and recovery saturates irrespective of the correction; on the two warm phantoms the updated-scatter strategies recover markedly more insert contrast, but at conspicuously higher background noise — the image-space counterpart of the operating curves in Figure **??**.

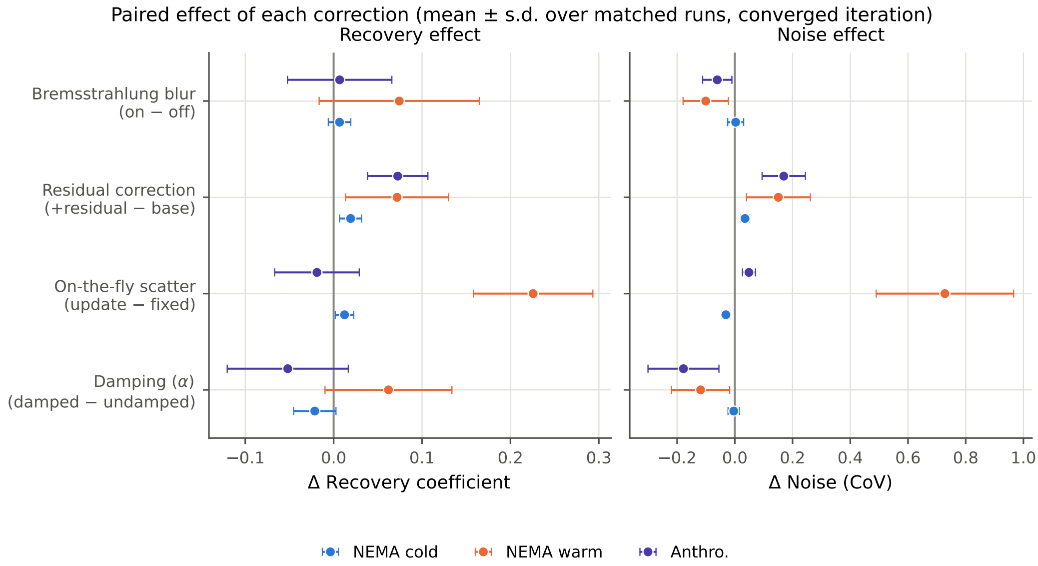


Figure 4.10: Paired effect of each correction on recovery (left) and noise (right), as the mean $\pm$ standard deviation over all matched runs at the converged iteration, split by dataset. Positive ΔRC is more recovery; positive ΔCoV is more noise. Residual correction and the online scatter update raise recovery (the latter only on warm backgrounds, at large noise cost); the range blur and damping reduce noise.

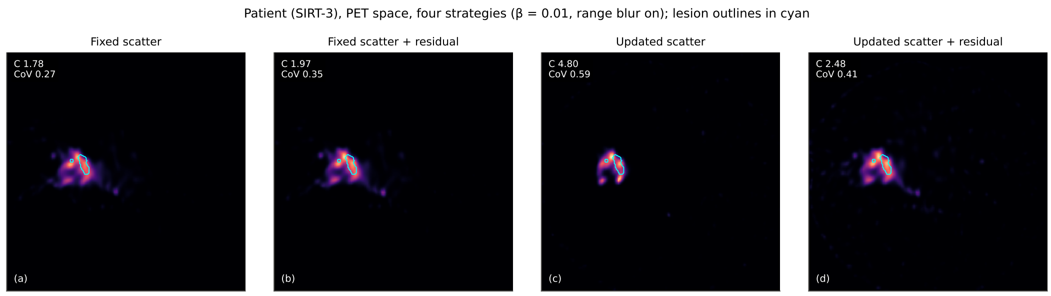


Figure 4.11: Patient (SIRT-3) reconstructions for the four correction strategies, resampled into PET space (single axial slice) at $\beta = 10^{-2}$ with the range blur on and at the converged iteration; the cyan outline is a PET-defined lesion. All panels share an intensity window; the inset gives the mean lesion-to-background contrast (C) and background noise (CoV). The online scatter update (c, d) suppresses the diffuse liver background and raises lesion contrast relative to the fixed-scatter reconstructions (a, b), at the cost of higher background noise — the clinical counterpart of Figure **??**.

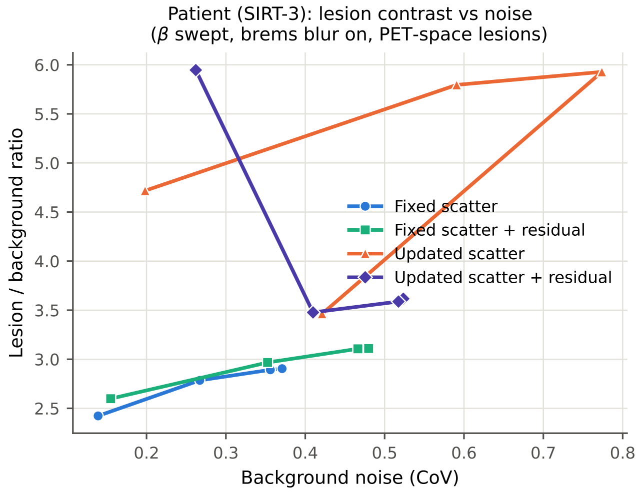


Figure 4.12: Patient (SIRT-3): lesion-to-background uptake ratio versus background noise, sweeping β (range blur on), averaged over five PET-space lesions. The online scatter update reaches substantially higher lesion contrast than a fixed scatter term, mirroring the warm-background phantom.

offers the largest recovery gains, but only where the background is warm enough to carry substantial scatter, and it destabilises the reconstruction; damping is what makes it usable, and in the most unstable configuration recovers both accuracy and noise simultaneously. No single strategy is universally optimal: the appropriate choice depends on the imaging task and on where along the recovery–noise trade-off one wishes to operate. For quantitative liver dosimetry, where warm backgrounds dominate, a damped online update combined with residual correction and the range blur, operated at a moderate RDP weight ($\beta \approx 10^{-2}$), offers the most favourable balance observed here.

Chapter 5

Anatomy-informed deconvolution for PET partial volume correction

5.1 Introduction and relationship to the thesis

Chapter **??** established the penalised-likelihood framework for emission tomography, reviewed representative methods for injecting prior knowledge into the optimisation problem (including anatomically guided and kernelised models), and motivated optimisation strategies that remain stable under regularisation. The core of this thesis concerns regularised and synergistic reconstruction under the full Poisson forward model $\bar{y}(x) = Ax + b$, where the ill-conditioning of A and the low-count regime make prior design, preconditioning, and choice of optimisation algorithm critical.

In this chapter we consider a deliberately simplified, image-domain inverse problem: post-reconstruction partial volume correction (PVC) via deconvolution with an estimated point spread function (PSF). Although this is not a substitute for incorporating resolution modelling within reconstruction, it provides a controlled sandbox in which the effects of anatomical guidance and regularisation can be studied without confounding factors such as subset noise, projector modelling error, and complex system geometry. In particular, the chapter serves two main purposes within the broader narrative:

1. it connects the priors reviewed in Chapter **??** (kernel methods, directional regularisation) to a tractable inverse problem where their qualitative behaviour

is easily interpreted;

2. it provides a practical PVC baseline that is straightforward to deploy on retrospectively reconstructed images.

Richardson–Lucy (RL) deconvolution is a purely image-domain analogue of MLEM under the Poisson model $\bar{y}(x) = Cx + b$: it is the same EM update, with the system matrix A replaced by the PSF convolution operator C [**sheppMaximumLikelihoodReconstruction1982**, **sawatzkyEMTVMethodsInverse2013**]. In this setting the "sensitivity image" simplifies substantially, since for a unit-sum PSF one has $C^{\top}\mathbf{1} \approx \mathbf{1}$, so the usual MLEM normalisation by $A^{\top}\mathbf{1}$ becomes approximately trivial. We extend Richardson–Lucy (RL) to a penalised deconvolution regime by incorporating a directional total variation prior (Equation **??**), and we construct kernelised deconvolution updates by substituting the kernel image model $x = K\alpha$, exactly as in KEM/HKEM. The resulting anatomy-aware deconvolution variants are referred to throughout as Kernelised Richardson–Lucy (KRL), Hybrid Kernelised Richardson–Lucy (HKRL), and Richardson–Lucy with Directional Total Variation (RL–DTV). Their basic update equations are derived in the following sections.

5.2 Image-domain inverse problem and objective

Let $x \in \mathbb{R}_+^J$ denote the unknown (sharper) PET activity image on a voxel grid, and let $y \in \mathbb{R}_+^J$ denote an observed blurred image. Although reconstructed PET images are not independent raw count data, their local noise is often treated as approximately Poisson in post-reconstruction restoration models [**sawatzkyEMTVMethodsInverse2013**]. We therefore use the following image-domain approximation:

$$y_j \sim \text{Poisson}(\bar{y}_j(x)), \qquad \bar{y}(x) = Cx + b, \qquad j = 1, \ldots, J, \tag{5.1}$$

where $C \in \mathbb{R}_+^{J \times J}$ is a discretised convolution operator representing a space-invariant PSF, and $b \in \mathbb{R}_+^J$ is an optional additive background term (set to 0 in the experiments unless stated otherwise). The operator C plays the same role as the system matrix A in (**??**), but in the image domain.

Within this approximation, assuming independence across voxels in the image-domain observation model, the Poisson negative log-likelihood is

$$\mathcal{L}(x) = \sum_{j=1}^{J} \left(\bar{y}_j(x) - y_j \log \bar{y}_j(x)\right), \qquad \bar{y}(x) = Cx + b, \tag{5.2}$$

dropping constants independent of x as in Chapter **??**.

A MAP (penalised-likelihood) deconvolution estimate is then

$$\hat{x} \in \arg\min_{x \in \mathcal{C}} \{\mathcal{L}(x) + \beta \mathcal{R}(x)\}, \qquad \mathcal{C} = \mathbb{R}_+^J, \tag{5.3}$$

with $\beta > 0$ and an appropriate regulariser $\mathcal{R}$.

Throughout, C is non-negative and (approximately) intensity preserving: for a unit-sum PSF kernel and negligible boundary effects, $C\mathbf{1} \approx \mathbf{1}$ and $C^\top \mathbf{1} \approx \mathbf{1}$. The adjoint $C^\top$ corresponds to convolution with the flipped PSF. For a symmetric PSFs common in PET, $C^\top \approx C$.

5.3 Baseline: Richardson–Lucy deconvolution

The RL algorithm is the EM algorithm for the Poisson model (**??**) with $b = 0$ [**sheppMaximumLikelihoodReconstruction1982**]. Introducing elementwise multiplication and division, denoted $\odot$ and $\oslash$, the unit-sensitivity RL update is

$$x^{(k+1)} = x^{(k)} \odot C^\top \left(y \oslash (Cx^{(k)})\right). \tag{5.4}$$

More generally, the right-hand side of (**??**) contains an elementwise division by $C^\top \mathbf{1}$; in this chapter this denominator is neglected because the PSF kernels are normalised and boundary effects are small. As with MLEM/OSEM under the full forward model, RL iterations tend to sharpen edges rapidly but amplify high-frequency noise as iterations increase. Consequently, unregularised RL is typically used with early stopping, which acts as an implicit regulariser but yields iteration-dependent bias.

5.3.1 Scaled-gradient interpretation and EM-type preconditioning

To connect with Chapter **??**, we rewrite (**??**) as an additive scaled-gradient step. The gradient of the Poisson negative log-likelihood (**??**) (with $b = 0$) is

$$\nabla_x \mathcal{L}(x) = C^\mathsf{T}(\mathbf{1} - y \oslash (Cx)). \tag{5.5}$$

Using the same rearrangement as for MLEM in Chapter **??**, and again using $C^\mathsf{T}\mathbf{1} \approx \mathbf{1}$, we obtain

$$x^{(k+1)} = x^{(k)} \odot \left(\mathbf{1} - \nabla_x \mathcal{L}(x^{(k)})\right). \tag{5.6}$$

This corresponds to an EM-type diagonal preconditioner analogous to $\mathcal{P}^{(k)}_{\mathrm{EM}} = \mathrm{diag}(x^{(k)}/(A^\mathsf{T}\mathbf{1}))$ in the full reconstruction problem.

5.4 Kernelised Richardson–Lucy deconvolution

Kernel methods were reviewed in Chapter **??** as an implicit regularisation strategy in which prior information enters through the image model $x = K\alpha$ rather than through an explicit penalty [**wangPETImageReconstruction2015**, **hutchcroftAnatomicallyaidedPETReconstruction2016**]. The same idea can be applied to image-domain deconvolution.

5.4.1 Kernel image model

Let u be a co-registered anatomical guidance image (magnetic resonance (MR) or CT), resampled to the PET grid. We represent

$$x = K\alpha, \qquad \alpha \in \mathbb{R}^J_+, \qquad K \in \mathbb{R}^{J\times J}_+, \tag{5.7}$$

where K is a sparse, row-normalised kernel matrix constructed from features extracted from u (and optionally spatial coordinates), as in Chapter **??**. Row normalisation ($K\mathbf{1} = \mathbf{1}$) ensures that constant coefficient images are mapped to constant activity images, preserving the intended intensity scale of the kernel representation.

Substituting into (**??**) gives the kernelised forward model

$$\bar{y}(\alpha) = C(K\alpha) + b. \tag{5.8}$$

5.4.2 KRL update

Maximising the Poisson likelihood with respect to α under (**??**) yields an EM-type multiplicative update analogous to KEM:

$$\alpha^{(k+1)} = \alpha^{(k)} \odot \frac{K^\top C^\top \left(y \oslash (CK\alpha^{(k)} + b)\right)}{K^\top C^\top \mathbf{1}}, \qquad x^{(k+1)} = K\alpha^{(k+1)}. \tag{5.9}$$

As in KEM, the denominator can be precomputed once, and the update preserves non-negativity given $\alpha^{(0)} \geq 0$. Under the same unit-sensitivity approximation used for vanilla RL, $C^\top \mathbf{1} \approx \mathbf{1}$, the denominator reduces to $K^\top \mathbf{1}$; this term is generally not equal to $\mathbf{1}$ and cannot be ignored.

5.4.3 Hybrid KRL (HKRL)

A known limitation of purely anatomy-derived kernels is suppression of PET-unique structures when they are not supported by the guidance image [**deiddaHybridPETMR2019**]. Hybrid kernels mitigate this by incorporating features derived from the evolving PET estimate [**deiddaHybridPETMR2019**].

Because iteration-dependent kernels can propagate noise back into the coupling structure at late iterations, we adopt a frozen-kernel strategy consistent with the motivation in HKEM, freezing after 5 iterations (although this proved to be unnecessary for the sphere phantom as the best iteration was usually before this point).

5.5 Directional-TV regularised RL deconvolution

directional total variation (DTV) was introduced in Chapter **??** as an anatomically guided regulariser that promotes edge alignment between the emission image and a structural guidance image [**ehrhardtPETReconstructionAnatomical2016**]. Here we use DTV as an explicit penalty for deconvolution.

5.5.1 Penalised objective and update

The penalised-likelihood deconvolution objective is

$$\hat{x} \in \arg\min_{x \in \mathbb{R}_+^J} \{\mathcal{L}(x) + \beta R_{\mathrm{sDTV}}(x;u)\}, \qquad \mathcal{L}(x) = \sum_j \left((Cx+b)_j - y_j \log(Cx+b)_j\right). \tag{5.10}$$

Here R_{sDTV} is a smoothed DTV penalty. A natural baseline solver is the scaled-gradient RL–DTV iteration

$$x^{(k+1)} = \left\{x^{(k)} \odot \left(\mathbf{1} - \varsigma_k \left[\nabla_x \mathcal{L}(x^{(k)}) + \beta \nabla_x R_{\mathrm{sDTV}}(x^{(k)};u)\right]\right)\right\}_{\geq 0}, \tag{5.11}$$

where $\{\}_{\geq 0}$ denotes projection onto non-negativity. Step sizes $\{\varsigma_k\}$ may be taken constant (with conservative tuning) or diminishing to improve stability. This update is the image-domain analogue of the scaled-gradient MAP schemes discussed in Chapter **??** (compare with BSREM-type forms), but without subsets.

In contrast to kernelised RL variants, RL–DTV is typically numerically stable once β is chosen, and the objective in (**??**) is smooth with simple bound constraints. This makes it well suited to a quasi-Newton solver: in the experiments of this chapter we therefore accelerate convergence by optimising (**??**) with the L-BFGS-B algorithm reviewed in Section **??**, using bound constraints to enforce non-negativity. In this setting, L-BFGS-B replaces the hand-tuned step-size sequence $\{\varsigma_k\}$ by an automatic line search and curvature-informed scaling, often reaching a stationary point of (**??**) in substantially fewer iterations than the scaled-gradient RL–DTV update (**??**).

For RL and KRL/HKRL, early stopping is essential and is treated as part of the regularisation. In experiments we report results across iteration counts and select a stopping point based on NRMSE (Equation **??**) for the phantom and qualitative assessment for the brain example. For RL–DTV, iteration sensitivity is reduced, but a stopping point is still selected to avoid unnecessary computation once the relative objective decrease satisfies

$$\frac{\mathcal{O}(x^{(k)}) - \mathcal{O}(x^{(k+1)})}{\max\left\{\left|\mathcal{O}(x^{(k)})\right|, \left|\mathcal{O}(x^{(k+1)})\right|, 1\right\}} \leq f_{\mathrm{tol}}, \qquad \mathcal{O}(x) = \mathcal{L}(x) + \beta R_{\mathrm{sDTV}}(x; u). \quad (5.12)$$

Here f_{tol} is chosen to be 10^{-6}. In L-BFGS-B implementations that expose the historical `factr` parameter, the equivalent objective tolerance is $f_{\mathrm{tol}} = \texttt{factr}\,\varepsilon_{\mathrm{mp}}$, where $\varepsilon_{\mathrm{mp}}$ is machine precision.

5.6 Data and experimental setup

Two datasets are used to characterise behaviour under controlled and realistic conditions.

1. **Spherical phantom:** a hot-sphere phantom enabling assessment of edge recovery and spill-over (see Figure **??**).

2. **τ-PET brain with magnetic resonance imaging (MRI) guidance:** a representative in vivo dataset (see Figure **??**).

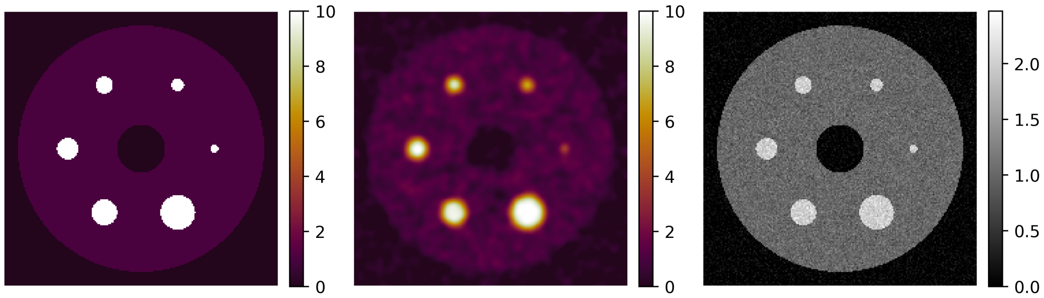


Figure 5.1: Left, ground truth phantom, centre blurred and noisy reconstruction and right MRI guidance image for the simulated sphere phantom. Units are arbitrary.

In both cases, the anatomical image u is registered to the PET grid and resampled to the reconstruction voxel size. The registered anatomy is used either to define the kernel K (KRL/HKRL) or the directional field $\{\xi_j\}$ (RL–DTV).

To obtain a controlled test case with known truth while retaining a realistic resolution/noise regime, the sphere dataset is simulated from a digital activity distribution by forward projecting with an isotropic Gaussian resolution model of

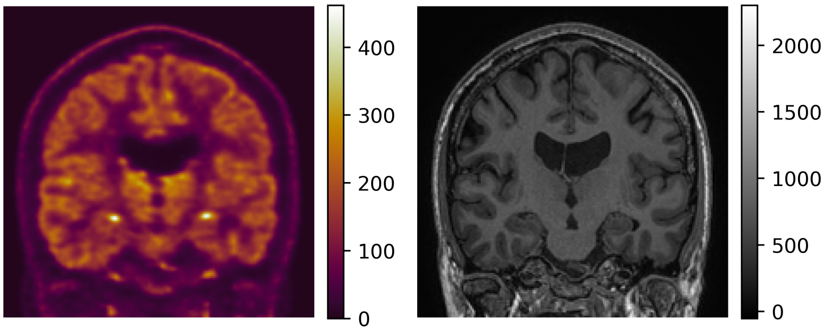


Figure 5.2: Left, reconstructed τ-PET brain, right registered MRI guidance image. Units are arbitrary.

FWHM $= 6\,$mm embedded in the forward projector, adding Poisson noise to the resulting sinograms, and then reconstructing using an operator without additional blurring. This deliberately introduces a resolution-modelling mismatch between simulation and reconstruction. The deconvolution operator used in the subsequent image-domain correction was the same 6 mm Gaussian model, implemented with the same boundary handling as the simulation blur.

Let $x^\star$ denote the digital ground-truth activity image and $\hat{x}$ the reconstructed image. For each sphere v, let $\mathcal{S}_v$ be the set of voxel indices belonging to that sphere, and let $\mathcal{S} = \bigcup_v \mathcal{S}_v$ denote the union over all spheres. We report an aggregate recovery coefficient computed by pooling all sphere voxels,

$$\mathrm{RC} \;=\; \frac{\frac{1}{|\mathcal{S}|}\sum_{i\in\mathcal{S}} \hat{x}_i}{\frac{1}{|\mathcal{S}|}\sum_{i\in\mathcal{S}} x_i^\star}. \tag{5.13}$$

To quantify background noise, let $\mathcal{B}$ denote the set of voxel indices in a warm background region. We compute the coefficient of variation (CoV) within $\mathcal{B}$ as

$$\mathrm{CoV}_{\mathcal{B}} \;=\; \frac{\sigma_{\mathcal{B}}}{\bar{x}_{\mathcal{B}}}, \qquad \bar{x}_{\mathcal{B}} \;=\; \frac{1}{|\mathcal{B}|}\sum_{i\in\mathcal{B}} \hat{x}_i, \qquad \sigma_{\mathcal{B}} \;=\; \sqrt{\frac{1}{|\mathcal{B}|-1}\sum_{i\in\mathcal{B}} (\hat{x}_i - \bar{x}_{\mathcal{B}})^2}. \tag{5.14}$$

Finally, overall fidelity to the truth is summarised by the normalised root mean squared error (NRMSE) over the full image grid Ω,

$$\mathrm{NRMSE} = \frac{\sqrt{\frac{1}{|\Omega|}\sum_{i\in\Omega}\left(\hat{x}_i - x_i^\star\right)^2}}{\sqrt{\frac{1}{|\Omega|}\sum_{i\in\Omega}\left(x_i^\star\right)^2}}. \tag{5.15}$$

The in vivo dataset is a τ-PET scan acquired on a healthy volunteer using the tracer ^{18}F-MK-6240, with a co-registered anatomical MRI serving as guidance. This dataset is used to qualitatively illustrate how the different priors behave in the presence of heterogeneous uptake patterns and fine anatomical structure, where over-regularisation can lead to anatomical "imprinting". The FWHM of the deconvolution operator was estimated from a Hoffman brain phantom scan acquired on the same Siemens Vision scanner. The phantom reconstruction used PSF modelling; after removing extra-cerebral activity, the reconstructed image was co-registered to a Hoffman template, and the template was smoothed over a grid of Gaussian FWHM values, independently in the in-plane and axial directions. The smoothing that minimised the sum-of-squared differences to the reconstructed phantom image was selected, giving FWHM $= (4.6\,\mathrm{mm}, 4.6\,\mathrm{mm}, 6.4\,\mathrm{mm})$.

5.6.1 Hyperparameter tuning

Hyperparameters were tuned by grid search.

For DTV reconstructions, we swept the regularisation weight

$$\beta \in \{0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1.0, 2.0, 5.0\}.$$

Kernel methods used a fixed number of neighbours (48 nearest neighbours) within a cubic $9\times9\times9$ neighbourhood [**hutchcroftAnatomicallyaidedPETReconstruction2016**]. Feature channels were normalised before computing Gaussian similarities, as in the kernel construction described in Chapter **??**; the reported σ_a and σ_f values are therefore interpreted in these normalised feature units. For KRL we swept the anatomical weighting σ_a. For HKRL we additionally swept the functional weighting σ_f. Because we used a k-nearest-neighbours strategy, distance weights $(\sigma_{df}, \sigma_{da})$ were not used.

Table 5.1: Hyperparameter sweeps for kernel-based deconvolution

Method	Hyperparameter	Values swept
KRL/HKRL	σ_a	$\{0.1, 0.2, 0.5, 1.0, 2.0, 5.0\}$
HKRL	σ_f	$\{0.1, 0.2, 0.5, 1.0, 2.0, 5.0\}$

The requirement for early stopping in RL, KRL and HKRL means that the initialising image plays a pivotal role in regularisation. We investigate the effect that initialisation could have on image quality by running identical sweeps over the sphere dataset initialising with a uniform image of ones and the blurred PET image.

τ-PET brain data reconstructions were initialised using a uniform image of ones for RL/KRL/HKRL and the blurred PET image for RL–DTV.

5.7 Results

5.7.1 Spheres

The results for the best reconstructions are summarised in Table **??** and visualised in Figure **??**.

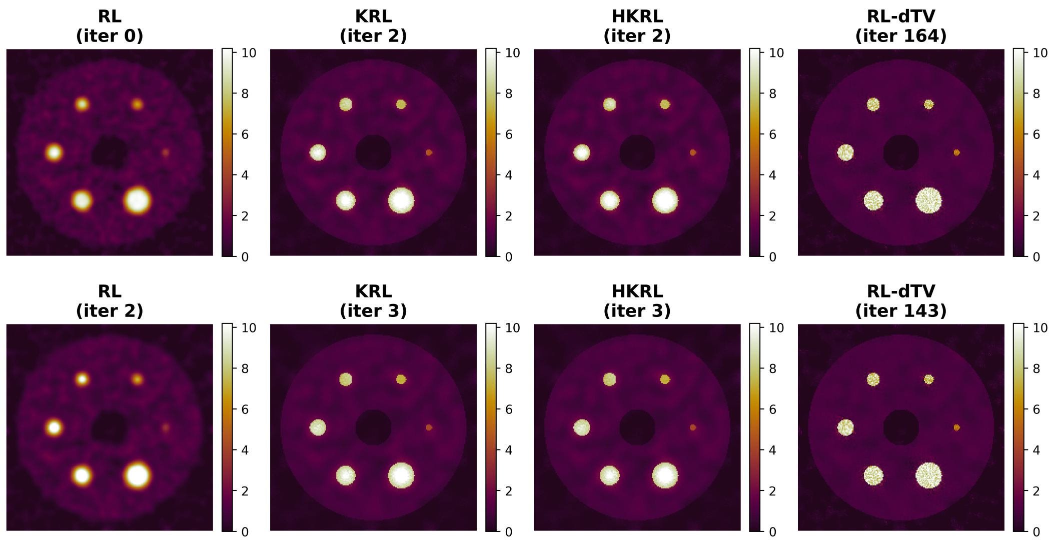


Figure 5.3: Figures showing central slice of reconstructions with best NRMSE. Top, initialised by the blurred PET image and bottom, initialised by a uniform image.

Across both initialisation strategies, RL–DTV achieved the lowest NRMSE (0.0177), the highest aggregate recovery coefficient (approximately 0.889), and

Uniform initialisation							
Method	Min NRMSE	Iter.	RC	Bkg CoV	σ_a	σ_f	β
RL	0.0327	2	0.7645	0.2106	–	–	–
KRL	0.0190	3	0.8781	0.1546	0.5	–	–
HKRL	0.0191	3	0.8776	0.1546	0.5	5	–
RL–DTV	0.0177	143	0.8890	0.1233	–	–	2

Blurred initialisation							
Method	Min NRMSE	Iter.	RC	Bkg CoV	σ_a	σ_f	β
RL	0.0338	0	0.7307	0.2278	–	–	–
KRL	0.0208	2	0.8841	0.1805	5	–	–
HKRL	0.0212	2	0.8805	0.1805	5	5	–
RL–DTV	0.0177	164	0.8893	0.1233	–	–	2

Table 5.2: Tables summarising the hyperparameters and metric values of the best (measured by NRMSE) reconstructions of the various methods for (top) a uniform initialisation of ones and (bottom) the blurred PET image.

the lowest background CoV (approximately 0.123). KRL and HKRL substantially improved on unregularised RL, but were almost indistinguishable from one another in this simple sphere phantom. With blurred initialisation, the best RL result occurred at iteration 0, indicating that unconstrained RL deconvolution immediately increased the NRMSE once applied to this noisy reconstruction.

The effect of initialisation on the quality of reconstructions can be seen in Figures **??** and **??**.

5.7.2 τ-PET brain

τ-PET brain data were only qualitatively analysed. Optimal images were chosen by visual quality, which is an obvious limitation of the study design. The chosen images are shown in Figure **??**.

In the brain example, RL produced sharp but visibly noisy cortical structure, while KRL and HKRL suppressed noise but showed stronger dependence on the anatomical guidance image. RL–DTV gave the best qualitative compromise in this dataset: cortical definition was improved relative to the blurred image and the kernel methods, while noise amplification was lower than for unregularised RL. This

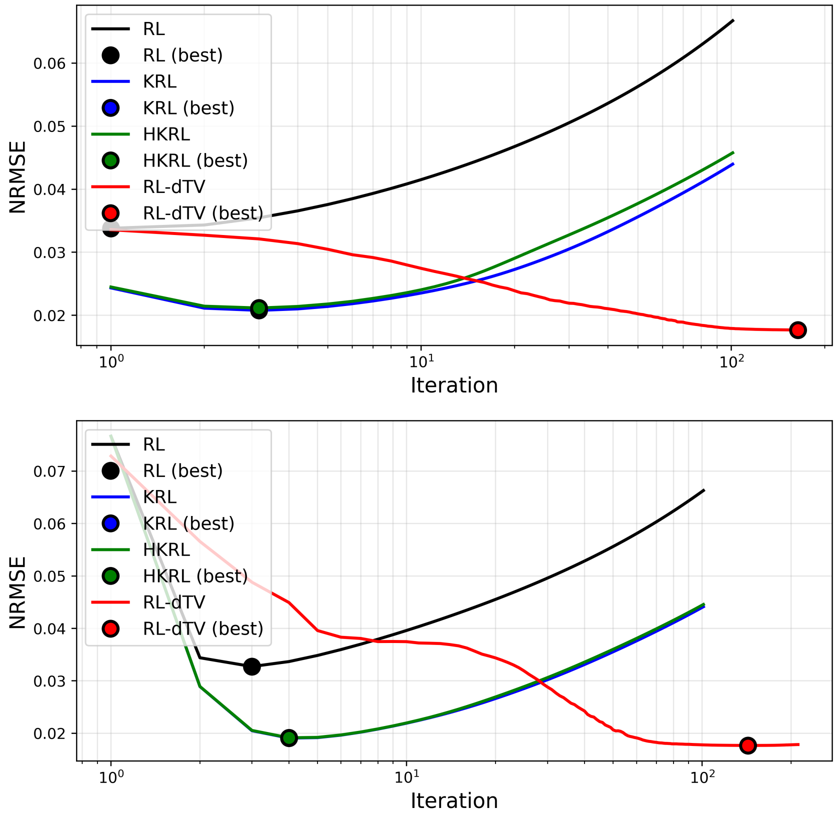


Figure 5.4: Figures showing iteration versus global NRMSE for the best performing reconstructions. Top, initialised by the blurred PET image and bottom, initialised by a uniform image.

visual assessment supports the quantitative sphere result, but should be interpreted cautiously because it is based on a single healthy volunteer scan and does not include a reference truth.

5.8 Discussion

Deblurring is inherently ill-posed: the PSF attenuates high frequencies, so naive inversion magnifies noise. Anatomical guidance provides a structured way to preferentially preserve edges that are plausible given u while damping noise-like oscillations, either by restricting the solution space (kernel methods) or by penalising anatomically inconsistent gradients (DTV).

All guidance-based methods risk bias when the guidance image does not reflect tracer uptake boundaries. KRL is the most restrictive because it enforces $x \in \mathrm{range}(K)$; RL–DTV is less restrictive but can still encourage alignment where

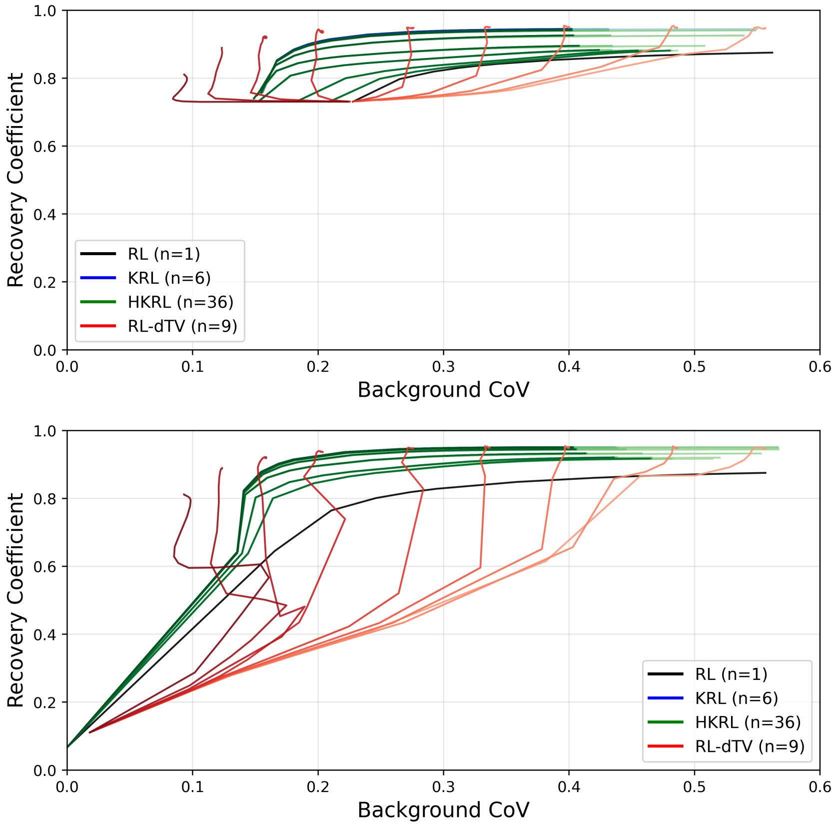


Figure 5.5: Figures showing RC versus CoV for all reconstructions. Top, initialised by the blurred PET image and bottom, initialised by a uniform image. KRL and HKRL are visually indistinguishable in these plots.

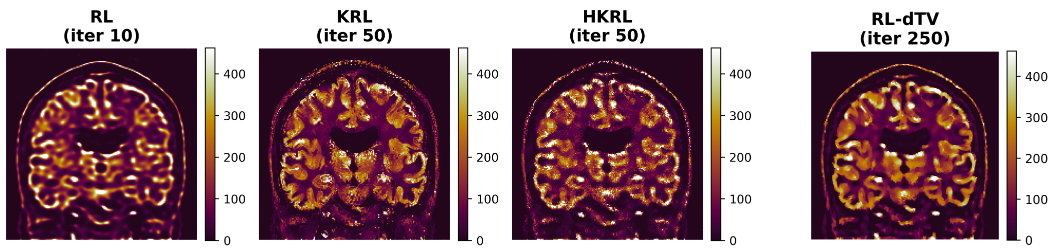


Figure 5.6: Figures showing visually best reconstructions for the different reconstructions.

the data term is weak (low counts, low contrast). HKRL partially mitigates mismatch by allowing the coupling structure to adapt to PET features, but must be controlled (e.g. freezing) to avoid noise-driven feedback.

The behaviours observed here anticipate challenges in the full reconstruction setting: strong priors induce spatially varying curvature and require effective scaling (preconditioning), and guidance mechanisms must be designed to avoid suppressing modality-unique information. These considerations directly motivate the synergy-aware prior and optimisation strategy developed later in the thesis.

The present study should therefore be viewed as a proof-of-principle rather than a definitive validation. A stronger evaluation would require a larger set of patient scans and more demanding controlled simulations, for example BrainWeb-style digital brain phantoms with inserted PET-unique lesions or tumours. Such work is ongoing, supported by additional clinical datasets, and would allow the qualitative observations here to be tested with population-level variability and known ground truth.

5.9 Summary

This chapter formulated post-reconstruction PET PVC as an approximate image-domain Poisson inverse problem $\bar{y}(x) = Cx + b$ and studied RL deconvolution and three anatomy-aware regularised variants. KRL and HKRL implement implicit regularisation through kernelised image models, consistent with the kernel framework reviewed in Chapter **??** [**wangPETImageReconstruction2015**, **hutchcroftAnatomicallyaidedPETReconstruction2016**, **deiddaHybridPETMR2019**]. RL–DTV implements explicit anatomically guided regularisation via a DTV penalty [**ehrhardtPETReconstructionAnatomical2016**] using the same directional notation as Chapter **??**. Together, these methods provide a practical PVC baseline and a controlled setting in which to diagnose guidance-driven bias and noise amplification, informing the design choices for the synergistic priors and optimisation algorithms developed in subsequent chapters.

Chapter 6

Introduction to Weighted Directional Total Nuclear Variation

In this chapter, we introduce a smoothed weighted directional total nuclear variation (w-dTNV) prior for synergistic emission tomography. The chapter focuses on the regulariser itself: its relationship to total nuclear variation, its differentiability after singular-value smoothing, the use of anatomical information through directional operators, and the modality-wise weighting needed when coupling heterogeneous emission images.

The optimisation machinery used to apply this prior in large-scale PET/SPECT reconstruction, including prior-aware preconditioning, subset selection, and multi-bed PET handling, is developed separately in Chapter **??**.

6.1 Total Nuclear Variation Revisited

This section builds on the definitions in Section **??**. For convenience, we restate the key definitions here:

- Registered images: $\boldsymbol{z} := \{z_m\}_{m=1}^{M}$, where $z_m := W_m x_m$ and $W_m : \mathbb{R}_+^{J_m} \to \mathbb{R}_+^{N}$.

- Finite-difference operator: for $\ell = 1, \ldots, d$,

$$\big((Dx)_n\big)^{(\ell)} := \frac{x_{n+\delta_\ell} - x_n}{\Delta_\ell}.$$

- Voxel-wise Jacobian:

$$\mathcal{G}_n(z) := \begin{bmatrix} Dz_{1,n} & Dz_{2,n} & \cdots & Dz_{M,n} \end{bmatrix} \in \mathbb{R}^{d\times M}.$$

- Nuclear norm: for a matrix Y with singular values $\{\sigma_k(Y)\}_{k=1}^{r}$ and $r = \operatorname{rank}(Y)$,

$$\|Y\|_* := \sum_{k=1}^{r} \sigma_k(Y).$$

- Total nuclear variation:

$$\mathcal{R}_{\mathrm{TNV}}(z) := \sum_{n=1}^{N} \|\mathcal{G}_n(z)\|_*.$$

6.1.1 Mathematical properties and differentiability

Briefly revisiting total variation, optimisation of $\mathcal{R}_{\mathrm{TV}}$ (cf. (**??**)) is inconvenient in gradient-based methods because these functionals are not globally smooth: the TV gradient is undefined when $(Dx_m)_j = 0$. A standard remedy is to introduce a small smoothing parameter $\varepsilon > 0$.

If we define

$$\tilde{\mathcal{R}}_{\mathrm{TV}}(x_m) \; := \; \sum_{j=1}^{J_m} \|(Dx_m)_j\|_{2,\varepsilon}, \qquad \|v\|_{2,\varepsilon} := \sqrt{\|v\|_2^2 + \varepsilon^2}, \tag{6.1}$$

then the image-space gradient is

$$\nabla_{x_m} \tilde{\mathcal{R}}_{\mathrm{TV}}(x_m) \; = \; -\operatorname{div}\left(\frac{Dx_m}{\|Dx_m\|_{2,\varepsilon}} \right), \tag{6.2}$$

where the division is understood pointwise over voxels.

For nuclear-norm vectorial total variation, the unsmoothed singular-value map is not differentiable at zero singular values, and hence the corresponding nuclear norm is not C^1 at rank-deficient points. We therefore apply the Charbonnier function to the singular values:

$$g(s)=\sqrt{s^2+\varepsilon^2}-\varepsilon,\qquad g'(s)=\frac{s}{\sqrt{s^2+\varepsilon^2}},\qquad g''(s)=\frac{\varepsilon^2}{(s^2+\varepsilon^2)^{3/2}}. \tag{6.3}$$

The constant $-\varepsilon$ matches the implemented objective value and does not affect gradients or optimisation.

The nuclear norm is orthogonally invariant: for any orthogonal $U\in\mathbb{R}^{d\times d}$ and $V\in\mathbb{R}^{M\times M}$,

$$\|Y\|_* \;=\; \|UYV^\top\|_*. \tag{6.4}$$

More generally, any function of the form

$$\phi(Y)\;:=\;f(\sigma(Y)),\qquad f(s)=\sum_{\ell=1}^{r} g(s_\ell),\qquad r=\min\{d,M\}, \tag{6.5}$$

is orthogonally invariant and depends on Y only through its singular values $\sigma(Y)=(\sigma_1(Y),\dots,\sigma_r(Y))$. Since $g\in C^\infty([0,\infty))$ for $\varepsilon>0$, the symmetric function f is smooth and hence ϕ is differentiable, with gradient given by the standard formula for orthogonally invariant matrix functions [**lewisConvexAnalysisUnitarily1995**, **lewisDerivativesSpectralFunctions1996**].

Jacobian-space gradient. Let $\phi(\mathcal{G})=\sum_{\ell=1}^{\min(d,M)} g(\sigma_\ell(\mathcal{G}))$, where $\sigma_\ell(\mathcal{G})$ are the singular values of the voxel-wise Jacobian matrix $\mathcal{G}\in\mathbb{R}^{d\times M}$. A thin singular value decomposition (SVD)

$$\mathcal{G}\;=\;U\Sigma V^\top,\qquad U\in\mathbb{R}^{d\times r},\;\Sigma=\operatorname{diag}(\sigma)\in\mathbb{R}^{r\times r},\;V\in\mathbb{R}^{M\times r}, \tag{6.6}$$

with $r=\min\{d,M\}$ gives

$$\nabla_{\mathcal{G}}\phi(\mathcal{G})\;=\;U\operatorname{diag}\big(g'(\sigma)\big)\,V^\top,\qquad g'(\sigma_\ell)=\frac{\sigma_\ell}{\sqrt{\sigma_\ell^2+\varepsilon^2}}. \tag{6.7}$$

In practice, it is unnecessary to form U if we exploit the small Gram matrix

$$G:=\mathcal{G}^\top\mathcal{G}\in\mathbb{R}^{M\times M},\qquad G\succeq 0. \tag{6.8}$$

Let $G = V\Lambda V^\top$ with $\Lambda = \operatorname{diag}(\lambda_1,\dots,\lambda_M)$. Then $\lambda_\ell = \sigma_\ell^2$ for the nonzero spectrum and V coincides with the right singular vectors of $\mathcal{G}$. On the range of $\mathcal{G}$ we have $U = \mathcal{G}V\Sigma^{-1}$, so substituting into (**??**) yields

$$\nabla_{\mathcal{G}}\phi(\mathcal{G}) \;=\; \mathcal{G}V\operatorname{diag}\!\left(\frac{g'(\sigma_\ell)}{\sigma_\ell}\right)V^\top, \tag{6.9}$$

with the continuous extension at $\sigma_\ell = 0$. For Charbonnier smoothing,

$$\frac{g'(\sigma_\ell)}{\sigma_\ell} = \frac{1}{\sqrt{\sigma_\ell^2+\varepsilon^2}} = \frac{1}{\sqrt{\lambda_\ell+\varepsilon^2}}, \tag{6.10}$$

hence

$$\nabla_{\mathcal{G}}\phi(\mathcal{G}) \;=\; \mathcal{G}V\operatorname{diag}\!\left(\frac{1}{\sqrt{\lambda_\ell+\varepsilon^2}}\right)V^\top \;=\; \mathcal{G}\,(G+\varepsilon^2\mathbb{I})^{-1/2}. \tag{6.11}$$

Equation (**??**) shows that the Jacobian-space gradient is obtained by applying the inverse square-root of the $M\times M$ symmetric positive-definite (SPD) matrix $G+\varepsilon^2\mathbb{I}$ to the columns of $\mathcal{G}$.

For $M = 2$ modalities, even an eigendecomposition of G can be avoided. Writing

$$G = \begin{pmatrix} a & c \\ c & b \end{pmatrix}, \qquad a = \|\mathcal{G}_{:,1}\|_2^2,\;\; b = \|\mathcal{G}_{:,2}\|_2^2,\;\; c = \langle \mathcal{G}_{:,1}, \mathcal{G}_{:,2}\rangle, \tag{6.12}$$

define $S := G+\varepsilon^2\mathbb{I} = \left(\begin{smallmatrix} s_{11} & s_{12} \\ s_{12} & s_{22}\end{smallmatrix}\right)$ with $s_{11} = a+\varepsilon^2$, $s_{22} = b+\varepsilon^2$, $s_{12} = c$, and the invariants

$$t := \operatorname{tr}(S) = s_{11}+s_{22}, \qquad s := \sqrt{\det(S)} = \sqrt{s_{11}s_{22}-s_{12}^2}. \tag{6.13}$$

Since $\varepsilon > 0$ implies $S \succ 0$, the following identity holds for 2×2 SPD matrices:

$$S^{-1/2} = \sqrt{t+2s}\,(S+s\mathbb{I})^{-1} = \frac{\sqrt{t+2s}}{2\det(S)+st}\begin{pmatrix} s_{22}+s & -s_{12} \\ -s_{12} & s_{11}+s \end{pmatrix}. \tag{6.14}$$

Therefore, for two modalities the Jacobian-space gradient can be evaluated via

$$\nabla_{\mathcal{G}}\phi(\mathcal{G}) = \mathcal{G}S^{-1/2}, \qquad S = G + \varepsilon^2\mathbb{I}, \tag{6.15}$$

using only dot products and scalar square-roots per voxel.

Image-space gradient. Define voxel-wise

$$Q_n \;:=\; \nabla_{\mathcal{G}}\phi\big(\mathcal{G}_n(\boldsymbol{z})\big) \in \mathbb{R}^{d\times M}, \tag{6.16}$$

and let $Q = \{Q_n\}_{n=1}^N$. Applying the chain rule through the linear map $\boldsymbol{z} \mapsto D\boldsymbol{z}$ yields

$$\delta\tilde{\mathcal{R}}_{\mathrm{TNV}}(\boldsymbol{z})[\delta\boldsymbol{z}] = \sum_{n=1}^{N} \langle Q_n,\, (D\,\delta\boldsymbol{z})_n\rangle_F = \langle Q,\, D\,\delta\boldsymbol{z}\rangle\,. \tag{6.17}$$

By definition of the adjoint D^*,

$$\langle Q,\, D\,\delta\boldsymbol{z}\rangle = \langle D^*Q,\, \delta\boldsymbol{z}\rangle\,. \tag{6.18}$$

Defining the discrete divergence as $\operatorname{div} := -D^*$, we obtain

$$\nabla_{\boldsymbol{z}}\tilde{\mathcal{R}}_{\mathrm{TNV}}(\boldsymbol{z}) = -\operatorname{div} Q, \tag{6.19}$$

where div acts column-wise on Q.

6.2 Weighted Directional Total Nuclear Variation (w-dTNV)

This section extends TNV [**rigieJointReconstructionMultichannel2015**, **holtTotalNuclearVariation**
in two orthogonal ways:

- high-resolution anatomical information is incorporated directionally rather than as an additional coupled intensity channel;
- voxel-wise, modality-wise diagonal weighting compensates for heterogeneous scaling and spatially varying reliability across modalities.

The starting point is the standard TNV construction: penalising the nuclear norm of a multi-modality Jacobian promotes joint sparsity and alignment of edges, since a low nuclear norm favours a locally low-rank Jacobian.

6.2.1 Directional operators from guidance

To incorporate guidance information, such as a high-resolution anatomical image, we borrow the operator from directional total variation (Equation **??**). The key design choice is that the guidance image is used only to steer the penalty, not to add an extra column to the joint Jacobian. This avoids directly coupling an anatomical channel, with distinct units, resolution, and artefacts, to the emission channels through the singular spectrum.

If a guidance image were appended as an additional modality, then a single global scaling would have data-dependent and spatially non-uniform consequences. Where emission gradients are weak, even modest guidance gradients can dominate the local Jacobian and hence the singular values; where emission gradients are strong, the same scaling may become negligible. Since the nuclear norm couples modalities through the eigenstructure of the local Gram matrix, such reweighting can also transfer structured guidance artefacts into the coupled reconstruction. By contrast, a directional operator modifies which gradient components are penalised, without forcing the presence of edges where the emission data do not support them.

Applying $\Pi_{m,n}$ to each modality gradient defines the directional multi-modality Jacobian

$$\mathcal{G}_n^{\mathrm{dir}}(\boldsymbol{z}) = \left[\Pi_{1,n}(Dz_1)_n \mid \cdots \mid \Pi_{M,n}(Dz_M)_n\right] \in \mathbb{R}^{d\times M}. \tag{6.20}$$

Here D denotes the chosen discrete directional-difference operator and each $\Pi_{m,n} \in \mathbb{R}^{d\times d}$ is constructed from the guidance field at voxel n. In the implementation used in this work, a guidance image u defines

$$\xi_n = \frac{(Du)_n}{\sqrt{\|(Du)_n\|_2^2+\eta^2}}, \qquad \Pi_{m,n} = \mathbb{I}_d - \gamma_m\,\xi_n\xi_n^\top, \qquad \gamma_m \in [0,1]. \tag{6.21}$$

When η is not specified explicitly it is set from a robust scale estimate of $\|(Du)_n\|_2$, namely 0.01 times the difference between its 99th and 1st percentiles. In homogeneous regions of the guidance image, ξ_n is small and $\Pi_{m,n}$ is close to the identity, so the regulariser reduces to an approximately isotropic TNV-type coupling.

Using CT only to construct the directional operators decouples anatomical steering from emission-to-emission weighting. Modality weights determine how strongly emission gradients are coupled to one another through the nuclear norm, while $\Pi_{m,n}$ determines how strongly the penalty prefers guidance-consistent edge directions. Retaining a two-modality Jacobian also preserves the efficient 2×2 Gram-matrix structure derived in Section **??**.

6.2.2 Modality-wise weighting and the full w-dTNV prior

PET and SPECT images of the same administered Y-90 activity differ substantially in dynamic range: the effective count level, sensitivity, and noise characteristics vary by orders of magnitude between the two modalities, and in clinical imaging the two reconstructions may also operate at different spatial resolutions. If the directional emission gradients $\Pi_{m,n}(Dz_m)_n$ are used directly without normalisation, the nuclear-norm coupling is dominated by the modality with the larger gradient magnitudes.

To compensate, we introduce a voxel-wise, modality-wise diagonal weight matrix

$$B_n := \operatorname{diag}(\rho_{1,n}, \rho_{2,n}) \in \mathbb{R}^{2\times 2}, \qquad \rho_{m,n} > 0. \tag{6.22}$$

The weighted directional Jacobian is

$$\tilde{\mathcal{G}}_n(\boldsymbol{z}) := \mathcal{G}_n^{\mathrm{dir}}(\boldsymbol{z})\, B_n = \left[\Pi_{1,n}(Dz_1)_n \,\middle|\, \Pi_{2,n}(Dz_2)_n\right] \operatorname{diag}(\rho_{1,n}, \rho_{2,n}) \in \mathbb{R}^{d\times 2}. \tag{6.23}$$

The columns of $\tilde{\mathcal{G}}_n$ are $\tilde{g}_{m,n} := \rho_{m,n}\Pi_{m,n}(Dz_m)_n \in \mathbb{R}^d$.

The *weighted directional Total Nuclear Variation* (w-dTNV) prior is then

$$\mathcal{R}_{\text{w-dTNV}}(\boldsymbol{z}) := V_{\mathrm{vox}} \sum_{n=1}^{N} \phi\left(\tilde{\mathcal{G}}_n(\boldsymbol{z})\right), \qquad \phi(Y) := \sum_{\ell=1}^{\min(d,M)} g(\sigma_\ell(Y)), \tag{6.24}$$

where V_{vox} is the voxel volume and g is the Charbonnier smoothing from (**??**). Choosing equal weights $\rho_{m,n} \equiv 1$ recovers unweighted dTNV. Choosing $\rho_{m,n}$ proportional to the inverse of a characteristic modality scale brings both channels to a comparable range before the nuclear-norm coupling is evaluated. The PET/SPECT normalisation used in the application study is specified in Chapter **??**.

Effect on the singular spectrum. Right-multiplying by B_n scales the m-th column of $\tilde{\mathcal{G}}_n$ by $\rho_{m,n}$. This scales the singular values of $\tilde{\mathcal{G}}_n$ in a non-trivial way, since the singular values of YB are not simply $\sigma_\ell(Y)\rho_m$. The intended effect is instead on the Gram matrix: $\tilde{\mathcal{G}}_n^\top \tilde{\mathcal{G}}_n$ contains modality self-terms and cross-terms after scaling, so suitable $\rho_{m,n}$ values allow both modalities to contribute comparably to the coupled singular spectrum. The 2×2 Gram-matrix algebra in Section **??** applies directly to this weighted case.

6.3 Conclusion

This chapter introduced a smoothed, weighted, and directional Total Nuclear Variation (w-dTNV) prior for synergistic reconstruction. Charbonnier smoothing of the singular values yields a differentiable spectral functional whose Jacobian-space gradient can be written in Gram-matrix form. For two coupled emission modalities this reduces the key spectral operation to a 2×2 inverse square-root, allowing the prior and its gradient to be evaluated robustly without a general per-voxel SVD.

The chapter also separated two roles that are easily confounded in multimodality reconstruction. Anatomical information is used directionally through $\Pi_{m,n}$, steering the penalty without adding CT as another coupled intensity channel. Modality scaling is handled separately through the diagonal weight matrix B_n, allowing PET and SPECT gradients to enter the nuclear-norm coupling on comparable scales. Chapter **??** uses these definitions to construct practical preconditioners and subset-selection strategies for large-scale w-dTNV optimisation.

Chapter 7

Preconditioning and Subset Selection for Synergistic Reconstruction

Chapter **??** defined the smoothed weighted directional total nuclear variation prior. Applying this coupled, edge-preserving prior at scale raises three linked optimisation questions. First, overlapping PET bed positions must be reconstructed into a common image if the prior is to act consistently across their overlap. Second, PET and SPECT subsets can be paired or treated as separate stochastic functions. Third, the very different data and prior curvature scales require an effective preconditioner. This chapter addresses these questions in turn and uses the resulting evidence to motivate the optimisation protocol carried forward into Chapter **??**. That application protocol had already been frozen before the wider preconditioner comparison reported here.

7.1 Multi-Bed PET Acquisition

Whole-liver PET commonly requires multiple partially overlapping bed positions. Reconstructing each bed independently and fusing the resulting images is awkward for an edge-preserving prior because the regularisation decisions on either side of the overlap are then made independently. A joint formulation instead estimates one activity image over the combined field of view.

7.1.1 Notation

Let x be the PET activity image on the combined reconstruction grid. For bed position $\text{bed} = 1, \ldots, N_{\text{bed}}$, let R_{bed} restrict the combined image to the axial support reconstructed by that bed. The expected PET data for bed position bed are

$$\bar{y}_{\text{bed}}(x) = A_{\text{bed}} R_{\text{bed}} x + r_{\text{bed}}, \tag{7.1}$$

where A_{bed} is the bed-specific PET system matrix and r_{bed} collects additive corrections. The joint PET data term is

$$\mathcal{D}_{\text{PET}}^{\text{multi}}(x) = \sum_{\text{bed}=1}^{N_{\text{bed}}} \mathcal{D}_{\text{bed}}\left(y_{\text{bed}} \mid A_{\text{bed}} R_{\text{bed}} x + r_{\text{bed}}\right), \tag{7.2}$$

with gradient

$$\nabla \mathcal{D}_{\text{PET}}^{\text{multi}}(x) = \sum_{\text{bed}=1}^{N_{\text{bed}}} R_{\text{bed}}^{\top} A_{\text{bed}}^{\top} \nabla_{\bar{y}_{\text{bed}}} \mathcal{D}_{\text{bed}}\left(y_{\text{bed}} \mid A_{\text{bed}} R_{\text{bed}} x + r_{\text{bed}}\right). \tag{7.3}$$

The corresponding PET sensitivity image on the combined grid is

$$s_{\text{PET}}^{\text{multi}} = \sum_{\text{bed}=1}^{N_{\text{bed}}} R_{\text{bed}}^{\top} A_{\text{bed}}^{\top} \mathbf{1}. \tag{7.4}$$

This sensitivity is used in the EM/BSREM data-curvature term in exactly the same way as the single-bed sensitivity in Equation (**??**).

7.1.2 Why Not Stitch Independent Reconstructions?

A common practical alternative is to reconstruct each PET bed independently and fuse the images using sensitivity-weighted stitching in the overlap. With an edge-preserving prior, however, each reconstruction makes its own noise-suppression and edge decisions before fusion. The qualitative example in Figure **??** shows overlap-region noise and discontinuities after separate reconstruction and fusion. It is included to motivate the joint formulation; no quantitative comparison was performed. **TODO:** Add the Zenodo-managed citation for the multi-bed PET work.

7.1.3 Joint Reconstruction

The joint formulation in Equation (**??**) estimates a single combined PET image. Each bed contributes its own data term and sensitivity, while the image-domain prior is

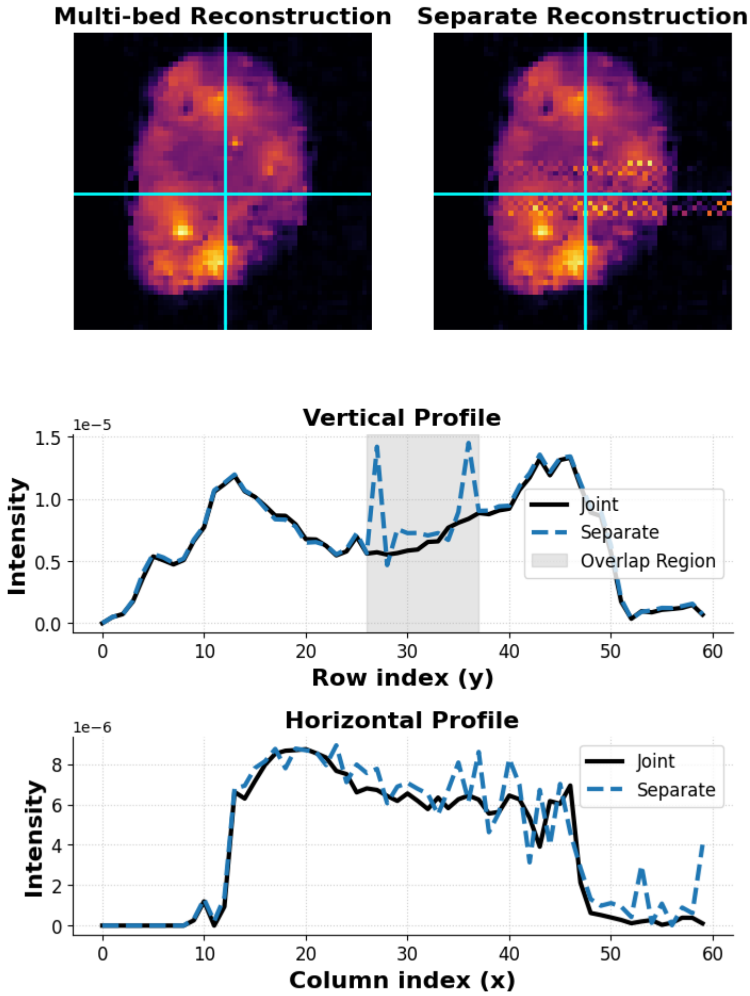


Figure 7.1: Qualitative comparison of separately reconstructed and fused PET bed positions with a joint multi-bed reconstruction using CT-guided edge-preserving regularisation. The shaded region marks the bed overlap and the cyan lines mark the displayed profiles. This example illustrates overlap-region behaviour only; no quantitative comparison was performed.

evaluated over the complete combined field of view. The combined reconstruction therefore permits the edge-preserving prior to act consistently across the overlap, which is the behaviour required for the dual-bed application in Chapter **??**. This construction does not itself guarantee an artefact-free reconstruction; the evidence here is qualitative.

7.2 Stochastic MAP Optimisation

Let $\boldsymbol{z} = (z_1, z_2)$ denote the PET and SPECT images represented on a common reconstruction grid. For N_d sampled data functions, define $\mathcal{D}(\boldsymbol{z}) = \sum_{i=1}^{N_\mathrm{d}} \mathcal{D}_i(\boldsymbol{z})$. The objective is

$$\mathcal{O}(z) = \mathcal{D}(z) + \mathcal{R}_{\text{w-dTNV}}(z) + \iota_{\mathbb{R}_+}(z), \tag{7.5}$$

where $\mathcal{D}_i$ is a Poisson negative log-likelihood term composed with the relevant projector, image-space registration operator, and optional resolution model; $\mathcal{R}_{\text{w-dTNV}}$ is the prior from Chapter **??**; and $\iota_{\mathbb{R}_+}$ is the indicator function of the non-negative orthant,

$$\iota_{\mathbb{R}_+}(z) = \begin{cases} 0, & z \geq 0, \\ +\infty, & \text{otherwise}, \end{cases} \tag{7.6}$$

with the inequality understood componentwise. As $\iota_{\mathbb{R}_+}$ is non-smooth, it is kept out of the gradient estimator $g^{(k)}$ below and instead handled by proximal (projected) gradient splitting: the smooth terms $\mathcal{D} + \mathcal{R}$ are advanced by a stochastic gradient step, and the result is passed through $\operatorname{prox}_{\varsigma^{(k)}\iota_{\mathbb{R}_+}}$, which for an indicator function reduces to Euclidean projection onto $\mathbb{R}_+$, i.e. clipping negative voxels to zero. This is exactly the $\{\cdot\}_{\geq 0}$ operation in the update Equation (**??**).

The stochastic experiments use SVRG [**johnsonAcceleratingStochasticGradient2013**, **twymanInvestigationStochasticVariance2022**, **ehrhardtFastPETReconstruction2025**]. Writing $\mathcal{R} = \mathcal{R}_{\text{w-dTNV}}$, the implemented estimator keeps the complete prior outside the sampler. At anchor point $\widetilde{z}$ it is

$$g^{(k)} = \nabla\mathcal{R}(z^{(k)}) + \nabla\mathcal{D}(\widetilde{z}) + N_{\text{d}}\left[\nabla\mathcal{D}_{i_k}(z^{(k)}) - \nabla\mathcal{D}_{i_k}(\widetilde{z})\right], \qquad i_k \sim \text{Unif}\{1, \dots, N_{\text{d}}\}. \tag{7.7}$$

The data functions are sampled uniformly with replacement. The complete prior gradient is evaluated at every stochastic update, and the full data-gradient snapshot is refreshed once per epoch.

The image update is a projected, preconditioned gradient step,

$$z^{(k+1)} = \left\{ z^{(k)} - \varsigma^{(k)}\mathcal{P}^{(k)}g^{(k)} \right\}_{\geq 0}, \tag{7.8}$$

where $\mathcal{P}^{(k)}$ is either a diagonal preconditioner or a voxel-wise 2×2 block preconditioner coupling the PET and SPECT update components. The step size follows the linear decay rule

$$\varsigma^{(k)} = \frac{\varsigma^{(0)}}{1+\eta_{\text{relax}}k}, \qquad \varsigma^{(0)} = 1, \qquad \eta_{\text{relax}} = 0.002, \tag{7.9}$$

for the 20-pass comparisons described below.

7.2.1 Common Evaluation Protocol

The convergence experiments used an anthropomorphic phantom scanned on the Mediso AnyScan at the National Physical Laboratory. PET and SPECT were each partitioned into 18 subsets. Each plotted comparison curve comprises five stochastic realisations and covers the first 20 complete data passes. A single 1000-epoch reconstruction of the same objective, run with more conservative optimisation settings, provided the reference image $x^{\star}$; it is a final long-run iterate rather than a proven global minimiser. Appendix **??** reports the objective and iterate diagnostics supporting its effective stationarity relative to the best observed iterate.

For each stochastic run and each saved iteration, the normalised root-mean-square error (NRMSE) in region R is

$$\mathrm{NRMSE}(x_k, x^{\star}; R) = \frac{\sqrt{N^{-1}\sum_{j\in R}(x_{k,j} - x_j^{\star})^2}}{\sqrt{N^{-1}\sum_{j\in R}(x_j^{\star})^2}}, \qquad N = |R|. \tag{7.10}$$

The regions are evaluated separately over the whole image, the hot-sphere mask, and the cold-sphere mask. The plotted centre line is the arithmetic mean of the five per-run NRMSE values at each iteration, not the NRMSE of an averaged reconstruction; the envelope is their pointwise minimum–maximum. The long-run reference itself was generated once.

7.3 Subset Selection

Subset methods accelerate early iterations in emission tomography by replacing full data passes with stochastic or ordered blocks of projection data [**hudsonAcceleratedImageReconstruction1994**]. Joint PET/SPECT reconstruction introduces a further choice: corresponding modality subsets can be added to form paired stochastic functions, or each modality subset can remain a separate function. The experiment below compares only these two organisations. In both cases the complete prior remains outside the sampler and is evaluated at every update, as in Equation (**??**). **TODO:** Add the Zenodo-managed citation for IEEE Xplore document 10654962.

7.3.1 Data-Subset Organisation

The one-bed phantom data were divided into 18 PET and 18 SPECT subsets. Two organisations were tested.

Separate subsets. PET and SPECT subsets are treated as distinct stochastic functions. For a one-bed acquisition with $N_{\mathrm{PET}} = 18$ and $N_{\mathrm{SPECT}} = 18$, this gives

$$\{\mathcal{D}_i\}_{i=1}^{36} = \{\mathcal{D}_{\mathrm{PET},1}, \ldots, \mathcal{D}_{\mathrm{PET},18}, \mathcal{D}_{\mathrm{SPECT},1}, \ldots, \mathcal{D}_{\mathrm{SPECT},18}\}. \tag{7.11}$$

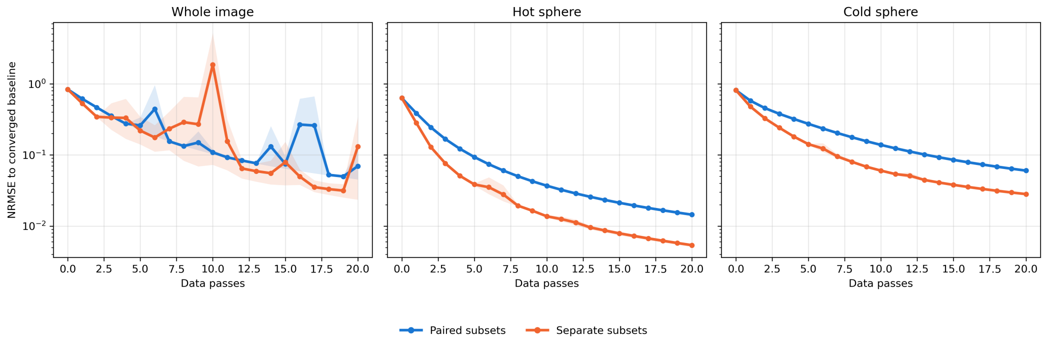
Paired subsets. PET and SPECT subsets with matching indices are summed into one stochastic function. For one bed this gives

$$\mathcal{D}_j^{\mathrm{pair}} = \mathcal{D}_{\mathrm{PET},j} + \mathcal{D}_{\mathrm{SPECT},j}, \qquad j = 1, \ldots, 18. \tag{7.12}$$

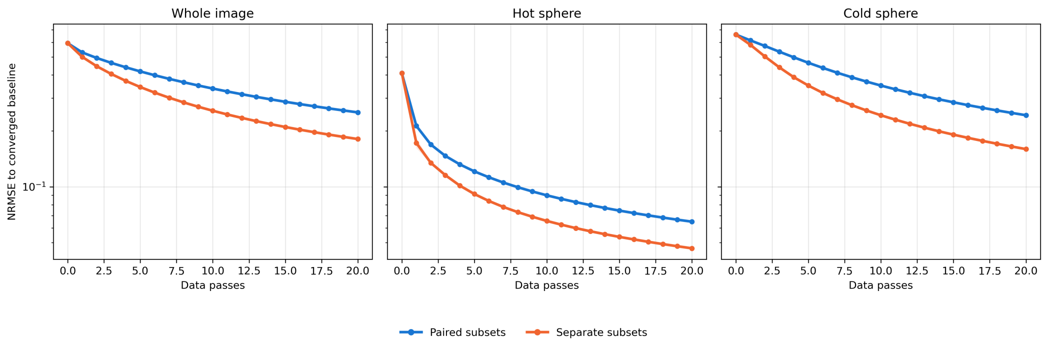
7.3.2 Experiment Design and Work Accounting

All data functions were sampled uniformly with replacement. To compare the organisations on a common horizontal axis, one matched data pass is defined by 36 modality-specific data-subset evaluations. Paired organisation reaches this count after 18 joint PET+SPECT updates: each update evaluates one PET and one SPECT subset, and therefore the 18 updates comprise 36 data-subset evaluations and 18 complete prior-gradient evaluations. Separate organisation reaches the same data-projection count after 36 modality-specific updates, comprising the same 36 data-subset evaluations but 36 complete prior-gradient evaluations.

This accounting matches data-projection work, but it does not match parameter-update count, prior-gradient work, or elapsed computation. Both organisations used the same objective and the MM local-diagonal curvature preconditioner, and were assessed with the common 20-pass protocol in Section **??**.



(a) PET.



(b) SPECT represented on the PET grid.

Figure 7.2: Convergence for paired and modality-separated subsets using the MM local-diagonal curvature preconditioner. Lines show the mean per-run NRMSE over five stochastic realisations and shading shows the pointwise minimum–maximum over the first 20 matched data passes. The complete prior gradient was evaluated at every update.

7.3.3 Results

Figure **??** shows faster convergence with modality-separated subsets for both PET and SPECT, with the clearest differences in the hot- and cold-sphere regions. A plausible explanation is that the separate organisation performs 36 parameter updates per matched data pass, compared with 18 paired updates, and therefore makes more frequent progress for the same number of data-subset projections. The experiment does not isolate modality grouping from update frequency: the separate configuration also evaluates the complete prior twice as often. It should therefore be interpreted as a data-projection-matched comparison of the two implemented organisations, not as a compute-matched speed benchmark or a causal test of modality splitting alone.

7.4 Prior-Aware Preconditioning

Preconditioning is critical for subset-based MAP reconstruction because the data term and the edge-preserving prior can have very different local curvature scales [**clinthornePreconditioningMethodsImproved1993**, **ahnGloballyConvergentOrdered2002**, **depierroFastEMlikeMethods2001**, **ehrhardtFastPETReconstruction2025**]. The comparison below includes data-only BSREM scaling and four prior-aware curvature models for w-dTNV.

7.4.1 Data Curvature

For modality m, let s_m denote the sensitivity image on the shared grid,

$$s_m = W_m A_m^\top \mathbf{1}, \tag{7.13}$$

where A_m is the native-grid system matrix and W_m maps native images to the shared prior grid. The data-only EM/BSREM preconditioner is

$$\mathcal{P}_{\mathrm{BSREM},m} = (z_m + \varepsilon_x) \oslash s_m, \tag{7.14}$$

with element-wise division. This scaling models the data term only; it contains no prior curvature. Equivalently, the corresponding diagonal data Hessian surrogate is

$$\widehat{H}_{\mathrm{data},m} = s_m \oslash (z_m + \varepsilon_x). \tag{7.15}$$

The implementation optionally smooths the numerator image used in the EM preconditioner with a Gaussian filter. In the experiments below the smoothing FWHM is 5 mm in each direction.

7.4.2 Prior Curvature Models

Four prior-curvature constructions were considered: three based on a local majorisation–minimisation (MM)/iteratively reweighted least squares (IRLS) surrogate and one based on the Lewis–Sendov Hessian of the smoothed spectral penalty. The term MM describes the derivation of the first family; none of the implemented preconditioners has been established as a majoriser of the complete objective, and no monotone-descent guarantee is claimed.

For the MM family, at voxel n let

$$\tilde{\mathcal{G}}_n = [\rho_{1,n}\Pi_{1,n}(Dz_1)_n \mid \rho_{2,n}\Pi_{2,n}(Dz_2)_n] \in \mathbb{R}^{d\times 2} \tag{7.16}$$

be the weighted directional Jacobian from Chapter **??** (Equation (**??**)). The active code forms the modality-space MM weight

$$C_n = \left(\tilde{\mathcal{G}}_n^\top \tilde{\mathcal{G}}_n + \varepsilon^2 \mathbb{I}_2\right)^{-1/2} = \begin{pmatrix} c_{11,n} & c_{12,n} \\ c_{12,n} & c_{22,n} \end{pmatrix}. \tag{7.17}$$

This matrix is the same 2×2 inverse square-root used in the gradient calculation in Chapter **??** (Equation (**??**)), now interpreted as the local quadratic-surrogate weight.

For diagonal preconditioning, the finite-difference stencil and directional projectors are collapsed to a per-voxel participation factor $\tau_{m,n}$. This factor includes the number of valid forward/backward stencil incidences at voxel n, the squared finite-difference scale factors, and the diagonal entries of $\Pi_{m,n}^\top \Pi_{m,n}$. The local diagonal w-dTNV curvature approximation is

$$\widehat{H}^{\text{loc-diag}}_{\mathcal{R},m,n} = V_{\text{vox}}\, \rho_{m,n}^2\, c_{mm,n}\, \tau_{m,n}. \tag{7.18}$$

This is the diagonal extraction of the local MM curvature surrogate. A more conservative Gershgorin-inflated diagonal surrogate adds the magnitude of the cross-modal term,

$$\widehat{H}^{\text{diag-G}}_{\mathcal{R},1,n} = V_{\text{vox}}\, \rho_{1,n}^2 \left(c_{11,n} + |c_{12,n}|\right) \tau_{1,n}, \qquad \widehat{H}^{\text{diag-G}}_{\mathcal{R},2,n} = V_{\text{vox}}\, \rho_{2,n}^2 \left(c_{22,n} + |c_{12,n}|\right) \tau_{2,n}. \tag{7.19}$$

For block preconditioning, the PET and SPECT update components at each voxel are coupled by a 2×2 block. Let $\mathcal{N}(n)$ be the set of target voxels whose finite-difference stencil depends on source voxel n. For target voxel $r \in \mathcal{N}(n)$, let $c_{r\leftarrow n} \in \mathbb{R}^d$ be the finite-difference impulse vector produced in the directional channels when source voxel n is perturbed. The local voxel-block curvature approximation uses

$$\alpha_{r\leftarrow n}^{mm'} = c_{r\leftarrow n}^{\top} \Pi_{m,r}^{\top} \Pi_{m',r} c_{r\leftarrow n}, \tag{7.20}$$

and

$$\widehat{H}_{\mathcal{R},n}^{\text{loc-block}} = V_{\text{vox}} \sum_{r\in\mathcal{N}(n)} [B_r C_r B_r] \circ A_{r\leftarrow n}, \qquad A_{r\leftarrow n} = \begin{pmatrix} \alpha_{r\leftarrow n}^{11} & \alpha_{r\leftarrow n}^{12} \\ \alpha_{r\leftarrow n}^{21} & \alpha_{r\leftarrow n}^{22} \end{pmatrix}, \tag{7.21}$$

where $B_r = \operatorname{diag}(\rho_{1,r}, \rho_{2,r})$ and $\circ$ denotes the Hadamard product.

The fourth construction uses the exact Lewis–Sendov Hessian of the smoothed spectral penalty [**lewisDerivativesSpectralFunctions1996**]. If $H_{\mathcal{R}} = \nabla^2 \mathcal{R}$ denotes the resulting image-space Hessian, the implemented approximation retains only its voxel-wise diagonal blocks,

$$\widehat{H}_{\mathcal{R}}^{\text{LS-block}} = \operatorname{blkdiag}\left([H_{\mathcal{R}}]_{11}, \ldots, [H_{\mathcal{R}}]_{NN}\right), \qquad [H_{\mathcal{R}}]_{nn} \in \mathbb{R}^{2\times 2}. \tag{7.22}$$

It therefore retains PET/SPECT cross-curvature within each voxel but discards spatial off-diagonal blocks. Evaluating the smoothed spectral Hessian requires the voxel-wise singular-value decompositions used by the Lewis–Sendov construction. Recomputing this block model once per epoch consequently has a non-negligible cost.

7.4.3 Combining Data and Prior Curvature

Let $\mathcal{P}_{\text{D}} = \widehat{H}_{\text{D}}^{-1}$ denote the data preconditioner and $\mathcal{P}_{\mathcal{R}} = \widehat{H}_{\mathcal{R}}^{-1}$ the preconditioner obtained from one of the prior-curvature models above. For block methods, $\widehat{H}_{\text{D}}$ is the 2×2 diagonal matrix containing the modality-specific data curvatures at each voxel. The principal comparison combines the operands through the inverse-sum, or parallel-sum, rule

$$\mathcal{P}_{\parallel} = \left(\mathcal{P}_{\text{D}}^{-1} + \mathcal{P}_{\mathcal{R}}^{-1}\right)^{-1} = \left(\widehat{H}_{\text{D}} + \widehat{H}_{\mathcal{R}}\right)^{-1}. \tag{7.23}$$

The same expression applies to diagonal and voxel-block operands, with numerical floors used before inversion. It is a combination of approximate curvature preconditioners, not a statement that either operand or their sum majorises the full objective.

The preconditioner is updated once per epoch and held fixed between updates. This limits curvature-computation cost, particularly for the Lewis–Sendov construction, but means the preconditioner becomes progressively stale within each epoch.

Table 7.1: Preconditioner models used in the principal comparison. The four prior-aware models use the inverse-sum combination in Equation (**??**).

Preconditioner	Prior-curvature construction	Combination
Data-only BSREM	None; EM data scaling only	Not applicable
MM local diagonal	Diagonal extraction of the local MM curvature surrogate	Inverse sum
MM Gershgorin-inflated diagonal	Local diagonal inflated by the magnitude of cross-modal coupling	Inverse sum
MM local voxel-block	Voxel-wise 2×2 local MM curvature surrogate	Inverse sum
Lewis–Sendov block-diagonal	Voxel-block diagonal of the exact smoothed spectral Hessian	Inverse sum

7.4.4 Preconditioner Comparison Methods

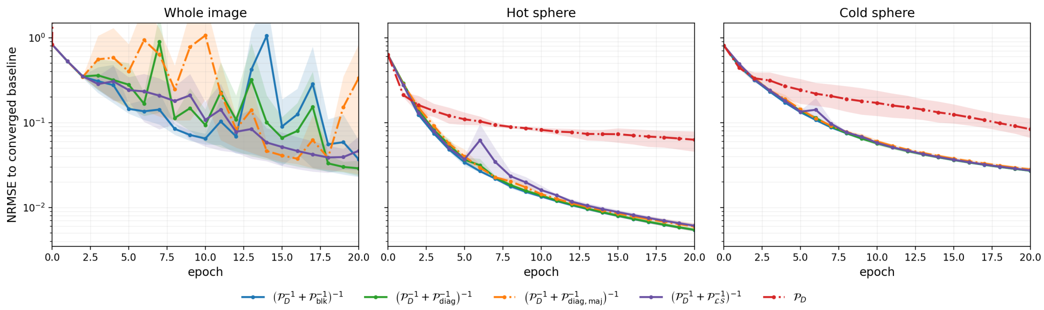
Five settings are compared: data-only BSREM, MM local diagonal, MM Gershgorin-inflated diagonal, MM local voxel-block, and Lewis–Sendov block-diagonal preconditioning. These settings are summarised in Table **??**. The four prior-aware settings use the inverse-sum combination.

The comparison uses the one-bed anthropomorphic phantom, modality-separated subsets, and the complete prior gradient at every update. All methods use the same objective, initialisation, long-run reference, initial step $\varsigma^{(0)} = 1$, and relaxation parameter $\eta_{\text{relax}} = 0.002$. Five stochastic realisations are evaluated for the first 20 matched data passes. Sensitivity-derived support masks are enabled with relative threshold 10^{-3}, and the data component is smoothed with a 5 mm Gaussian FWHM.

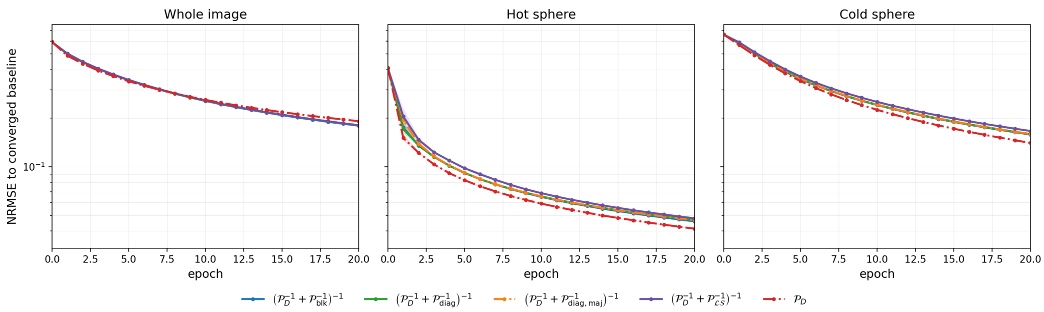
The reference is separate from this five-method comparison: it is a single 1000-

epoch run with the same objective, a Gershgorin-diagonal MM preconditioner, and more conservative optimisation settings, including $\eta_{\text{relax}} = 10^{-4}$. It provides $x^\star$ for the NRMSE calculation rather than an additional candidate method.

7.4.5 Preconditioner Comparison



(a) Count-limited PET.



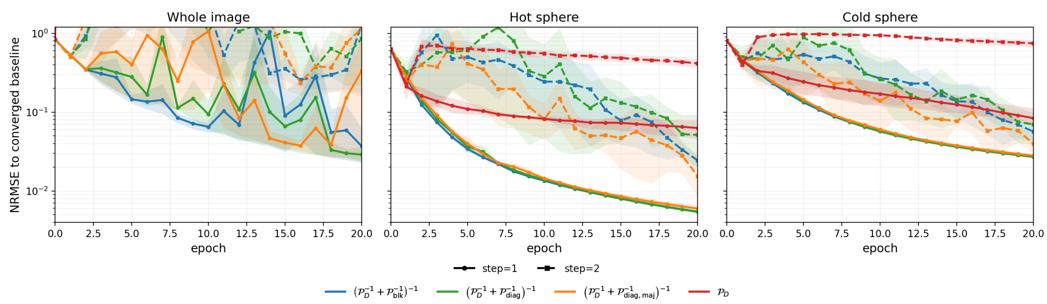
(b) SPECT represented on the PET grid.

Figure 7.3: Convergence of data-only BSREM and the four prior-aware preconditioners in Table **??**. All methods use the same objective and long-run reference within each modality. Lines show the mean per-run NRMSE over five stochastic realisations, and shading shows the pointwise minimum–maximum over the first 20 matched data passes.

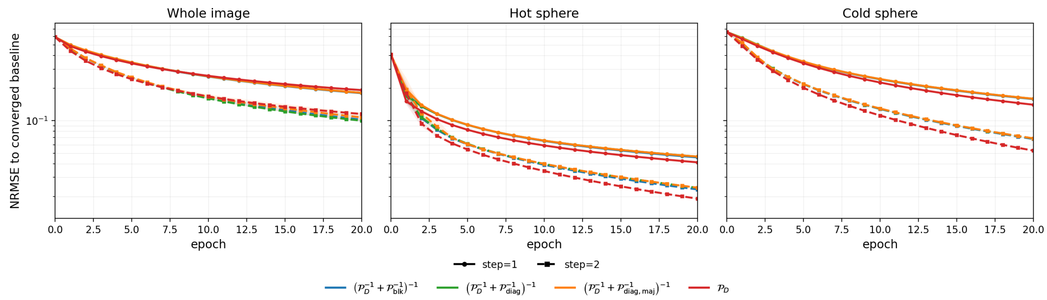
The four prior-aware constructions in Figure **??** behave broadly similarly. Their ordering varies between regions, and the differences are small compared with the distinction between prior-aware and data-only scaling; there is no universal winner. For count-limited PET, the whole-image data-only BSREM trajectory leaves the displayed range, indicating divergence under the tested schedule, whereas the prior-aware methods remain bounded and continue to reduce the hot- and cold-sphere errors. Prior curvature was therefore required for stable PET convergence under this common step-size schedule.

SPECT is more robust. All five methods remain stable, and data-only BSREM is competitive with, or slightly faster than, the prior-aware alternatives in the sphere regions. This contrast suggests that the tested prior-aware scalings can be conservative for SPECT even though they are needed to stabilise the limited-count PET case.

7.4.6 Sensitivity to the Initial Step Size



(a) Count-limited PET.



(b) SPECT represented on the PET grid.

Figure 7.4: Sensitivity to the common initial step size for the displayed preconditioners. Solid curves use $\varsigma^{(0)} = 1$ and dashed curves use $\varsigma^{(0)} = 2$. Lines and envelopes again show the five-run mean and pointwise minimum–maximum over 20 matched data passes.

Doubling the common initial step improves SPECT convergence across the three regions in Figure **??**, but destabilises the limited-count PET reconstruction. A single shared relaxation schedule is therefore not equally well calibrated to both modalities. Modality-dependent initial steps or relaxation rates are a natural direction for future work, but were not tested here.

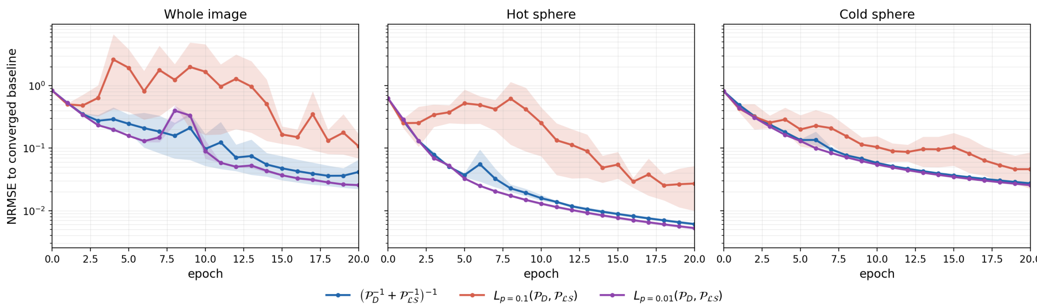
7.4.7 Lehmer Combination

The inverse sum is one member of a broader matrix Lehmer family. For the voxel-wise symmetric positive-definite preconditioners $\mathcal{P}_{\mathcal{R}}$ and $\mathcal{P}_{\mathrm{D}}$, define

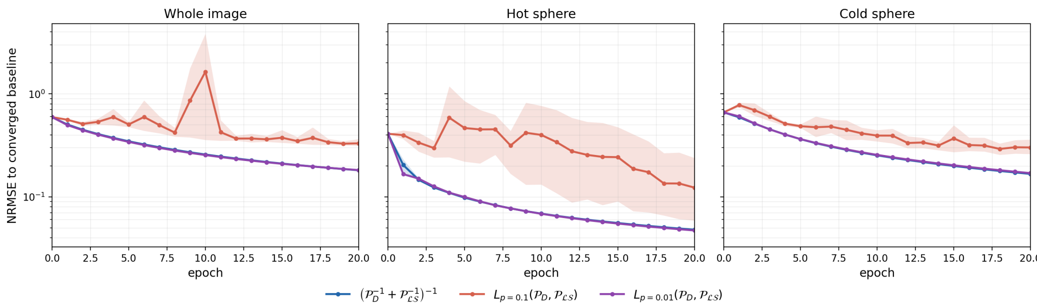
$$C = \mathcal{P}_{\mathrm{D}}^{-1/2}\mathcal{P}_{\mathcal{R}}\mathcal{P}_{\mathrm{D}}^{-1/2}, \qquad L_p(\mathcal{P}_{\mathcal{R}},\mathcal{P}_{\mathrm{D}}) = \mathcal{P}_{\mathrm{D}}^{1/2} f_p(C)\mathcal{P}_{\mathrm{D}}^{1/2}, \qquad f_p(t) = \frac{t^p+1}{t^{p-1}+1}, \tag{7.24}$$

where $f_p(C)$ is evaluated spectrally. The implemented preconditioner is $\frac{1}{2}L_p(\mathcal{P}_{\mathcal{R}},\mathcal{P}_{\mathrm{D}})$. At $p=0$ this reduces exactly to the parallel sum,

$$\tfrac{1}{2}L_0(\mathcal{P}_{\mathcal{R}},\mathcal{P}_{\mathrm{D}}) = \left(\mathcal{P}_{\mathrm{D}}^{-1}+\mathcal{P}_{\mathcal{R}}^{-1}\right)^{-1}. \tag{7.25}$$



(a) Count-limited PET.



(b) SPECT represented on the PET grid.

Figure 7.5: Parallel-sum and Lehmer combinations for the Lewis–Sendov block-diagonal prior model. This experiment changes only the combination rule; lines and shading show the five-run mean and pointwise minimum–maximum over 20 matched data passes.

For the Lewis–Sendov block model, $p=0.01$ is marginally better than the inverse-sum result in these curves, but the difference is small. This comparison does not establish that the same order is preferable for the MM-derived preconditioners. Chapter **??** had already frozen the Lewis–Sendov block-diagonal model with $p = 0.01$ before the broader benchmark in this chapter. Retaining it therefore provides protocol continuity, not a post-hoc claim that Lewis–Sendov or $p = 0.01$ is universally optimal.

7.5 Conclusion

Joint reconstruction of the PET bed positions is required for the edge-preserving prior to act consistently across their overlap; the qualitative example showed the noise and discontinuities that can remain after separate reconstruction and fusion. For the one-bed convergence study, modality-separated subsets converged faster than paired subsets under matched data-projection work, although they also performed twice as many parameter updates and prior-gradient evaluations per pass.

Prior-aware preconditioning was required for stable convergence of the count-limited PET reconstruction under the tested shared step schedule. SPECT was more robust, and data-only BSREM was sometimes competitive, suggesting that the prior-aware methods can be conservative for that modality. The four prior-curvature models behaved similarly overall, with no universal winner. Increasing the common initial step improved SPECT but destabilised PET, motivating modality-dependent relaxation as future work.

The Lewis–Sendov Lehmer experiment found only a small advantage for $p = 0.01$ over the parallel sum. Chapter **??** retains the previously frozen Lewis–Sendov block-diagonal/$p = 0.01$ configuration for protocol continuity. Its use in that chapter should not be interpreted as selection of the best method after the wider comparison reported here.

Chapter 8

Weighted Directional Total Nuclear Variation for Joint ^{90}Y PET/SPECT Reconstruction with CTAC-derived Guidance

This chapter applies the weighted directional total nuclear variation (w-dTNV) prior developed in Chapter **??** to the clinically relevant problem of post-treatment ^{90}Y activity imaging after SIRT. We consider joint reconstruction of low-count ^{90}Y PET and bremsstrahlung SPECT, using computed tomography attenuation correction (CTAC) information directionally (via the CTAC-derived attenuation map μ) to stabilise edge geometry without enforcing intensity correlation.

Relative to Chapter **??**, the focus here is validation and comparison with other methods: we specify modality-dependent forward models, co-registration operators, practical normalisation for heterogeneous PET/SPECT scales, and a reproducible evaluation protocol on a National Electrical Manufacturers Association (NEMA) International Electrotechnical Commission (IEC) phantom with PET bootstrapping and on a post-SIRT patient cohort.

8.1 Motivation and clinical context

Quantitative post-treatment imaging is required for personalised dosimetry following ^{90}Y SIRT [**levillainInternationalRecommendationsPersonalised2021**]. Pre-treatment planning based on ^{99m}Tc-macroaggregated albumin (MAA) often mispredicts microsphere deposition, motivating post-treatment verification of delivered activity distributions [**hasteCorrelationTechnetium99mMacroaggregated2017**].

Two ^{90}Y modalities are clinically available:

- **^{90}Y PET**: detects the rare internal pair-production branch, providing comparatively high spatial resolution but extremely low count statistics [**selwynNewInternalPair2007**].
- **Bremsstrahlung SPECT**: has substantially higher corrected counts (about 15$\times$ for the data studied here) but suffers from severe blur, septal penetration, and scatter due to the continuous emission spectrum [**rongDevelopmentEvaluationImproved2012**, **minarikEvaluationQuantitative90Y2008**].

Both modalities are typically acquired alongside CT for attenuation correction and anatomical localisation. Since PET and SPECT image the same underlying microsphere distribution while CTAC provides structural rather than activity information, independent reconstruction discards both a strong physical coupling and useful anatomical context.

The sequential hybrid kernelised expectation maximisation (SHKEM) method of Deidda *et al.* improves lesion contrast by reconstructing one modality to guide another [**deiddaTripleModalityImage2023**, **deiddaHybridKernelisedExpectation2022**]. However, the sequential construction remains sensitive to iteration choice and noise amplification. This motivates a truly joint MAP formulation with a coupled regulariser that promotes shared geometry while allowing modality-specific contrast.

8.2 Joint MAP formulation for PET/SPECT

We use the joint MAP framework introduced in Section **??**. For the present application we set $M = 2$ modalities, with $m = 1$ denoting PET and $m = 2$ denoting SPECT.

Let $z_m \in \mathbb{R}_+^N$ denote the image for modality m represented on a shared grid and let $y_m \in \mathbb{N}^{I_m}$ denote the corresponding projection data.

The native image grid for modality m may differ from the shared grid. Once registration is fixed, let $W_m : \mathbb{R}_+^{J_m} \to \mathbb{R}_+^N$ denote the discrete resampling operator from the native grid to the shared grid, and let $W_m^* : \mathbb{R}_+^N \to \mathbb{R}_+^{J_m}$ denote its discrete adjoint with respect to the chosen image-space inner products. The adjoint maps shared-grid variables back to the native grid before application of the system matrix. For each modality we therefore use the affine Poisson model

$$\bar{y}_m(z_m) = A_m W_m^* z_m + b_m, \tag{8.1}$$

where A_m is the system matrix and b_m collects additive corrections. For example, randoms and scatter for PET and simulated scatter/penetration terms for SPECT (see Sections **??** and **??**).

The joint objective is

$$\hat{\mathbf{z}} = \arg\min_{z \geq 0} \left\{ \sum_{m=1}^{2} \mathcal{D}_m(y_m \mid A_m W_m^* z_m + b_m) + \mathcal{R}(\mathbf{z}) \right\}, \tag{8.2}$$

where $\mathcal{D}_m$ is the Poisson negative log-likelihood (up to an additive constant),

$$\mathcal{D}_m(y_m \mid \bar{y}_m) = \sum_{i=1}^{I_m} \left((\bar{y}_m)_i - (y_m)_i \log(\bar{y}_m)_i \right), \tag{8.3}$$

and $\mathcal{R}$ is the coupled prior. In this chapter, $\mathcal{R}$ is the smoothed w-dTNV prior $\tilde{\mathcal{R}}_{\text{w-dTNV}}$ defined in Section **??** (Chapter **??**), specialised to PET/SPECT with CTAC guidance and practical global modality normalisation. The overall regularisation strength is carried by the modality weights introduced below, so no additional scalar λ is used in the implementation.

8.3 w-dTNV instantiated for ^{90}Y PET/SPECT with CTAC guidance

8.3.1 Directional guidance from the CTAC-derived μ-map

As in Section **??**, we use CT information directionally rather than appending CT as a third coupled channel. Here the guidance image is the CTAC-derived attenuation map μ (resampled to the PET grid). We pre-smooth μ with a small blurring kernel (FWHM $= 2\,$mm) to reduce high-frequency CT noise while retaining high guidance edge resolution. We then compute a stabilised direction field in the discrete gradient space,

$$\xi_n = \frac{(D\mu)_n}{\sqrt{\|(D\mu)_n\|_2^2 + \eta^2}}, \qquad \eta = 0.05 \cdot \max_n \|(D\mu)_n\|_2, \tag{8.4}$$

where D is the chosen directional finite-difference operator (specified in Section **??**). The stabiliser η prevents amplification of small-magnitude μ differences in homogeneous regions.

We can then form the modality-wise directional operators $\Pi_{m,n} \in \mathbb{R}^{d\times d}$ using the construction already adopted in Chapter **??**:

$$\Pi_{m,n} = \mathbb{I}_d - \gamma_m \, \xi_n \xi_n^\top, \qquad \gamma_m \in [0,1]. \tag{8.5}$$

Unless stated otherwise, we set $\gamma_1 = \gamma_2 = 1$. Since ξ_n is stabilised and generally has norm less than one, this should be interpreted as anisotropic shrinkage of gradient components parallel to the local μ-edge normal direction rather than as a hard projection.

8.3.2 Global modality normalisation as a constant diagonal weighting

Chapter **??** introduced a general voxel-wise diagonal modality weighting $B_n = \mathrm{diag}(\rho_{1,n}, \dots, \rho_{M,n})$. In the present study we use a spatially uniform normalisation to mitigate the dominant scale disparity between ^{90}Y PET and bremsstrahlung SPECT:

$$B_n \equiv B := \mathrm{diag}(\rho_1, \rho_2), \qquad \rho_1, \rho_2 > 0. \tag{8.6}$$

Weights are computed from early-stopped unregularised reconstructions represented on the shared grid, z_m^{init}, via robust dynamic-range (percentile) scaling over a CTAC-derived foreground mask Ω':

$$\rho_m = \frac{\beta_m}{q_{0.99}\big(z_m^{\text{init}}|_{\Omega'}\big) - q_{0.01}\big(z_m^{\text{init}}|_{\Omega'}\big)}, \qquad m \in \{1,2\}, \tag{8.7}$$

where $q_p(\cdot|_{\Omega'})$ denotes the pth percentile over voxels in Ω', and β_m are user-controlled regularisation weights used for operating-point selection (Section **??**). Percentile scaling provides a robust dynamic-range estimate and reduces sensitivity to outliers. This matches the intent of Section **??** while avoiding an additional spatially varying weight for this investigation [**tsaiBenefitsUsingSpatiallyVariant2020**].

8.3.3 Directional finite-difference operator for PET/SPECT coupling

We use the generic finite-difference definition from Chapter **??** (Section **??**) but specify the set of offsets to better accommodate SPECT blur and modest registration error. Let $\Delta = (\Delta_x, \Delta_y, \Delta_z)$ denote the voxel sizes in SI units. We define a nine-direction half-stencil

$$\mathcal{E}_1 = \{(1,0,0),(0,1,0),(0,0,1)\} \cup \{(\pm 1,1,0),(\pm 1,0,1),(0,1,\pm 1)\}, \tag{8.8}$$

and use both one- and two-voxel steps, $\mathcal{E} = \mathcal{E}_1 \cup \{2e : e \in \mathcal{E}_1\}$, yielding $d = 18$ directions. For a set of offsets $\mathcal{E} = \{e_\ell\}_{\ell=1}^d$, define the d-directional forward difference operator D by

$$\big((Dz)_n\big)^{(\ell)} := \frac{z_{n+e_\ell} - z_n}{\|e_\ell\|_\Delta}, \qquad \|e\|_\Delta := \sqrt{(e_x\Delta_x)^2 + (e_y\Delta_y)^2 + (e_z\Delta_z)^2}, \tag{8.9}$$

with Neumann boundary conditions. This follows the "extended Jacobian" motivation for TNV-type coupling [**holtTotalNuclearVariation2014**] and empirically improved stability in the present PET/SPECT setting.

8.3.4 The *w*-*d*TNV Jacobian and regulariser (two modalities)

Using the notation of Chapter **??**, we form the guided gradients $\Pi_{m,n}(Dz_m)_n \in \mathbb{R}^d$ and define the weighted directional Jacobian

$$\tilde{\mathcal{G}}_n := \tilde{\mathcal{G}}_n^{\text{w-dTNV}}(\boldsymbol{z}) = \left[\Pi_{1,n}(Dz_1)_n \,\middle|\, \Pi_{2,n}(Dz_2)_n\right] B \in \mathbb{R}^{d\times 2}. \tag{8.10}$$

The regulariser is the smoothed spectral potential introduced in Chapter **??**, weighted by the voxel volume V_{vox},

$$\tilde{\mathcal{R}}_{\text{w-dTNV}}(\boldsymbol{z}) = V_{\text{vox}} \sum_{n=1}^{N} \phi(\tilde{\mathcal{G}}_n), \qquad \phi(Y) = \sum_{\ell=1}^{2} g(\sigma_\ell(Y)), \tag{8.11}$$

where g is the Charbonnier smoothing (Equation **??**) and $\sigma_\ell(Y)$ are singular values. We set the smoothing parameter ε relative to initial joint-gradient magnitudes:

$$\varepsilon = 0.05 \cdot \text{median}_{n\in\Omega'}\, \sigma_1\left(\tilde{\mathcal{G}}_n^{\text{init}}\right), \qquad \tilde{\mathcal{G}}_n^{\text{init}} := \tilde{\mathcal{G}}_n^{\text{w-dTNV}}(\boldsymbol{z}^{\text{init}}), \tag{8.12}$$

with $\boldsymbol{z}^{\text{init}} = (W_1 x_1^{\text{init}}, W_2 x_2^{\text{init}})$.

If $\Pi_{2,n}(Dz_2)_n \approx 0$ at some voxel (e.g. SPECT locally uniform after guidance), then $\tilde{\mathcal{G}}_n$ has an effectively single nonzero column and $\phi(\tilde{\mathcal{G}}_n)$ reduces to a TV-type penalty on the remaining modality column. This is the same "no forced edges" behaviour discussed in Chapter **??**.

8.3.5 Ablation comparators

To separate the contributions of PET/SPECT coupling and CTAC-derived directional guidance, we compare *w*-dTNVwith two variational ablations. The first is weighted TNV (*w*-TNV), which keeps the two-modality Jacobian and the same modality normalisation but removes directional CTAC guidance by setting $\Pi_{m,n} = \mathbb{I}_d$ for both modalities:

$$\tilde{\mathcal{G}}_n^{\text{w-TNV}}(\boldsymbol{z}) = \Big[(Dz_1)_n \,\Big|\, (Dz_2)_n\Big] B. \tag{8.13}$$

This tests PET/SPECT coupling without anatomical anisotropy.

The second is directional TV (dTV), which keeps CTAC-derived directional guidance but removes PET/SPECT coupling. It is obtained by applying the same directional operator only to the PET gradient,

$$\tilde{\mathcal{R}}_{\text{dTV}}(z_1) = V_{\text{vox}} \sum_{n=1}^{N} \phi\left(\rho_1 \Pi_{1,n} (Dz_1)_n\right), \tag{8.14}$$

which reduces to a smoothed single-channel directional-TV penalty. This tests anatomical anisotropy without using SPECT as a coupled functional channel. All variational methods use the same forward models, registration, foreground mask, dynamic-range normalisation, and optimisation framework; only the regulariser structure is changed.

8.4 Optimisation and implementation

We minimise the joint objective (**??**) using a subset stochastic variance-reduced gradient (SVRG) method [**johnsonAcceleratingStochasticGradient2013**], consistent with the stochastic framework studied in Chapter **??**. Let N_{d} be the total number of PET and SPECT data functions in the reconstruction being described, including separate functions for each PET bed position where applicable. The application implementation distributes the prior equally across these functions,

$$\mathcal{O}(\boldsymbol{z}) = \sum_{i=1}^{N_{\text{d}}} \mathcal{O}_i(\boldsymbol{z}), \qquad \mathcal{O}_i(\boldsymbol{z}) = \mathcal{D}_i(\boldsymbol{z}) + \frac{1}{N_{\text{d}}} \tilde{\mathcal{R}}_{\text{w-dTNV}}(\boldsymbol{z}), \tag{8.15}$$

so that summing all stochastic functions recovers one complete prior term. This differs organisationally from the Chapter **??** experiments, in which the complete prior was held outside the data-function sampler. Here every stochastic function includes the scaled prior contribution; the prior is not sampled as a separate subset. Data functions are formed using staggered, view-interleaved partitioning and are split by modality, so each $\mathcal{D}_i$ contains either PET or SPECT data.

At iteration k, with anchor point $\tilde{\boldsymbol{z}}$ refreshed once per epoch, we use the SVRG estimator

$$g^{(k)} = \nabla\mathcal{O}(\tilde{\boldsymbol{z}}) + N_{\mathrm{d}}\left[\nabla\mathcal{O}_{i_k}(\boldsymbol{z}^{(k)}) - \nabla\mathcal{O}_{i_k}(\tilde{\boldsymbol{z}})\right], \qquad i_k \sim \mathrm{Unif}\{1,\ldots,N_{\mathrm{d}}\}, \tag{8.16}$$

and apply a projected voxel-block-preconditioned update

$$\boldsymbol{z}^{(k+1)} = \left\{\boldsymbol{z}^{(k)} - \varsigma^{(k)}\mathcal{P}^{(k)}g^{(k)}\right\}_{\geq 0}, \tag{8.17}$$

with a decaying step size

$$\varsigma^{(k)} = \frac{\varsigma^{(0)}}{1+\delta k}, \qquad \varsigma^{(0)} = 1, \qquad \delta = 0.005. \tag{8.18}$$

Reconstructions were run for 250 epochs, corresponding to approximately 500 full data passes because the full-gradient snapshot is recomputed once per epoch. This fixed stopping point was used for all methods rather than a case-specific convergence criterion.

8.4.1 Block-diagonal preconditioning

For modality m, the data component is the EM-type scaling [**depierroFastEMlikeMethods2001**, **ahnGloballyConvergentOrdered2002**]

$$\mathcal{P}_{\mathrm{data},m} = z_m \oslash \left(W_m A_m^\top \mathbf{1}\right), \tag{8.19}$$

represented on the shared grid. At voxel n, these modality-specific values form the diagonal block

$$\mathcal{P}_{\mathrm{D},n} = \begin{pmatrix} \mathcal{P}_{\mathrm{data},1,n} & 0 \\ 0 & \mathcal{P}_{\mathrm{data},2,n} \end{pmatrix}. \tag{8.20}$$

The prior component $\mathcal{P}_{\mathcal{R},n}$ is obtained by inverting the Lewis–Sendov voxel-block diagonal of the exact smoothed spectral Hessian described in Section **??**. It retains the PET/SPECT cross-curvature within each 2×2 voxel block while discarding spatial off-diagonal blocks. The prior blocks are recomputed once per epoch using the smoothed spectral Hessian and its voxel-wise singular-value decompositions, then held fixed within that epoch.

After applying an eigenvalue floor $\lambda_{\min} = 10^{-6}$ to both operands, the application combines them using the scaled matrix Lehmer construction from Equation (**??**). Specifically,

$$C_n = \mathcal{P}_{\mathrm{D},n}^{-1/2} \mathcal{P}_{\mathcal{R},n} \mathcal{P}_{\mathrm{D},n}^{-1/2}, \qquad \mathcal{P}_n = \frac{1}{2} \mathcal{P}_{\mathrm{D},n}^{1/2} f_p(C_n) \mathcal{P}_{\mathrm{D},n}^{1/2}, \qquad f_p(t) = \frac{t^p + 1}{t^{p-1} + 1}, \tag{8.21}$$

with $p = 0.01$. At $p = 0$ this expression is the parallel sum. The Lewis–Sendov/$p = 0.01$ setting was frozen for this application before the broader optimisation benchmark in Chapter **??**; its use here reflects protocol continuity rather than selection as the universally best preconditioner.

8.4.2 Comparator: sequential hybrid kernel EM

We compare against the sequential hybrid kernel expectation–maximisation (SHKEM) method of Deidda *et al.* [**deiddaTripleModalityImage2023**], using the reconstruction order and hyperparameter settings reported as optimal in that work. The sequential order is CT-guided SPECT followed by CT/SPECT-guided PET: SPECT is reconstructed first using CT-derived anatomical features and hybrid kernel updates, and the resulting SPECT reconstruction is then used together with CT-derived features and the current PET iterate to construct the PET hybrid kernel [**deiddaHybridKernelisedExpectation2022**]. To reduce noise amplification we use frozen kernels, halting kernel updates after 27 PET subiterations and 72 SPECT subiterations, matching the comparator configuration in [**deiddaTripleModalityImage2023**]. All kernel parameters followed the original implementation from Deidda *et al.* For clinical PET data, the SHKEM implementation is applied in the same joint multi-bed setting as the variational methods.

8.4.3 Software implementation

All reconstructions were implemented in the Synergistic Image Reconstruction Framework (SIRF) [**ovtchinnikovSIRFSynergisticImage2017**] using Software for Tomographic Image Reconstruction (STIR) projector backends [**thielemansSTIRSoftwareTomographic2012**, **schrammPARALLELPROJOpensourceFramewor** **fusterIntegrationAdvanced3D2013**] and the Core Imaging Library [**jorgensenCoreImagingLibrary** with custom code for the proposed regulariser. The SHKEM implementation follows [**deiddaTripleModalityImage2023**], extended here to the same joint multi-bed PET reconstruction setting used by the variational methods.

8.5 Forward models, corrections, and co-registration

8.5.1 PET forward model

PET list-mode data were processed using the vendor toolchains to generate normalisation, attenuation, and additive corrections (randoms and scatter), which were held fixed during reconstruction [**stearnsRandomCoincidenceEstimation2003**, **iatrou3DImplementationScatter2006**]. Reconstructions use a voxel size of $3.27 \times 2.13 \times 2.13$ mm^3 (z,y,x) on a grid of $83 \times 255 \times 255$.

Patient PET data were acquired across two partially overlapping bed positions. For all methods, these bed positions were reconstructed jointly in a single combined-bed image space, with separate forward- and back-projections for each bed position; for the variational methods, the image-domain prior is evaluated over the combined volume. This is the multi-bed formulation derived in Chapter **??**. It permits one edge-preserving prior to act over the combined field of view and was motivated by the qualitative overlap comparison in Figure **??**. Bed positions overlapped by 11 voxels. Phantom PET data were reconstructed using a single bed position.

Projection data are partitioned into staggered, view-interleaved subsets. For SHKEM we use $S = 9$ PET subsets, following the comparator study. For SVRG-based reconstructions we use 18 PET subsets for phantom data and 18 subsets per PET bed position for clinical data, giving 36 PET subsets in total for two-bed patient reconstructions.

8.5.2 SPECT forward model

For bremsstrahlung SPECT we use a geometric projector with an analytic collimator–detector response and a simulated additive correction term. SPECT images are discretised on a $128 \times 128 \times 128$ grid with isotropic voxel size 4.18 mm.

The collimator response is modelled as a depth-dependent Gaussian with linearly varying width, obtained by fitting an analytic collimator model [**angerSensitivityResolutionLinearity1966**]. Bremsstrahlung blur is modelled as an isotropic Gaussian with energy-dependent width fitted to ^{90}Y range simulations [**raultFastSimulationYttrium902010**]; where multiple energy windows are used, effective blur parameters are obtained by count-weighted averaging across windows.

Scatter and septal penetration are incorporated via an additive term estimated by an outer reconstruct–simulate loop using SIMIND [**ljungbergSIMINDMonteCarlo2015**, **rongDevelopmentEvaluationImproved2012**]. We initialise this additive term using a learned estimate from a neural network [**xiangDeepNeuralNetwork2020**]. The additive estimate is then refined by alternating reconstruction and SIMIND simulation for 10 outer iterations, using a damped update

$$b^{(k+1)} = \nu \hat{b}^{(k+1)} + (1-\nu) b^{(k)}, \qquad \nu = 0.2, \tag{8.22}$$

where $\hat{b}^{(k+1)}$ is the newly simulated additive estimate and $b^{(k)}$ is the previous estimate. The final additive estimate is fixed during the stochastic optimisation.

SPECT projection data are partitioned into staggered subsets, with $S = 12$ for both SVRG-based reconstructions and SHKEM.

8.5.3 Co-registration and warp operators

CTAC provides the anatomical reference for co-registration and for guidance via the μ-map. The shared reconstruction grid is the PET grid. The PET CTAC is therefore downsampled to the PET reconstruction grid and used both to define the attenuation map μ and as the anatomical reference space for the coupled prior, so W_1 is the identity on the shared grid.

The PET and SPECT CTAC volumes were retained in their native acquisition grids before registration. Since patient arm positioning differed between modalities (arms down in PET, arms up in SPECT), registration was restricted to the trunk. A trunk mask was obtained automatically from each CTAC using TotalSegmentator [**TODO˙totalsegmentator˙2023**]**TODO:** Zenodo-managed bibliography entry pending for TotalSegmentator; replace `TODO_totalsegmentator_2023` once imported., and the native-grid SPECT CTAC was registered to the PET CTAC using a rigid transformation followed by deformable registration initialised from the rigid solution. The resulting native-SPECT-to-PET deformation defines W_2. The corresponding operator W_2^* in the data-fidelity term is the discrete adjoint of this resampling operator, not merely a matrix transpose; linear interpolation was used so that the forward and adjoint warp pair were implemented consistently.

8.6 Data and evaluation protocol

All quantitative image analysis is performed on PET reconstructions to preserve direct comparability with SHKEM, where PET is the target modality and SPECT provides guidance. SPECT reconstructions are shown for completeness but are not included in the quantitative endpoints. Reconstruction hyperparameters are tuned once using the clinical training patient (SIRT3) and then fixed for all phantom and clinical reconstructions. This single-patient tuning strategy avoids repeated adjustment on the evaluation cohort, but it does not constitute a robustness study of the selected weights. Throughout, dTV, w-TNV, and SHKEM are compared against the reference method w-dTNV.

8.6.1 Phantom data

A NEMA IEC Body Phantom with a sphere-to-background ratio of 8 was scanned at The Christie Hospital using a Siemens Biograph mCT PET/CT system and a GE Discovery 670 SPECT/CT system. PET was acquired over 15 h and SPECT over 15 min. From the 15 h PET acquisition, $N_{\text{bs}} = 20$ statistically independent PET datasets equivalent to 15 min were generated by sinogram bootstrapping [**markiewiczAssessmentBootstrapResampling2014**]. The SPECT energy window was 39.84–128.16 keV.

We quantify PET reproducibility across bootstraps via voxel-wise coefficient of variation (CoV), and PET quantification via recovery coefficients (RCs) for spheres of diameters 10 mm, 13 mm, 22 mm, 28 mm, and 37 mm. Three VOIs were defined

(Figure **??**): a high-count (HC) VOI comprising five filled spheres (one sphere was excluded due to filling issues), a warm-background (BG) VOI comprising the phantom volume excluding the spheres plus a 10 mm margin and excluding the lung insert plus a 5 mm margin, and the cold lung insert (LI).

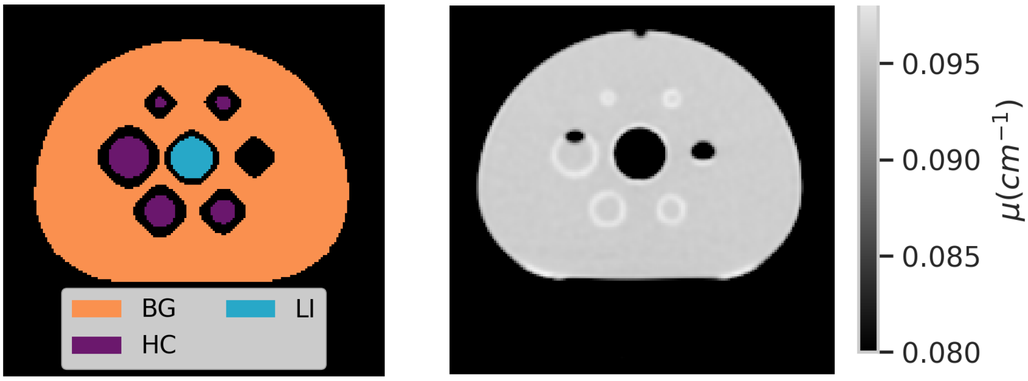


Figure 8.1: NEMA phantom analysis regions. Left: warm-background (BG), lung insert (LI), and high-count (HC) VOIs. Right: CTAC-derived attenuation map showing sphere edges, the lung insert, and the under-filled sphere excluded from the HC VOI.

Let $x^{(c)}_{\mathrm{bs},j}$ be the reconstructed PET intensity at voxel j for method c in bootstrap realisation $\mathrm{bs} \in \{1,\ldots,N_{\mathrm{bs}}\}$. Define the inter-bootstrap mean and standard deviation

$$\bar{x}^{(c)}_j := \frac{1}{N_{\mathrm{bs}}} \sum_{\mathrm{bs}=1}^{N_{\mathrm{bs}}} x^{(c)}_{\mathrm{bs},j}, \qquad \mathrm{SD}^{(c)}_j := \sqrt{\frac{\sum_{\mathrm{bs}=1}^{N_{\mathrm{bs}}} \left(x^{(c)}_{\mathrm{bs},j} - \bar{x}^{(c)}_j\right)^2}{N_{\mathrm{bs}}-1}}, \tag{8.23}$$

and voxel-wise

$$\mathrm{CoV}^{(c)}_j = \frac{\mathrm{SD}^{(c)}_j}{\bar{x}^{(c)}_j}. \tag{8.24}$$

For a VOI $\mathcal{V}$ we report the VOI-averaged statistic

$$T^{(c)}_{\mathcal{V}} = \frac{1}{|\mathcal{V}|} \sum_{j\in\mathcal{V}} \mathrm{CoV}^{(c)}_j. \tag{8.25}$$

For sphere v, let $\bar{C}^{(c)}_{\mathrm{bs},v}$ denote the mean reconstructed intensity within the sphere and $\bar{C}^{(c)}_{\mathrm{bs,bkg}}$ the mean within BG for the same (c, bs). With true ratio $R_{\mathrm{true}} = 8$, define

$$\mathrm{RC}^{(c)}_{\mathrm{bs},v} = \frac{\bar{C}^{(c)}_{\mathrm{bs},v}}{\bar{C}^{(c)}_{\mathrm{bs,bkg}} R_{\mathrm{true}}}. \tag{8.26}$$

To assess statistical significance of inter-bootstrap CoV, we compare each method c with the reference $c_0 = w$-dTNV using a paired row-swap permutation test on $\log(T^{(c)}_{\mathcal{V}})$. Let $X^{(c)}_{\mathcal{V}} \in \mathbb{R}^{N_{\mathrm{bs}} \times |\mathcal{V}|}$ be the bootstrap-by-voxel stack restricted to VOI $\mathcal{V}$, and define the observed difference $d_{\mathrm{obs}} = \log(T^{(c_0)}_{\mathcal{V}}) - \log(T^{(c)}_{\mathcal{V}})$. Under the null of exchangeability, each bootstrap index is independently swapped between the paired stacks with probability $1/2$ to form permuted differences d_{perm}. We use $N_{\mathrm{perm}} = 5000$ permutations and compute the two-sided p-value

$$p = \frac{1 + \sum_{i=1}^{N_{\mathrm{perm}}} \mathbf{1}\big(|d_{\mathrm{perm},i}| \geq |d_{\mathrm{obs}}|\big)}{N_{\mathrm{perm}} + 1}, \tag{8.27}$$

with Benjamini–Hochberg (BH) adjustment across methods within each VOI. No thresholding was applied to exclude near-zero voxels, so CoV estimates in the cold lung insert should be interpreted as a noise-realisation stability measure rather than a conventional percentage variability around a substantial mean signal.

Inference for RC uses a linear mixed model on

$$y_{\mathrm{bs},c,v} := \log\Big(\mathrm{RC}^{(c)}_{\mathrm{bs},v}\Big), \qquad \mathrm{bs} = 1, \ldots, N_{\mathrm{bs}}, \tag{8.28}$$

with a random intercept for bootstrap replicate and fixed effects for method, sphere, and their interaction:

$$\begin{gathered} y_{\mathrm{bs},c,v} = \beta_0 + \alpha_c + \gamma_v + (\alpha\gamma)_{c,v} + u_{\mathrm{bs}} + \varepsilon_{\mathrm{bs},c,v}, \\ u_{\mathrm{bs}} \overset{\mathrm{iid}}{\sim} \mathcal{N}(0, \tau^2_{\mathrm{bs}}), \qquad \varepsilon_{\mathrm{bs},c,v} \overset{\mathrm{iid}}{\sim} \mathcal{N}(0, \tau^2). \end{gathered} \tag{8.29}$$

For each sphere size, many-to-one contrasts compare each method to the reference $c_0 = w$-dTNV, with Benjamini–Hochberg adjustment across planned comparisons within that sphere. Since the model is fitted on the log scale, $\exp(\Delta_{c,v}) < 1$ indicates lower recovery than w-dTNVfor sphere v.

8.6.2 Clinical data

Clinical data were acquired at Oxford University Hospitals from 10 patients undergoing ^{90}Y SIRT: one patient was used for hyperparameter selection and nine for testing. PET was acquired on a GE Discovery 710 PET/CT system and SPECT on a GE Discovery NM/CT 670 system. Informed consent is not necessary for retrospective image reviews of this nature at Oxford University Hospitals. PET was acquired with two bed positions (15 min each) and SPECT with 15 min. Energy windows were 50–150 keV.

Tumour VOIs were delineated once per patient on the vendor Q.Clear β-4000 PET reconstruction using an automatic gradient-based method [**mikellImpact90YPET2018**] and propagated unchanged across reconstructions so that differences reflected reconstruction effects rather than VOI definition. A homogeneous liver background VOI was drawn manually, avoiding motion and visible artefacts. The training-patient lesion and background VOIs are shown in Figure **??**.

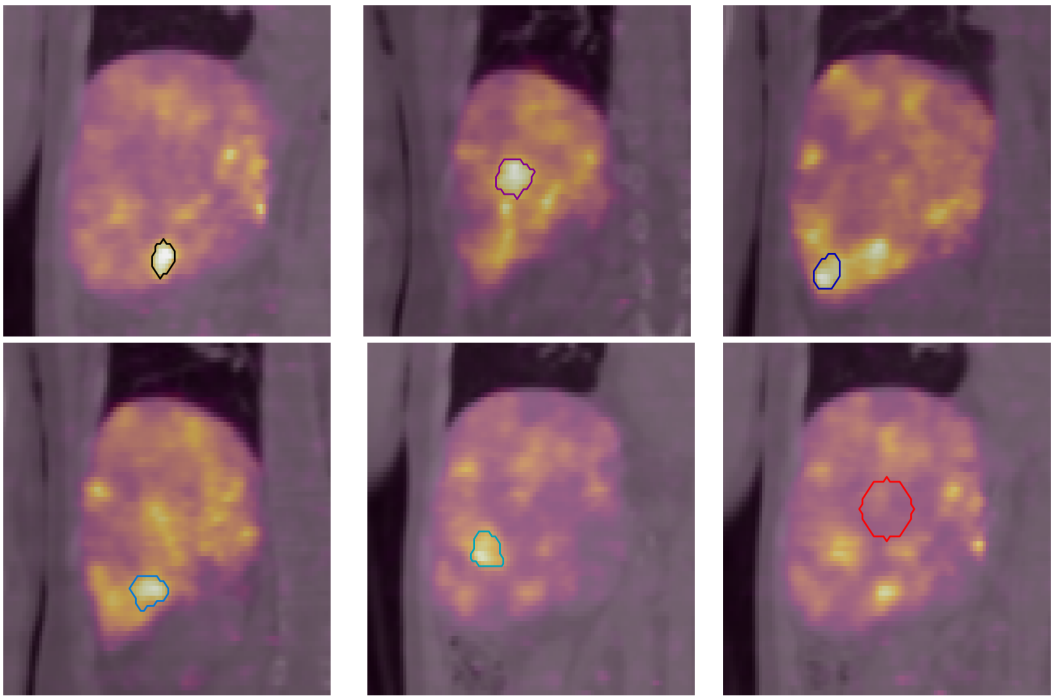


Figure 8.2: Training-patient VOIs. The five identified lesions and the background reference region are shown on the vendor PET reconstruction with registered attenuation maps. Top row: lesions 1–3. Bottom row: lesions 4–5 and background.

For patient p, lesion l, and method c, define tumour-to-background ratio (TBR)

$$\mathrm{TBR}_{p,l,c} = \frac{\bar{C}_{p,l,c}}{\bar{C}_{p,\mathrm{bkg},c}}, \tag{8.30}$$

and background noise (percentage CoV)

$$\mathrm{CoV}_{p,c}(\%) = 100\,\frac{\mathrm{SD}_{p,c}}{\bar{C}_{p,\mathrm{bkg},c}}, \tag{8.31}$$

where $\mathrm{SD}_{p,c}$ is the within-background-VOI standard deviation.

We analyse $\log(\mathrm{TBR})$ using a linear mixed-effects model with fixed method effects and random intercepts for patient and lesion-within-patient:

$$\begin{aligned} y_{p,l,c} &:= \log\big(\mathrm{TBR}_{p,l,c}\big) = \beta_0 + \alpha_c + u_p + v_{p,l} + \varepsilon_{p,l,c}, \\ u_p &\overset{\mathrm{iid}}{\sim} \mathcal{N}(0,\tau_{\mathrm{p}}^2), \qquad v_{p,l} \overset{\mathrm{iid}}{\sim} \mathcal{N}(0,\tau_{\mathrm{l}}^2), \qquad \varepsilon_{p,l,c} \overset{\mathrm{iid}}{\sim} \mathcal{N}(0,\tau^2). \end{aligned} \tag{8.32}$$

Background noise is analysed analogously at the patient level using

$$\begin{aligned} y_{p,c}^{\mathrm{CoV}} &:= \log(\mathrm{CoV}_{p,c}(\%)) = \beta_0 + \alpha_c + u_p + \varepsilon_{p,c}, \\ u_p &\overset{\mathrm{iid}}{\sim} \mathcal{N}(0,\tau_{\mathrm{p,CoV}}^2), \qquad \varepsilon_{p,c} \overset{\mathrm{iid}}{\sim} \mathcal{N}(0,\psi^2). \end{aligned} \tag{8.33}$$

Planned many-to-one contrasts compare dTV, w-TNV, and SHKEM to w-dTNV, with BH adjustment across the three comparisons within each endpoint. Results are reported as ratios $\exp(\Delta)$ on the original scale, where values below one indicate lower TBR or lower background CoV than w-dTNV.

8.7 Results

8.7.1 Hyperparameter selection

Regularisation weights β_{PET} and β_{SPECT} in (**??**) were tuned on a single training patient by sweeping the TBR–$\mathrm{CoV}_{\mathrm{bkg}}$ trade-off and selecting an operating point on the Pareto frontier [**paretoNewTheoriesEconomics1897**] with background CoV approximately matched to the vendor reconstruction. The selected weights were then fixed for all phantom and clinical experiments:

Table 8.1: Selected regularisation weights (tuned on a single training patient) used for all phantom and clinical experiments.

Method	β_{PET}	β_{SPECT}
w-dTNV	0.01	0.02
w-TNV	0.0075	0.02
dTV	0.02	–

For SHKEM, hyperparameters followed Deidda *et al.* [**deiddaTripleModalityImage2023**]; the PET subiteration displayed and analysed here (27) was chosen to approximately match the background CoV of the vendor reconstruction.

Figure **??** shows the training-patient trade-off curves for two representative lesions. Lesion 1 shows only a modest initial benefit from synergistic information and is largely insensitive to further increases in SPECT weighting until over-regularisation introduces negative bias. Lesion 2 shows a clearer synergistic benefit: increasing SPECT weighting shifts the TBR–CoV$_{\text{bkg}}$ curve towards higher TBR at comparable background CoV up to a distinct elbow region, after which excessive coupling reduces TBR and/or increases background CoV. For lesion 2, the PET image obtained with *w*-dTNValso exhibited features consistent with a CTAC artefact; this is discussed in Section **??**.

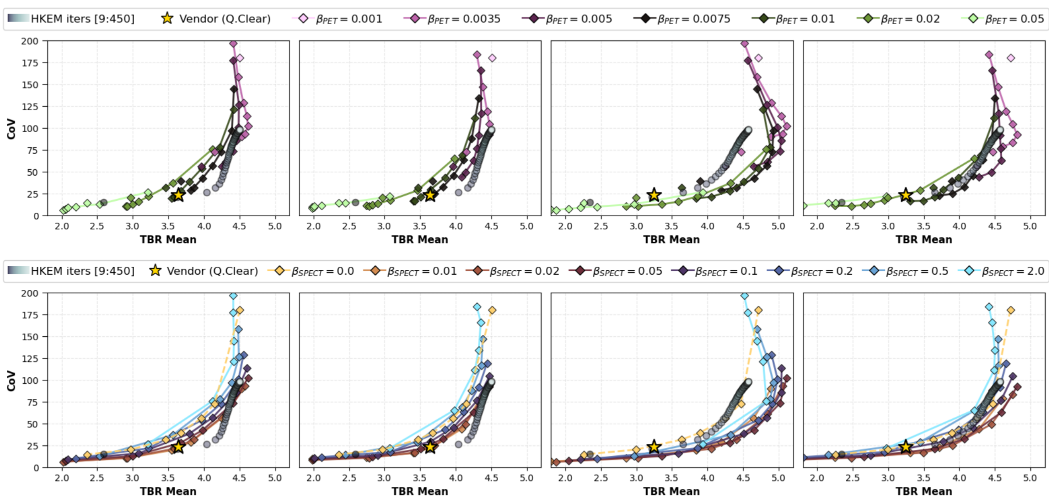


Figure 8.3: Training-patient hyperparameter sweep. Impact of varying PET weighting while holding SPECT regularisation constant (top row) and varying SPECT weighting while holding PET regularisation constant (bottom row) on TBR and CoV$_{\text{bkg}}$ for lesions 1 and 2. The first and third columns show *w*-dTNV; the second and fourth show *w*-TNV.

8.7.2 Phantom results

A representative reconstruction from a single bootstrapped PET realisation is shown in Figure **??** (PET and corresponding SPECT; for SHKEM, SPECT is the separately reconstructed guidance image; dTVdoes not reconstruct SPECT).

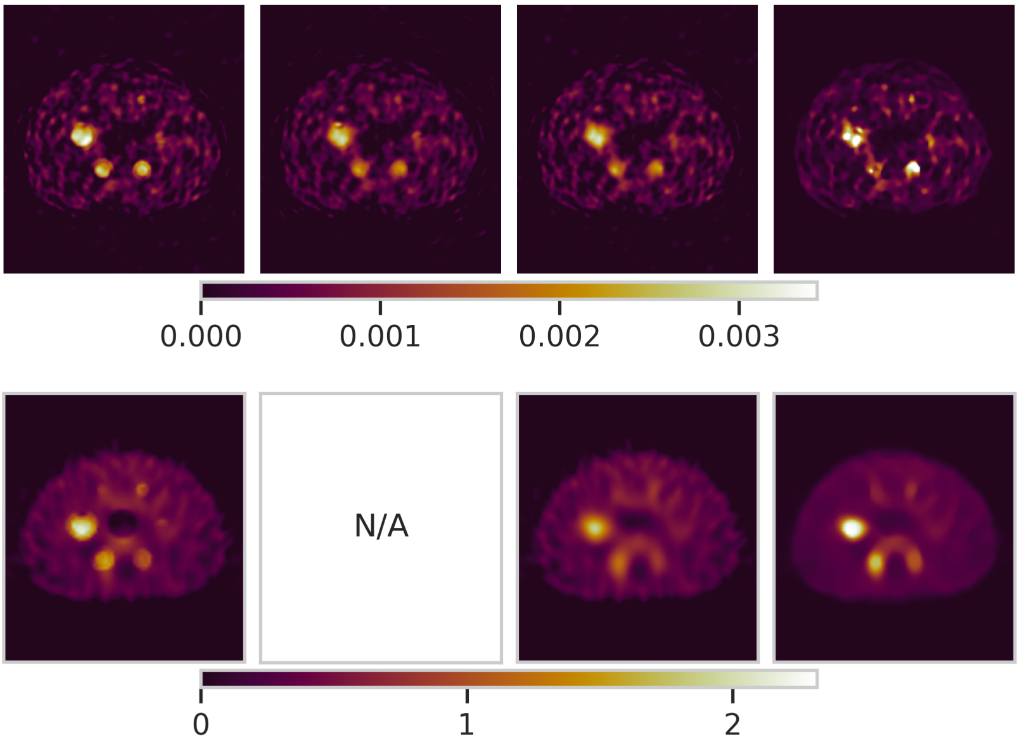


Figure 8.4: Visual comparison of NEMA phantom reconstructions for a randomly chosen bootstrap. From left to right: *w*-dTNV, dTV, *w*-TNV, SHKEM (subiteration 27). PET reconstructions are shown on the top row and SPECT on the bottom row (SPECT guidance for SHKEM). Images are displayed in reconstruction intensity units, proportional to activity concentration.

Figure **??** and Table **??** summarise sphere recovery. Quantitative analysis demonstrated that *w*-dTNVachieved significantly higher recovery than dTVand *w*-TNVfor all sphere sizes $\leq 22\,\mathrm{mm}$. *w*-dTNVshowed similar recovery to SHKEM in the larger spheres but significantly improved recovery for the two smallest spheres at the matched operating point.

Figure **??** summarises voxel-wise reproducibility across bootstraps. At the matched operating point, dTVand *w*-TNVyield lower voxel-wise CoV than *w*-dTNVin the evaluated VOIs, consistent with stronger effective smoothing. SHKEM exhibits higher variability in background and high-count regions but im-

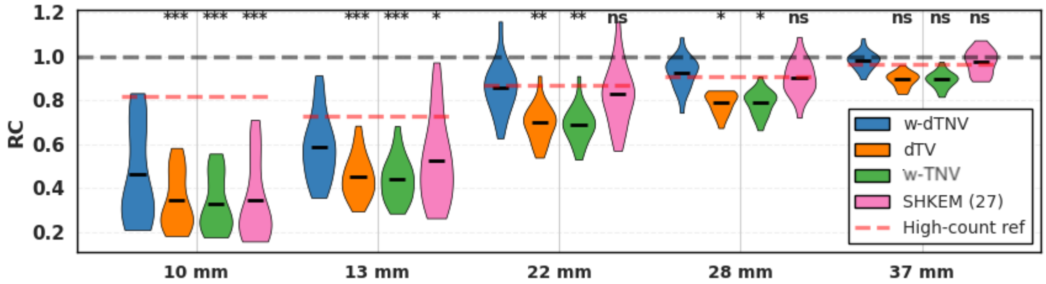


Figure 8.5: Recovery coefficient (RC) for NEMA phantom spheres for *w*-dTNV, dTV, *w*-TNVand SHKEM (subiteration 27). Also shown are the equivalent RCs for a high-count, long-OSEM reconstruction. The grey line indicates perfect recovery. Statistical significance is reported for paired comparisons versus *w*-dTNV(Benjamini–Hochberg adjusted): * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Method	v	$\exp(\Delta_{c,v})$	95% CI	p_{BH}
dTV	37 mm	0.9113	[0.8001, 1.0380]	0.2472
	28 mm	0.8543	[0.7500, 0.9730]	0.0267
	22 mm	0.8165	[0.7168, 0.9300]	0.0034
	13 mm	0.7819	[0.6865, 0.8906]	0.00043
	10 mm	0.7706	[0.6766, 0.8777]	$< 10^{-4}$
w-TNV	37 mm	0.9108	[0.7997, 1.0374]	0.2472
	28 mm	0.8544	[0.7502, 0.9732]	0.0267
	22 mm	0.8090	[0.7103, 0.9215]	0.0034
	13 mm	0.7598	[0.6671, 0.8654]	0.0001
	10 mm	0.7384	[0.6483, 0.8410]	$< 10^{-4}$
SHKEM (27)	37 mm	0.9931	[0.8720, 1.1312]	0.9175
	28 mm	0.9739	[0.8551, 1.1093]	0.6906
	22 mm	0.9616	[0.8443, 1.0953]	0.5559
	13 mm	0.8723	[0.7659, 0.9936]	0.0397
	10 mm	0.7381	[0.6480, 0.8407]	$< 10^{-4}$

Table 8.2: Effect ratios $\exp(\Delta_{c,v})$ (comparator vs. baseline *w*-dTNV) with 95% confidence intervals and BH-adjusted p-values, reported by sphere diameter v. Values below one indicate lower recovery coefficient than *w*-dTNV.

proved stability in the cold lung insert. Sphere-level CoV trends were consistent with the voxel-wise analysis, with *w*-TNVand dTVgenerally providing the lowest variability across the evaluated sphere diameters and SHKEM the highest.

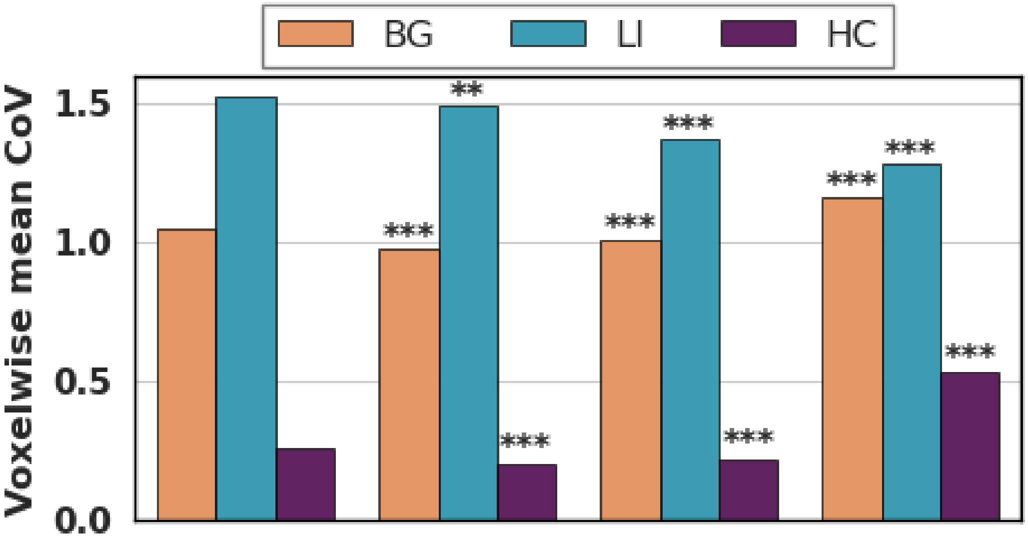
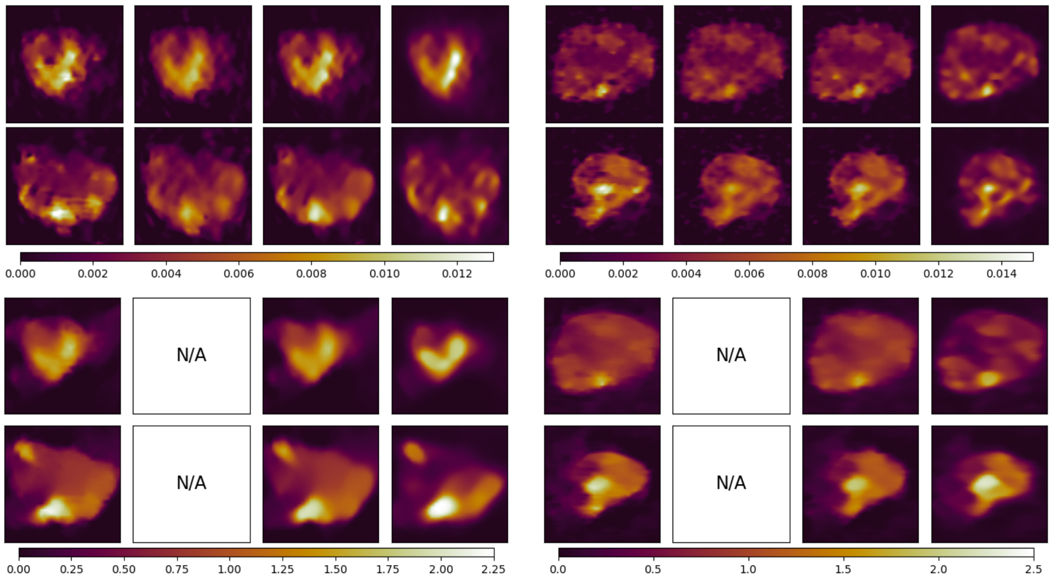


Figure 8.6: Mean voxel-wise coefficient of variation (CoV) across 20 bootstrapped PET realisations within each VOI. Statistical significance is reported for paired comparisons versus *w*-dTNV(BH-adjusted): * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

8.7.3 Clinical results

Figure **??** shows representative PET and SPECT reconstructions for the training patient. These images illustrate the operating-point behaviour used to select the fixed regularisation weights: *w*-dTNVimproves lesion contrast where the SPECT and CTAC-derived guidance contain useful structural information, while the response is weaker where there is no clear anatomical correlate.

Across the test cohort, *w*-dTNVyields higher lesion TBR than dTV, *w*-TNV, and SHKEM at the matched operating point, while background CoV differences are smaller and method-dependent. SHKEM reconstructions appeared visually smoother and, as in the phantom, appeared more effective at suppressing cold-background noise:



Axial views. **Coronal views.**

Figure 8.7: Visual comparison of reconstruction methods for training patient SIRT3. Columns show *w*-dTNV, dTV, *w*-TNVand SHKEM (subiteration 27). The upper two quadrants show PET reconstructions; the lower two show SPECT reconstructions or guidance images. Images are displayed in reconstruction intensity units, proportional to activity concentration.

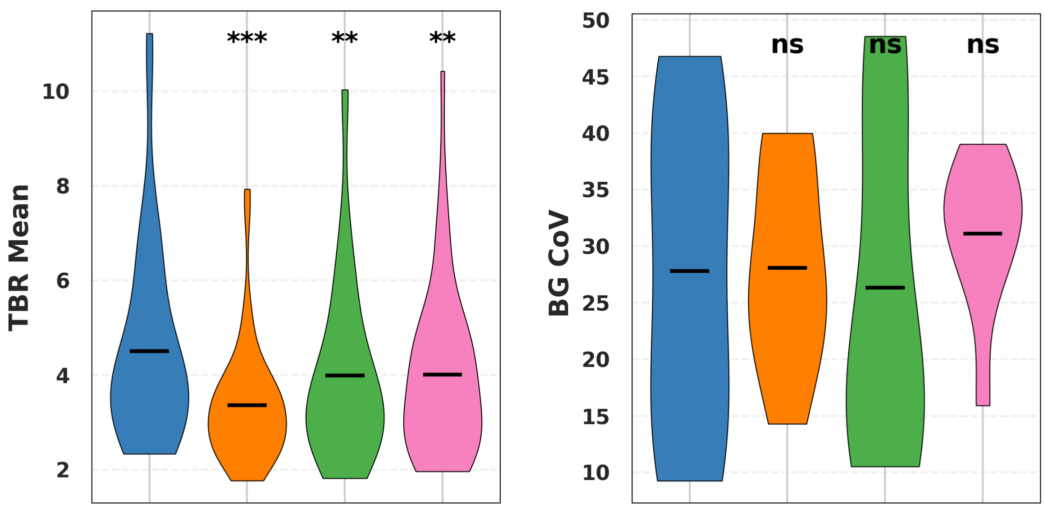


Figure 8.8: Clinical cohort summary: lesion-wise mean TBR (left) and patient-wise background CoV (right). Statistical significance is reported for comparisons versus *w*-dTNV(BH-adjusted): * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Method	$\exp(\Delta)$	95% CI	p_{BH}
dTV	0.802	$[0.758, 0.849]$	$< 10^{-4}$
w-TNV	0.915	$[0.865, 0.968]$	0.004
SHKEM (27)	0.935	$[0.884, 0.990]$	0.020

Table 8.3: Effect ratios $\exp(\Delta)$ with 95% confidence intervals and BH-adjusted p-values for mean tumour-to-background ratio (TBR), comparing each method to w-dTNV. Values below one indicate lower TBR than w-dTNV.

Method	$\exp(\Delta)$	95% CI	p_{BH}
dTV	1.183	$[0.903, 1.549]$	0.35
w-TNV	1.007	$[0.833, 1.167]$	0.984
SHKEM (27)	1.339	$[0.933, 1.913]$	0.093

Table 8.4: Effect ratios $\exp(\Delta)$ with 95% confidence intervals and BH-adjusted p-values for background CoV, comparing each method to w-dTNV. Values above one indicate higher background CoV than w-dTNV.

8.8 Discussion

The phantom results indicate that functional coupling alone (w-TNV) is insufficient to recover small-sphere boundaries at the matched operating point, and that anatomical guidance (explicit in w-dTNV, implicit in SHKEM through CT-informed kernel features) is beneficial for boundary localisation. The recovery gain for w-dTNVrelative to w-TNVand dTVis most pronounced for the smallest spheres, consistent with guidance stabilising edge geometry when PET statistics are poor and SPECT resolution is limited.

The reproducibility analysis highlights a trade-off: dTVand w-TNVprovide lower voxel-wise CoV than w-dTNVat the selected operating point, consistent with stronger effective smoothing. All three variational methods outperform SHKEM in background and high-count repeatability, whereas SHKEM is most stable in the cold lung insert. A plausible mechanism is that SHKEM kernel weights in near-zero regions favour local averaging among consistently low voxels, so "zeros reinforce zeros" under positivity constraints, improving cold-region stability while not necessarily improving reproducibility elsewhere.

Notably, dTVand w-TNVproduced broadly similar PET appearances in the phantom, despite dTVincluding μ-directional guidance and w-TNVincluding

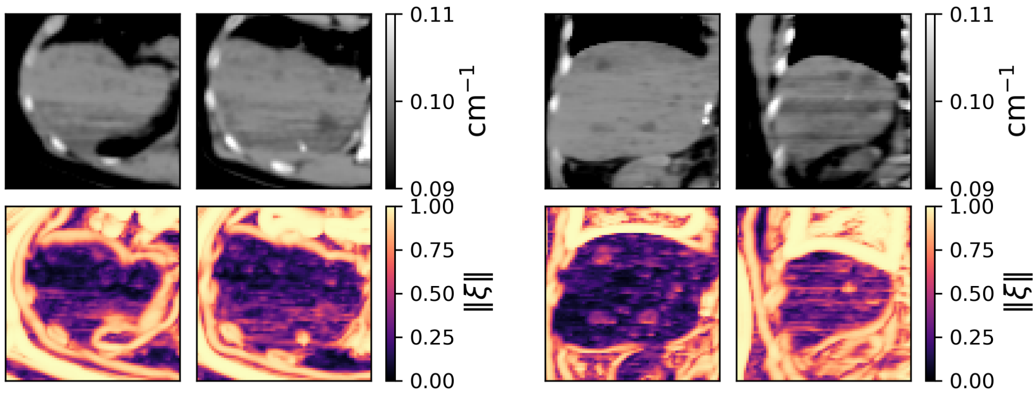
PET/SPECT coupling. This suggests that, in this experiment, neither CTAC-derived directional guidance alone nor isotropic PET/SPECT coupling alone was sufficient to recover the smallest structures. The improved recovery observed with w-dTNVis therefore consistent with a complementary mechanism: the higher-count SPECT image provides additional functional support for the location of activity transitions, while the μ-derived directional field reduces penalisation of gradients aligned with anatomical boundaries.

Clinically, lesion responses to synergy and directional guidance are heterogeneous. Lesions with clear anatomical correlates (e.g. calcification or liver capsule boundaries visible in the μ-map) tend to benefit more from μ-guided anisotropy and PET/SPECT coupling, while lesions lacking such correlates may show limited benefit. This heterogeneity is consistent with the design goal of w-dTNV: CTAC guidance steers *orientation* preferences, but cannot invent edges unsupported by the functional data.

For example, lesion 2 in the training patient shows substantial improvement in the TBR–CoV trade-off as SPECT weighting is increased (Figure **??**). Post-hoc examination indicates that this lesion lies adjacent to a calcified region visible in the μ-map, providing a sharp anatomical edge that coincides with the functional boundary. Lesion 3 also improves under both w-dTNVand w-TNV; this lesion abuts the liver on three sides, where the soft-tissue interface generates reasonably strong anatomical contrast. For w-dTNV, the μ-map liver boundary provides explicit directional guidance aligned with the functional uptake margin, while for w-TNVthe SPECT boundary may be sufficiently well defined to support PET/SPECT coupling even without directional CTAC guidance.

Conversely, lesion 1 shows little benefit from increasing SPECT weighting until excessive weighting introduces negative bias. Visual inspection shows no obvious anatomical correlate at the tumour margin, and the PET and SPECT images appear already well aligned. The reason for this lesion's limited response to synergistic reconstruction remains unclear.

A practical limitation is sensitivity to CTAC artefacts. Since ξ_n in (**??**) depends



Axial slice. **Coronal slice.**

Figure 8.9: Training-patient CTAC-derived guidance. The CTAC-derived attenuation map μ (top row; cm^{-1}) and magnitude of the w-dTNVdirectional weighting field $||\xi||$ (bottom row) are shown for lesions 1 and 2. CTAC streak artefacts are visible in μ and are reproduced in $||\xi||$, providing a pathway for artefact transfer into the emission reconstruction.

on $D\mu$, structured CT artefacts (e.g. arms-down streaking) can imprint into $||\xi||$ and therefore into the directional weighting field, providing a pathway for artefact transfer into the emission reconstructions. In the training patient, the low-dose PET CTAC was acquired with arms alongside the torso, introducing streaking and beam-hardening artefacts in the μ-map. The stabiliser η mitigates amplification of small gradients but cannot reliably suppress coherent high-contrast artefacts. SHKEM anatomical guidance was derived from the SPECT attenuation map, acquired with arms raised, rather than from the low-dose PET CTAC used to construct the w-dTNVdirectional weights; this difference in guidance input may contribute to the absence of streak-related artefacts in the SHKEM reconstructions.

The contrast with the NEMA phantom is instructive (Figure **??**). In the phantom, the CTAC is largely free of coherent streak artefacts and the resulting $||\xi||$ follows the true boundaries and insert edges. In this setting, directional smoothing attenuates small high-frequency fluctuations without introducing structured bias.

These observations motivate future work on robust direction-field estimation and artefact-aware guidance strategies. From a practical perspective, if a low-dose CTAC is used to provide directional information, acquisition protocols that minimise artefacts, such as arms raised where feasible, are preferable.

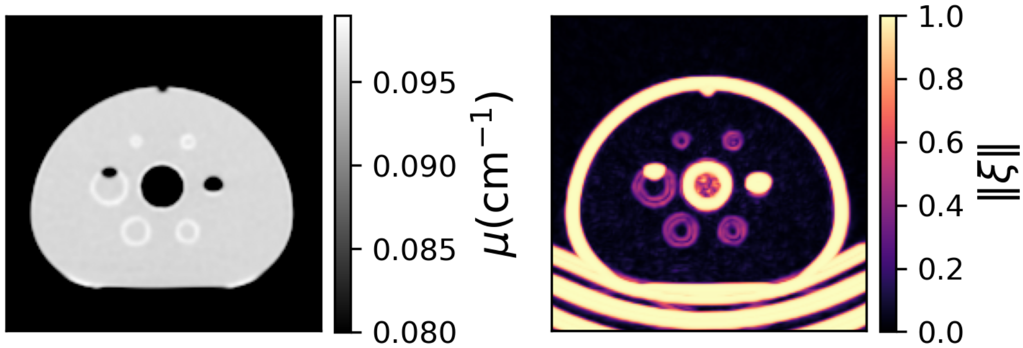


Figure 8.10: NEMA phantom, central axial slice. CTAC-derived attenuation map μ (left; cm^{-1}) and magnitude of the w-dTNVdirectional weighting field $\|\xi\|$ (right). With a high-quality CTAC free of prominent streak artefacts, $\|\xi\|$ concentrates around true object boundaries and insert edges, enabling stable μ-map-guided directional smoothing.

To preserve direct comparability with SHKEM, regularisation weights were selected once on the clinical training case and quantitative evaluation focused on PET. This means the robustness of the chosen weights across acquisition protocols and lesion presentations was not fully characterised. The jointly reconstructed SPECT images were assessed qualitatively only. Although they appeared visually less noisy while maintaining lesion conspicuity, a rigorous assessment of SPECT quantification would require modality-specific lesion delineation and evaluation protocols. The sensitivity of w-dTNVto residual PET/SPECT/CT registration error also remains an important topic for future work.

More broadly, this chapter focuses on the favourable case in which PET and SPECT image the same underlying ^{90}Y microsphere distribution. The same framework may be extendable to settings with non-identical functional distributions but shared anatomy, such as planning–treatment image pairs (^{99m}Tc-MAA versus post-treatment ^{90}Y), multi-nuclide imaging, or multi-timepoint studies. In such cases, the directional/anatomical component remains well motivated, but the inter-modality coupling strength may need to be relaxed or made spatially adaptive to avoid transferring modality-specific or temporally evolving uptake features.

8.9 Conclusion

This chapter validated the w-dTNV prior of Chapter **??** in joint ^{90}Y PET/SPECT reconstruction with CTAC-derived directional guidance. The key modelling choice is to use CTAC information directionally (via $\Pi_{m,n}$) while preserving a two-modality Jacobian, retaining the computational advantages of the 2×2 Gram structure (Chapter **??**) and avoiding unstable intensity coupling to CT.

On a bootstrapped NEMA IEC phantom, w-dTNVimproves small-object recovery relative to w-TNV, dTV, and a sequential hybrid kernel baseline at a matched noise operating point, with significant gains over the isotropic and non-synergistic variational baselines. Repeatability improves relative to SHKEM in the background and high-count regions, although it is reduced relative to the more strongly smoothing w-TNVand dTVbaselines and SHKEM remains most stable in the cold lung insert. In a post-SIRT patient cohort of 45 lesions in nine patients, w-dTNVyields significantly higher lesion TBR than the comparators at approximately matched background CoV. Lesion responses are heterogeneous: lesions with clear anatomical correlates, such as calcification or liver boundaries, benefit most from synergy and μ-map-informed edge alignment, whereas lesions without corresponding anatomical structure show more limited improvement. These findings support CTAC-guided synergistic variational coupling as a practical route towards more reliable post-SIRT ^{90}Y activity imaging for personalised dosimetry, while highlighting the need for careful handling of CT artefacts and PET/SPECT/CT registration in clinical translation.

Chapter 9

Discussion, Conclusion & Future Work

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Appendix A

`simind-python-connector`: Python Workflows for SIMIND Simulations

This appendix documents the `simind-python-connector` package used to support reproducible SIMIND-based simulation workflows in this thesis. The inspected package version was `1.0.1`, released on 16 April 2026 and archived on Zenodo with DOI 10.5281/zenodo.19605413. The public repository is available at GitHub.

TODO: Replace the DOI URL above with the Zenodo-managed citation key for DOI 10.5281/zenodo.19605413 once the automatic bibliography import has added it.

A.1 Purpose and Scope

SIMIND is a Monte Carlo code for simulating clinical SPECT scintillation-camera imaging and related system effects [**ljungbergSIMINDMonteCarlo2012**, **ljungbergSIMINDMonteCarlo2015**]. It is especially useful in this thesis because ^{90}Y bremsstrahlung SPECT requires careful modelling of scatter, attenuation, collimator response, and septal penetration (Chapters **??** and **??**). The role of `simind-python-connector` is not to replace SIMIND, but to make SIMIND easier to drive from Python and easier to connect to reconstruction software.

The package is independent of SIMIND and is not affiliated with, endorsed by, or maintained by the SIMIND project or Lund University. SIMIND itself is not distributed with the Python package; a separately licensed SIMIND installation must be available on the local machine or inside the configured container environment. This separation is useful for reproducibility because the open Python workflow can be version controlled while the licensed simulation executable remains an external dependency.

The package provides two main capabilities. First, `SimindPythonConnector` offers a direct Python interface for configuring voxel phantoms, writing SIMIND input files, setting runtime switches, running SIMIND, and loading projection outputs as NumPy arrays. Second, adaptor classes translate the resulting data into reconstruction-library-native representations for STIR, Synergistic Image Reconstruction Framework (SIRF), and PyTomography workflows [**thielemansSTIRSoftwareTomographic2012**, **ovtchinnikovSIRFSynergisticImage2020**]. The package therefore sits between simulation and reconstruction: SIMIND produces projection data, while reconstruction is still performed by the target reconstruction framework.

TODO: Add Zenodo-managed BibTeX entry for PyTomography if the PyTomography adaptor is retained in the final cited text.

TODO: Add Zenodo-managed BibTeX entry for the Schneider CT-density conversion paper if this utility is discussed beyond software documentation.

A.2 Connector Workflow

The core workflow is deliberately explicit. A user constructs a connector from a base SIMIND configuration, output directory, and output prefix; supplies a source image and attenuation or density information; specifies energy windows; applies runtime switches such as radionuclide, collimator, photon multiplier, and random seed; and then calls `run()`. The connector writes SIMIND-compatible source and density files, saves the generated configuration, executes SIMIND, and loads the resulting Interfile projection data.

For simple Python use, the package can be installed from PyPI with `pip install simind-python-connector`; optional plotting dependencies are provided through `simind-python-connector[examples]`. The base package de-

Table A.1: Main `simind-python-connector` interfaces relevant to thesis workflows.

Interface	Role
`SimindPythonConnector`	Backend-agnostic connector that runs SIMIND and returns projection outputs as NumPy arrays with header and data-file paths.
`RuntimeOperator`	Container for per-run SIMIND runtime switches and optional orbit-file selection.
`StirSimindAdaptor`	Converts connector inputs and outputs for STIR-based reconstruction workflows.
`SirfSimindAdaptor`	Provides the analogous adaptor path for SIRF, which builds on STIR for PET/SPECT reconstruction.
`PyTomography SimindAdaptor`	Accepts torch tensors, executes the SIMIND connector path, and returns PyTomography-compatible projection tensors and headers.
`hu_to_density _schneider`	Converts CT Hounsfield units to material density using the packaged Schneider-style tissue lookup.

pends only on standard Python scientific and medical-image utilities such as NumPy, pydicom, and PyYAML. STIR, SIRF, PyTomography, and SIMIND are optional or external dependencies and are needed only for the corresponding adaptor or simulation paths.

A.3 Geometry and Data Conventions

A central difficulty in connecting simulation and reconstruction software is that array-order, projection-axis, and unit conventions differ between packages. The connector documentation therefore makes these conventions explicit. STIR and SIRF image arrays are handled as `(z, y, x)`. PyTomography object-space tensors are handled as `(x, y, z)`, while SIMIND input image files are written internally in `(z, y, x)` order. Projection arrays may also differ: raw Interfile loading is exposed as `(views, bins, axial)` when no singleton time-of-flight axis is present, whereas PyTomography SIMIND projections are used in `(theta, r, z)` order.

The connector also handles the common unit conversion between reconstruction libraries and SIMIND. SIMIND geometry parameters are configured in centimetres, whereas STIR/SIRF and many Python reconstruction workflows express voxel sizes in millimetres. The connector therefore derives the SIMIND voxel-size switch from the supplied voxel size and writes corresponding source and density geometry values into the generated configuration.

These details matter for the work in this thesis because an apparently small axis flip or millimetre-to-centimetre error would change the simulated collimator response, attenuation path lengths, or reconstructed orientation. The package examples and tests therefore include explicit geometry guardrails: fixed toy configurations, centre-of-mass checks, view-index checks, and backend-specific adaptor examples.

A.4 Validation and Reproducibility

The package separates lightweight tests from tests that need heavyweight external dependencies. Pure Python unit tests cover configuration handling, connector behaviour, Interfile parsing, density conversion, and optional-backend guards. Backend-dependent tests are marked separately for STIR, SIRF, PyTomography, and SIMIND, and are skipped automatically when the corresponding dependency is unavailable. This is important for continuous integration because most public CI runners cannot include a licensed SIMIND executable.

Dedicated Docker environments are supplied for STIR, SIRF, and PyTomography. These containers isolate reconstruction-library dependencies and can run either connector-only validation or SIMIND-dependent validation when a local SIMIND executable is mounted under the expected repository layout. The repository helper scripts look for the executable at `./simind/simind` and the SIMIND data directory at `./simind/smc_dir/`. This mirrors the thesis requirement that the simulation code remains external while the workflow around it remains inspectable.

The included examples cover basic NumPy simulations, runtime-switch comparisons, multi-window simulations, SCATTWIN/PENETRATE scoring comparisons, Schneider-style density conversion, adaptor-based OSEM reconstructions, and DICOM-driven setup paths. Together these examples define the intended reproducibility contract: the connector should make SIMIND runs scriptable, recordable, and portable across reconstruction frameworks, while leaving reconstruction algorithm choices to the software stack used downstream.

Appendix B

Convergence Diagnostics for the Long-Run Reference Reconstruction

The NRMSE comparisons in Chapter **??** use the final iterate of one long-run reconstruction as their reference image. This one-bed phantom reconstruction used the same objective as the 20-pass comparisons, but was continued for 1000 epochs with 36 stochastic updates per epoch, a Gershgorin-diagonal MM preconditioner, and a more conservative relaxation rate of $\eta_{\text{relax}} = 10^{-4}$. Only one such reference run was performed.

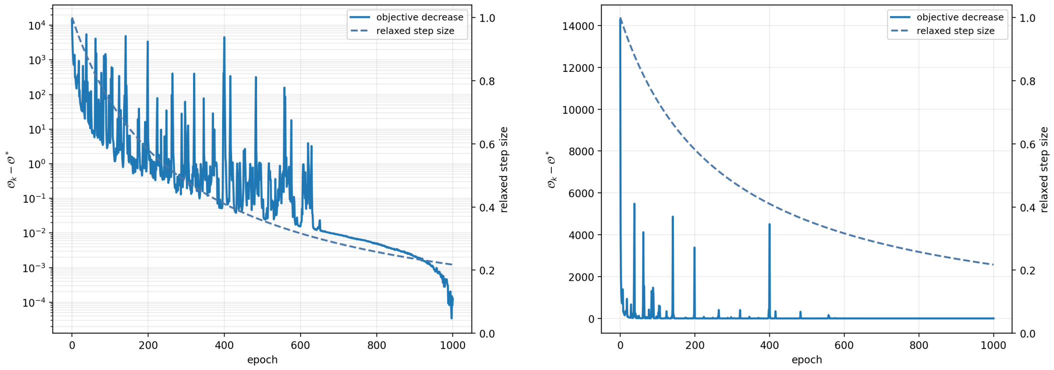
Let $\mathcal{O}_k = \mathcal{O}(\boldsymbol{z}^{(k)})$ be the recorded objective after epoch k, and define

$$\mathcal{O}^\star = \min_{0 \le k \le 1000} \mathcal{O}_k. \tag{B.1}$$

Here $\mathcal{O}^\star$ is the best objective observed during this run. It is not a proven global minimum, and the reference image should likewise be understood as a long-run numerical reference rather than ground truth.

The objective contains stochastic excursions early in the run, but its gap relative to the best observed value becomes negligible. Table **??** reports the final objective and iterate diagnostics. The run had achieved 99.9% of its total observed objective decrease by epoch 45, when the relaxed step remained 0.8606.

Together, the vanishing objective gap, relative objective change, and iterate velocity indicate that the reference was effectively stationary relative to the best



(a) Objective gap on a logarithmic scale. **(b)** Objective gap on a linear scale.

Figure B.1: Long-run objective diagnostics. The solid curve is $\mathcal{O}_k - \mathcal{O}^\star$ and the dashed curve is the relaxed step size. The linear view shows that most of the observed objective decrease occurs early, while the logarithmic view resolves the residual decrease over the complete 1000-epoch run.

Table B.1: Stationarity diagnostics for the 1000-epoch reference reconstruction.

Quantity	Value
Final objective gap / observed objective decrease	8.819×10^{-9}
Final relative objective change per epoch	1.717×10^{-11}
Final PET iterate velocity per epoch	7.285×10^{-8}
Final relaxed step size	0.2174
Epoch reaching 99.9% of observed decrease	45
Step size at that epoch	0.8606

observed iterate. The appreciable final step size makes step-size starvation an unlikely explanation for the plateau. These diagnostics support its use as a numerical reference for the short-run NRMSE comparisons, without asserting convergence to a global minimiser.

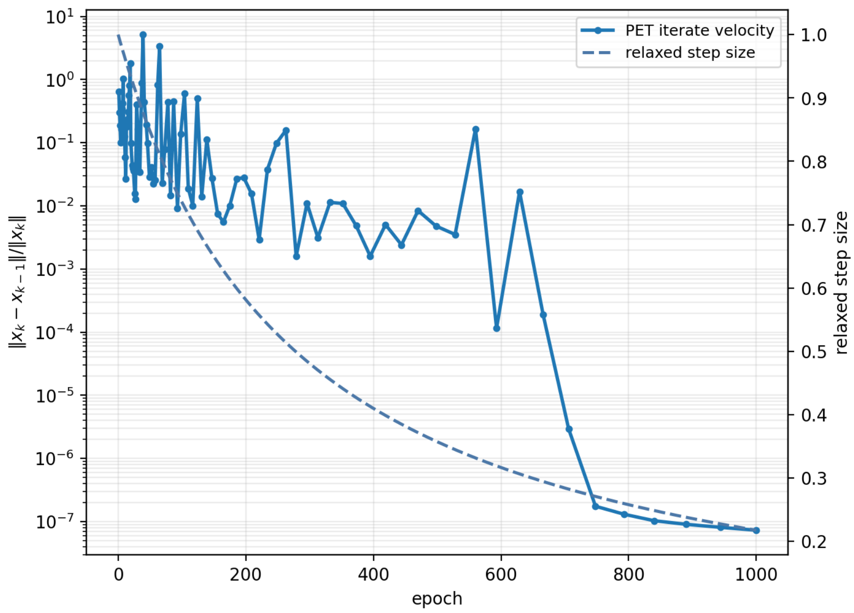


Figure B.2: Relative PET iterate velocity, $\|x^{(k)} - x^{(k-1)}\|/\|x^{(k)}\|$, and relaxed step size during the long-run reference reconstruction. The final velocity is 7.285×10^{-8} per epoch while the step size remains 0.2174.

Appendix C

Colophon

This is a description of the tools you used to make your thesis. It helps people make future documents, reminds you, and looks good.

(example) This document was set in the Times Roman typeface using LaTeX and BibTeX, composed with a text editor.